Ashok Leyland Pilot · vehicle dossier

Request

TEST – Bada Dost rear axle 5.13

Document status

ENGINEERING DRAFT — manufacturer and testing-agency acceptance pending

Configuration

N1 / diesel · revision 3

Comparison

Bada Dost i9 · supplied base

Readiness

0 tests supported by linked evidence; 88 test rows require verification or information.

Tables

Applicable set: 2, 3, 4E, 5, 6, 7, 8, 9, 10, 11, 12

Template status

Source editions and hashes are included in manifest.json. Current revision review is required. No unseen form is fabricated.

N1 regulatory register · review trail

Seed and source status

al-n1-regulatory-seed-4.0.0
Expanded known-topic inventory. Confirm current notified editions and clauses; no legal completeness claim is made.

CMVR 96; applicable IS 11852 / IS 15986 route · Service braking performance

CMVR rules: 96
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR 96; applicable IS 11852 / IS 15986 route · Parking braking performance

CMVR rules: 96
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR 96; applicable braking standard; N1 fitment / mandate review · ABS / electronic braking applicability and performance

CMVR rules: 96
Status: edition_review
Confirm ABS fitted / applicable. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-008 · Lighting installation and geometric visibility

CMVR rules: 102, 104, 105, 106, 107, 108, 109, 110, 111
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-008 (Rev.2):2019
Locator: Category-specific clause and applicable edition to be confirmed

Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-008 · Dipped-beam inclination / levelling verification

CMVR rules: 102, 104, 105, 106, 107, 108, 109, 110, 111
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-008 (Rev.2):2019
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-010 / AIS-012 / AIS-034; selected device standard · Lamp / light-source component approval

CMVR rules: 102, 104, 105, 106, 107, 108, 109, 110, 111
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: component
Source: AIS-017 (Part 6), published reference copy
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-002 (Part 1) · Indirect-vision installation and field of view

CMVR rules: 125
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-001 (Part 1) (Rev.2):2023 + Corrigendum 1 (12/2024)
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-001 (Part 1) · Mirror reflecting-surface dimensions

CMVR rules: 125
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: component
Source: AIS-001 (Part 1) (Rev.2):2023 + Corrigendum 1 (12/2024)
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-001 (Part 1) · Mirror radius of curvature

CMVR rules: 125
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: component

Source: AIS-001 (Part 1) (Rev.2):2023 + Corrigendum 1 (12/2024)
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-001 (Part 1) · Mirror reflectivity

CMVR rules: 125
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: component
Source: AIS-001 (Part 1) (Rev.2):2023 + Corrigendum 1 (12/2024)
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-001 (Part 1) · Mirror impact / mounting verification

CMVR rules: 125
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: component
Source: AIS-001 (Part 1) (Rev.2):2023 + Corrigendum 1 (12/2024)
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

IS 15139; safety-belt anchorages · Seat-belt anchorage strength

CMVR rules: 125
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: MoRTH Rajya Sabha answer 430, 22 July 2026
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

IS 15140; safety-belt assemblies and restraint systems · Seat-belt assembly approval and installation

CMVR rules: 125
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: component
Source: MoRTH Rajya Sabha answer 430, 22 July 2026
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-023; N-category seats, anchorages and head restraints; IS 15546 where applicable · Seats, seat anchorages and head restraints

CMVR rules: 125
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: MoRTH AIS-023 amendments, category-N seats / anchorages / head restraints
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-029; applicable N-category cab scope / notification · Cab structural strength applicability

CMVR rules: 124, 125
Status: edition_review
Confirm Cab-strength standard applies. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: AL workbook / standards register verification required
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-098; category / mass / edition review · Offset frontal collision applicability

CMVR rules: 125
Status: edition_review
Confirm Frontal-impact scope applies. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: AL workbook / standards register verification required
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

IS 2553 Part 2; CMVR 100 · Safety glazing approval and installation

CMVR rules: 100, 101
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: component_and_installation
Source: AIS-017 (Part 6), published reference copy
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

IS 15802 / IS 15804 / AIS-045; allocation and scope review · Wiping, washing and demisting applicability

CMVR rules: 100, 101
Status: edition_review
Confirm Windscreen fitted. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy

Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

IS 15633 / IS 15636; confirm tyre class · Tyre component approval and markings

CMVR rules: 94, 95
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: component
Source: AIS-017 (Part 6), published reference copy
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

Applicable tyre / vehicle installation provisions · Tyre load/speed rating and installation suitability

CMVR rules: 94, 95
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AL workbook / standards register verification required
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-110 · Temporary spare / run-flat configuration

CMVR rules: 94, 95
Status: edition_review
Confirm Temporary spare / run-flat fitted. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: AL workbook / standards register verification required
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR 98; IS 12222 / IS 11948 · Steering and manoeuvrability verification

CMVR rules: 98
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

IS 13942; applicable N-category external projection scope · External projections / construction review

CMVR rules: 124, 125
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AL workbook / standards register verification required
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

IS 12056 / applicable notified fuel-tank standard · Fuel tank / fire-risk and installation verification

CMVR rules: 115, 124
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AL workbook / standards register verification required
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

IS 1884 · Horn component approval

CMVR rules: 119, 124
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: component
Source: AIS-017 (Part 6), published reference copy
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

IS 15796 · Audible warning installation / sound level

CMVR rules: 119
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR 117; IS 11827; current edition review · Speedometer indication and installation

CMVR rules: 117
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: Government-hosted historical CMVR consolidation (not current consolidated law)

Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-137 (Part 3); route / fuel review · Type I tailpipe emissions

CMVR rules: 115, 116
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-137 (Part 3), amendments 1–9, including 08/2024
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-137 (Part 3) · Type III crankcase emissions applicability

CMVR rules: 115, 116
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-137 (Part 3), amendments 1–9, including 08/2024
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-137 (Part 3) · Type V durability / deterioration factors

CMVR rules: 115, 116
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-137 (Part 3), amendments 1–9, including 08/2024
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-137 (Part 3) · OBD monitoring and diagnostic demonstration

CMVR rules: 115, 116
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-137 (Part 3), amendments 1–9, including 08/2024
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-137 (Part 3), applicable amendments · Real-driving emissions / route applicability

CMVR rules: 115, 116
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Scope: vehicle
Source: AIS-137 (Part 3), amendments 1–9, including 08/2024
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

IS 14599; CMVR 124; current edition / route review · Engine net-power measurement

CMVR rules: 115, 116
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: Government-hosted historical CMVR consolidation (not current consolidated law)
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR 120; IS 3028; current edition review · Exterior / pass-by noise

CMVR rules: 120
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-004 (Part 3) · Vehicle / ESA electromagnetic emissions

CMVR rules: 124, 125
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-004 (Part 3) (Rev.1):2020
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-004 (Part 3) · Vehicle / ESA electromagnetic immunity

CMVR rules: 124, 125
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-004 (Part 3) (Rev.1):2020
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-065 · VIN, manufacturer plate and mandatory markings

CMVR rules: 122
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-065 with amendments 1–6
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-017 (Part 6) · Whole-vehicle safety conformity of production

CMVR rules: 126, 126A
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-017 / CMVR; confirm current notified procedure · Whole-vehicle type-approval dossier / extension

CMVR rules: 126, 126A
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AL workbook / standards register verification required
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR / VAHAN; administrative work item · Registration / approval identifiers

CMVR rules: 47, 48, 50, 51
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AL workbook / standards register verification required
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

Applicable notified CMVR / marking provisions · Mandatory labels and statutory markings applicability

CMVR rules: 126, 126A
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AL workbook / standards register verification required
Locator: Category-specific clause and applicable edition to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass

threshold is seeded.

CMVR 93 · Overall dimensions and overhang

CMVR rules: 93
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: Government-hosted historical CMVR consolidation (not current consolidated law)
Locator: Rule 93
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR 95; IS 11825; applicable axle-weight notifications · Permissible GVW, payload and axle loading

CMVR rules: 93, 95
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Annex C Appendix C-3, PDF pp.57–58
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR 98; IS 12222 · Turning circle and manoeuvrability

CMVR rules: 98
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: Government-hosted historical CMVR consolidation (not current consolidated law)
Locator: Rule 98
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR 96; applicable IS 11852 / IS 15986 route · Secondary / residual braking

CMVR rules: 96
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Annex C Appendix C-3, PDF pp.57–58
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR 96; applicable IS 11852 / IS 15986 route · Brake fade, recovery and laden/unladen performance

CMVR rules: 96
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Table 5; category-specific test selection required
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR 96; applicable braking standard · Brake circuit, warning, control and installation checks

CMVR rules: 96
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Annex C Appendix C-3, PDF pp.57–58
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR 96; IS 11852:2019; GSR 834(E), 11 Nov 2025 · 2027 braking standard transition

CMVR rules: 96
Status: edition_review
Confirm manufacture date and whether this is a new or existing model. Final notified route and edition still need confirmation.
Scope: vehicle
Source: MoRTH Rajya Sabha answer 430, 22 July 2026
Locator: Annex III (2)(xiii), PDF p.25; exact gazette clause review
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-057; device class / edition review · Retro-reflector component approval

CMVR rules: 102, 104, 105, 106, 107, 108, 109, 110, 111
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: component
Source: AIS-017 (Part 6), published reference copy
Locator: Table 5, lighting approval marks; verify device-specific standard
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-090; CMVR marking provisions · Conspicuity tape and rear marking applicability

CMVR rules: 102, 104, 105, 106, 107, 108, 109, 110, 111
Status: edition_review
Confirm Transport vehicle. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Table 5 rows 1–3, PDF pp.14–15

Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-089; heavy/long category and scope review · Heavy / long vehicle rear marking plate scope check

CMVR rules: 102, 104, 105, 106, 107, 108, 109, 110, 111
Status: not_applicable
Heavy-category scope is not assumed for N1. Enable the relevant scope fact if the actual classification / special route brings it into scope.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Table 5 row 3; combined category matrix is not blanket N1 applicability
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-012; selected signalling device standard · Direction, stop, position and registration lamps

CMVR rules: 102, 104, 105, 106, 107, 108, 109, 110, 111
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-008 (Rev.2):2019
Locator: Clause 6; device approval versus installation
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-010 / selected notified headlamp standard · Headlamp component approval and photometry

CMVR rules: 102, 104, 105, 106, 107, 108, 109, 110, 111
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: component
Source: AIS-008 (Rev.2):2019
Locator: Clause 6; selected lamp technology / class
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-012 / device-specific standard; AIS-008 · Front / rear fog lamp approval and installation

CMVR rules: 102, 104, 105, 106, 107, 108, 109, 110, 111
Status: edition_review
Confirm Fog lamps fitted. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: AIS-008 (Rev.2):2019
Locator: Clause 6; optional-fitment / class review
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

Selected notified lighting device standard; AIS-008 · Additional driving lamp approval and installation

CMVR rules: 102, 104, 105, 106, 107, 108, 109, 110, 111
Status: edition_review
Confirm Additional driving lamps fitted. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: AIS-008 (Rev.2):2019
Locator: Clause 6; optional-fitment review
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-034 / selected device and light-source standard · Replaceable filament / LED light-source approval

CMVR rules: 102, 104, 105, 106, 107, 108, 109, 110, 111
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-008 (Rev.2):2019
Locator: Clause 6; lamp approval compatibility
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

IS 16694; CMVR restraint provisions · Seat-belt and restraint installation

CMVR rules: 125
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: MoRTH Rajya Sabha answer 430, 22 July 2026
Locator: Annex III (2)(xii), PDF p.25; notified edition requires confirmation
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-099; applicable N1 scope / notified revision · Lateral collision scope and occupant protection

CMVR rules: 124, 125
Status: edition_review
Confirm Side-impact scope applies. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: Current notified standard / clause copy required
Locator: Obtain current notified standard scope; no assumed crash exemption
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

IS 14225 / applicable notified door-lock standard · Door latches and retention systems

CMVR rules: 124, 125
Status: input_gap

Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: Government-hosted historical CMVR consolidation (not current consolidated law)
Locator: Rule 125 / safety component provisions; verify current allocation
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-023; IS 15546 where applicable; N-category scope review · Seat head-restraint installation and strength

CMVR rules: 125
Status: edition_review
Confirm Head restraints fitted / required. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: MoRTH AIS-023 amendments, category-N seats / anchorages / head restraints
Locator: Current BIS edition and category scope required
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

IS 15223 / applicable notified scope · Cab interior fittings and injury protection scope

CMVR rules: 124, 125
Status: edition_review
Confirm Interior-fittings scope applies. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: Current notified standard / clause copy required
Locator: Current notification and N1 applicability required
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-084; applicable category and scope · Windscreen defrosting and demisting

CMVR rules: 100, 101
Status: edition_review
Confirm Defrost/demist scope applies. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: Current notified standard / clause copy required
Locator: Scope and notified edition required
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-021 / IS 15802 / IS 15804; scope review · Forward field of vision and wiping coverage

CMVR rules: 100, 101
Status: edition_review
Confirm Forward-vision scope applies. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Table 5 row 6; verify each standard allocation

Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

Applicable wheel/rim and tyre installation standard · Wheel / rim specifications and load suitability

CMVR rules: 94, 95
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Table 5 tyre checks; component allocation to be confirmed
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR 118; AIS-018 · Speed limiting function / device and tamper protection

CMVR rules: 118
Status: edition_review
Confirm Transport vehicle. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Table 5 row 22; Annex C Appendix C-3, PDF p.58
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-071; applicable category scope · Identification of controls, tell-tales and indicators

CMVR rules: 124, 125
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: Current notified standard / clause copy required
Locator: Current notified edition required
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR 124; notified fuel-system standard · Fuel leakage and installation safety

CMVR rules: 115, 124
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: Government-hosted historical CMVR consolidation (not current consolidated law)
Locator: Fuel system and safety component provisions
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-024 / AIS-025 / AIS-028; retrofit route · Gas-fuel conversion kit type approval and installation

CMVR rules: 115, 124
Status: edition_review
Confirm Gas-fuel retrofit / conversion. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Table 5 rows 19–20; approval scope must include conversion route
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

Applicable notified LNG requirements and PESO approval · LNG fuel-system and cylinder approval route

CMVR rules: 115, 124
Status: edition_review
Confirm LNG fuel route. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: Current notified standard / clause copy required
Locator: Route-specific official standards / notification required
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-137 Part 3; CMVR 115 · Emission family, road-load and worst-case selection

CMVR rules: 115, 116
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-137 (Part 3), amendments 1–9, including 08/2024
Locator: Applicable family / extension chapters and amendments
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-137 Part 3; CMVR 115 · Mass-emission conformity of production

CMVR rules: 115, 116
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-137 (Part 3), amendments 1–9, including 08/2024
Locator: Applicable emission CoP provisions
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR 115; AIS-137 applicable in-service provisions · In-service conformity / PUC applicability

CMVR rules: 115, 116
Status: input_gap

Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-137 (Part 3), amendments 1–9, including 08/2024
Locator: Type approval, in-service conformity and PUC are separate records
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR 115; AIS-137 applicable diesel provisions · Diesel free-acceleration smoke / opacity route

CMVR rules: 115, 116
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-137 (Part 3), amendments 1–9, including 08/2024
Locator: Fuel and applicable measurement route review
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

Applicable AIS-137 / notified consumption measurement procedure · Fuel consumption and CO2 declaration

CMVR rules: 115, 116
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-137 (Part 3), amendments 1–9, including 08/2024
Locator: Route-specific measurements; BEV uses the AIS-039 / AIS-040 work item
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-137 Part 3 · OBD interface, fault detection and calibration identification

CMVR rules: 115, 116
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-137 (Part 3), amendments 1–9, including 08/2024
Locator: Applicable OBD chapter and amendment
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-137 Part 3; applicable diesel provisions · SCR reagent / inducement and dosing system

CMVR rules: 115, 116
Status: edition_review
Confirm SCR fitted. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: AIS-137 (Part 3), amendments 1–9, including 08/2024
Locator: SCR fitment, reagent and applicable inducement requirements

Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-004 Part 3 · Electrical / electronic subassembly EMC approval

CMVR rules: 124, 125
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: component
Source: AIS-004 (Part 3) (Rev.1):2020
Locator: ESA requirements; Annex 21B change criteria
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-037; applicable notified component standards · Safety-critical component type approval and CoP

CMVR rules: 126, 126A
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: component
Source: AIS-017 (Part 6), published reference copy
Locator: Table 5 component approval marks; AIS-037 procedure requires current edition
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-017 Part 6; CMVR 126A · Vehicle safety CoP plan, test records and surveillance

CMVR rules: 126, 126A
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Table 5 and Annex C; approved plan and surveillance periods required
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-017; CMVR 126 / 126A · Manufacturer / production-plant extension review

CMVR rules: 126, 126A
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Change in manufacturer, model or production-site scope
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-017; CMVR 126; category-specific body scope · Incomplete / completed vehicle and body-builder approval

CMVR rules: 124, 125
Status: edition_review
Confirm Multi-stage / body-builder route. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: Current notified standard / clause copy required
Locator: Base chassis evidence does not automatically cover completed body
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

Applicable CMVR / notified load-body requirements · Goods-body, load restraint and body mounting review

CMVR rules: 124, 125
Status: input_gap
Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Scope: vehicle
Source: Current notified standard / clause copy required
Locator: Engineering inspection and applicable notified standards required
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

Applicable CMVR / notified coupling standard · Trailer coupling approval and installation scope

CMVR rules: 124, 125
Status: edition_review
Confirm Trailer coupling fitted. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: Current notified standard / clause copy required
Locator: Installed coupling class, trailer mass and notified N1 scope required
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR 129–137; applicable dangerous-goods standards · Dangerous-goods vehicle special requirements

CMVR rules: 129, 130, 131, 132, 133, 134, 135, 136, 137
Status: edition_review
Confirm Dangerous-goods use. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: Government-hosted historical CMVR consolidation (not current consolidated law)
Locator: Rules 129–137; tanker and goods-class-specific route
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-125; applicable CMVR notification · Road ambulance conversion and equipment scope

CMVR rules: 124, 125
Status: edition_review

Confirm Ambulance conversion. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: Current notified standard / clause copy required
Locator: Category/class conversion may change N1 classification
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-169; current final notification required · Adapted-vehicle approval and controls

CMVR rules: 124, 125
Status: edition_review
Confirm Adapted vehicle. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: Current notified standard / clause copy required
Locator: Do not treat draft GSR 416(E), June 2025, as a final notification
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

Applicable category-specific AIS / CMVR requirements · Other special-purpose / conversion route review

CMVR rules: 124, 125
Status: edition_review
Confirm Other special-purpose conversion. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: Current notified standard / clause copy required
Locator: Vehicle classification and conversion scope require review
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

IS 14812; heavy-vehicle scope · Rear underrun protection category scope check

CMVR rules: 124, 125
Status: not_applicable
Heavy-category scope is not assumed for N1. Enable the relevant scope fact if the actual classification / special route brings it into scope.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Table 5 row 14; not automatically required for N1
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

IS 14682; heavy-vehicle scope · Lateral protection category scope check

CMVR rules: 124, 125
Status: not_applicable
Heavy-category scope is not assumed for N1. Enable the relevant scope fact if the actual classification / special route brings it into scope.
Scope: vehicle

Source: AIS-017 (Part 6), published reference copy
Locator: Table 5 row 15; not automatically required for N1
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

AIS-069; heavy-vehicle scope · Front underrun protection category scope check

CMVR rules: 124, 125
Status: not_applicable
Heavy-category scope is not assumed for N1. Enable the relevant scope fact if the actual classification / special route brings it into scope.
Scope: vehicle
Source: AIS-017 (Part 6), published reference copy
Locator: Table 5 row 16; not automatically required for N1
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR 125H; AIS-140; transport-use and exemption review · Vehicle location tracking / emergency button applicability

CMVR rules: 124, 125
Status: edition_review
Confirm Location-tracking / emergency-button requirement applies. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: Current notified standard / clause copy required
Locator: Do not impose public-service vehicle rules on every N1 goods vehicle
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

CMVR fitness / permit / registration provisions · Transport fitness, permit and registration handover

CMVR rules: 62, 81
Status: edition_review
Confirm Transport vehicle. Unknown fitment / scope is retained for review.
Scope: vehicle
Source: Government-hosted historical CMVR consolidation (not current consolidated law)
Locator: Operational approval record; separate from homologation test
Acceptance: Use the confirmed edition and actual report acceptance criteria. No universal pass threshold is seeded.

Derived values · engineering review required

1st · Overall Ratio / AVTEC

21.44
Final-drive ratio changed. Source overall-ratio values need a reviewed formula; supplier-specific gear / reduction factors must be confirmed.

2nd · Overall Ratio / RSB

12.67
Final-drive ratio changed. Source overall-ratio values need a reviewed formula; supplier-specific gear / reduction factors must be confirmed.

2nd · Overall Ratio / AVTEC

12.71
Final-drive ratio changed. Source overall-ratio values need a reviewed formula; supplier-specific gear / reduction factors must be confirmed.

3rd · Overall Ratio / RSB

7.34
Final-drive ratio changed. Source overall-ratio values need a reviewed formula; supplier-specific gear / reduction factors must be confirmed.

3rd · Overall Ratio / AVTEC

7.25
Final-drive ratio changed. Source overall-ratio values need a reviewed formula; supplier-specific gear / reduction factors must be confirmed.

4th · Overall Ratio / RSB

4.56
Final-drive ratio changed. Source overall-ratio values need a reviewed formula; supplier-specific gear / reduction factors must be confirmed.

4th · Overall Ratio / AVTEC

4.56
Final-drive ratio changed. Source overall-ratio values need a reviewed formula; supplier-specific gear / reduction factors must be confirmed.

5th · Overall Ratio / RSB

3.64
Final-drive ratio changed. Source overall-ratio values need a reviewed formula; supplier-specific gear / reduction factors must be confirmed.

5th · Overall Ratio / AVTEC

3.67
Final-drive ratio changed. Source overall-ratio values need a reviewed formula; supplier-specific gear / reduction factors must be confirmed.

Reverse · Overall Ratio / RSB

20.46
Final-drive ratio changed. Source overall-ratio values need a reviewed formula; supplier-specific gear / reduction factors must be confirmed.

Reverse · Overall Ratio / AVTEC

20.44
Final-drive ratio changed. Source overall-ratio values need a reviewed formula; supplier-specific gear / reduction factors must be confirmed.

Table 2 of AIS-007 (Revision 5) TECHNICAL SPECIFICATION – M & N CATEGORY OF VEHICLES (FOUR WHEELERS AND ABOVE): PART A – GENERAL			
A1.0	**Details of Vehicle Manufacturer:**		
A1.1	Name & address of the manufacturer	Ashok Leyland Limited No.1, Sardar Patel Road, Guindy, Chennai 600 032	
A1.2	Telephone No	044-XXXXXXXX	
A1.3	Fax No.	044-XXXXXXXX	
A1.4	E-mail address	XXXXXXXX@ashokleyland.com	
A1.5	Contact person	XXXXXXXXXXX.XXX	
A1.6	Name of model and variants (Features differentiating the model and its variants to be given in a separate table)	BADA DOST i9 TNZX	
A1.6.1	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc)	Diesel (Maximum 7% bio-diesel blend)	
A1.7	Plant/(s) of manufacture		
A1.7.1	Name & address of vehicle manufacturing plant	**Ashok Leyland Limited** XXXXXX, XXXXXXX - XXXXXX	
A1.7.2	Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also)	**Ashok Leyland Limited** XXXXXX, XXXXXXX - XXXXXX	
A1.8	Importer's Name and address	NA	
A1.8.1	Telephone No	--	
A1.8.2	Fax No.	--	
A1.8.3	E-mail address	--	
A1.8.4	Contact person	--	
A2.0	Vehicle type:		
A2.1	Type of vehicle (Rigid / articulated / Tractor- Trailer combination / others)	Rigid	
A2.1.1	Usage (goods / passenger / others)	Goods	
A2.1.2	Control (Forward / semi-forward / normal / others)	Forward	
A2.1.3	Drive (4x2 or 4x4 or 6x2 or 6x4 or others)	4x2	
A2.1.4	Cab type (Fully built cab (Tiltable / Non-Tiltable) / sleeper cab / Front end structure / Cowl with wind shield / Cowl without wind shield)	Fully Built Cab, Non – Tiltable	
A2.1.5	Load body (fitted / not fitted)	Fitted - For Fully built variants Not Fitted - For CC variants	
A2.2	Category of vehicle as per IS 14272:2011	N1	
A2.3	Vehicle available modes		
A2.3.1	Vehicle Default mode Yes / No – If Yes then define	No	
A3.0	Vehicle Performance:	**GVW 2890 kg**	**GVW 3390 kg**
A3.1	Max. Recommended Gradeability (Stand-start) – in degrees	10 degs	8.5 degs
A3.2	Max. Design speed km/h		
A3.2.1	Vehicle Max Speed in unladen condition (km/h)	Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF	
A3.2.2	Vehicle Max Speed in laden condition (km/h)	Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF	

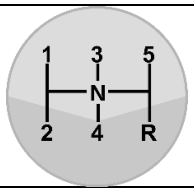
Manufacturer signature: _____ Test agency: _____

Table 3 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL				
B1.0	**Vehicle Dimensions:**		**2490 WB**	**2640 WB**
B1.1	Overall Length mm	**CC**	4750	4988
		FSD	4925	5225
		HSD		
B1.1.1	Total length (mm) (for articulated / combination vehicles)		NA	
B1.2	Width mm		1742	
B1.3	Height (Unladen) (mm)		1961	1948
B1.4	Wheel base (mm)		2490	2640
B1.4.1	Axle spacing in case of multi axle vehicles.		NA	
B1.5	Wheel track (mm)			
B1.5.1	Front		1480	
B1.5.2	Rear		1395	
B1.5.3	Other axles (for articulated/combination vehicles)		NA	
B1.6	Body overhang (mm)			
B1.6.1	Front end		1197	
B1.6.2	Rear end	CC	863	951
		FSD / HSD	1038	1188
B1.7	Frame overhang mm (in case of vehicles without complete body)			
B1.7.1	Front end		937	
B1.7.2	Rear end		808	958
B1.8	Inner dimensions of room or platform (For goods carriage vehicles only)		Refer Annexure T7 – C	
B1.8.1	Length (mm)			
B1.8.2	Width (mm)			
B1.8.3	Height (mm)			
B1.9	Lateral projection		≤2.5mm	
B1.10	Cargo box dimensions (mm)		Refer Annexure T7 – C	
B1.10.1	Length			
B1.10.2	Width			
B1.10.3	Height			
B2.0	**Weights:**			
B2.1	Vehicle kerb weight kg		Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C	
B2.1.1	Front axle 1			
B2.1.2	Front axle 2			
B2.1.3	Rear axle			
B2.1.4	Trailer axle (applicable for articulated / combination vehicles)			
B2.1.5	Total			
B2.2	Gross vehicle weight kg (for rigid vehicles) (Front, Rear & Total)			
B2.2.1	Maximum permissible axle weights (kg)			
B2.2.2	Front axle			
B2.2.3	Rear axle			
B2.2.4	Other axle		NA	
B2.3	Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles)		NA	
B2.3.1	Front axle		NA	
B2.3.2	Rear axle		NA	
B2.3.3	Trailer axle / other axles		NA	
B 2.3.4	If applicable, maximum mass of trailer whic5h may be coupled		NA	
B 2.3.5	Unbraked trailer		NA	
B2.4	Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.)		Refer Annexure T7-B	
B2.5	Seating capacity			

Manufacturer signature: _____ Test agency: _____

Table 3 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL					
B2.5.1	Maximum (Including driver) for completely built vehicles	D+2			
B3.0	**Tyres:**				
B3.1	Make	**APOLLO**	**CEAT**	**APOLLO**	**CEAT**
B3.1.1	Brand(s) name(s) and/or Trade description(s)	APOLLO Endumaxx LT	CEAT WINMILE X3-R	Endu Maxx LT	CEAT MILAZE LT
B3.1.2	Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142	Compliance to Wet Surfaces & Rolling Resistance as per AIS 142			
B3.1.3	Tyre rolling Resistance	7.84 N/kN	7.86 N/kN	8.13 N/kN	9.61 N/kN
B3.1.4	Tyre Class (C1 / C2 / C3) as per AIS:142				
B3.1.4.1	Front	C2			
B3.1.4.2	Rear				
B3.1.5	Category of use (Normal / Special / Snow/Severe Snow)	Normal			
B3.1.7	Tyre Tread Pattern along with Drawing	Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX	Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX	Refer - TC7719XX000003XXXF	Refer - SHL/16/2022-2023/300002XXXX/TA/XXXX
B3.1.8	Identification: TAC No. / BIS License No. / E- Marking	CM/L - 756XXXX	CM/L - 372XXXX	CM/L-756XXXX	CM/L – 670007XXXX CM/L - 372XXXX
B3.2	No. and arrangement of wheels				
B3.2.1	Front	2	2	2	2
B3.2.2	Rear	2	2	2	2
B3.2.3	Spare wheel	1	1	1	1
B3.2.4	Identification: TAC No. / BIS License No. / E- Marking	NA			
B3.2.5	Others (For articulated / combination vehicles)				
B3.3	Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.	7.00 R15 LT 12 PR 114/109 Q Radial Tube	7.00 R15 LT 12 PR 114/109 L Radial Tube	215/75 R15 LT 115/113 S Radial, Tubeless	215/75 R15 LT 115/113 S Radial, Tubeless
B3.3.1	Front wheel				
B3.3.2	Rear wheel				
B3.3.3	Spare wheel (Other than temporary use spare wheel)				
B3.3.4	Other (for articulated/combination vehicles)	NA			
B3.4	Rolling Radius				
B 3.4.1	Static rolling radius (only applicable for M2, M3, N and T category of vehicles.)	349 mm		325 mm	
B 3.4.2	Dynamic rolling radius, mm, as per IS 15633 / IS 15636	362 mm		339 mm	
B3.5	Inflation pressure – Unladen (kg/cm^2 / kPa)				
B3.5.1	Front, kg/cm^2	2.91 kg/cm2		2.56 kg/cm2	
B3.5.2	Rear, kg/cm^2	2.91 kg/cm2		2.56 kg/cm2	
B3.5.3	Other	NA			
B3.6	Inflation pressure – Laden (kg/cm^2 / kPa)				
B3.6.1	Front, kg/cm^2	3.61 kg/cm2		2.56 kg/cm2	
B3.6.2	Rear, kg/cm^2	6.77 kg/cm2		3.26 kg/cm2	
B3.6.3	Other	NA			
B3.7	Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided)	NA			
B3.7.1	Make (If Applicable)	NA			
B3.7.2	Brief Description of the system	NA			
B3.8	Temporary Spare Wheel / RFT- as per AIS-110 (If Provided)	NA			
B3.8.1	Type as per AIS-110	NA			
B3.8.2	Make	NA			
B3.8.3	Size	NA			

Manufacturer signature: _____ Test agency: _____

Table 3 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL						
B3.8.4	Load and speed rating		NA			
B3.8.5	Recommended max speed in Unladen condition		Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF			
B3.8.6	Recommended max speed Laden condition		Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF			
B4.0	**Transmission:**					
B4.1	Type (Manual / Automatic / semi-automatic) (Note: If automatic give all pertinent data)		Manual			
B4.2	Clutch Type (wet / Dry / Single plate / Multiplate / Hydraulic)		Dry / Single plate /Mechanical actuation			
B4.3	**Gear box**					
B4.3.1	Type		Fully synchronized 5 - forward gears with a Constant mesh reverse gear			
B4.3.2	Model name (if any)		XXX-XXX	YYYY-YY		
B4.3.3	Gear shifting control system (Sketch showing gear shifting arrangement)					
B4.3.4	No. of gears		Forward 5 + 1 Rev			
B4.4	Stall ratio of torque converter		NA			
B4.5.	**Sub transmission**		NA			
B4.5.1	Type		NA			
B4.5.2	Control system		NA			
B4.5.3	Gear ratio	High	4.62	4.60		
		Low	0.70	0.70		
B4.6	Final Drive (4x2 or 4x4 or 6x2 or 6x4 or others) Front / Rear / Both		(4 x 2) Rear			
B4.6.1	Type		Single Reduction, Hypoid, Semi-floating Banjo type			
B4.6.2	Reduction ratio		5.13			
B4.6.3	Differential type		Single Reduction, Hypoid, Semi-floating Banjo type			
B4.6.4	Final Drive ratio		5.13			
B4.7	Gear ratio / Gear box ratio / Overall ratio		**Gear Ratio**		**Overall Ratio**	
			RSB	**AVTEC**	**RSB**	**AVTEC**
	1st		4.62	4.60	23.70	21.44
	2nd		2.68	2.69	12.67	12.71
	3rd		1.51	1.49	7.34	7.25
	4th		1.0	1.0	4.56	4.56
	5th		0.70	0.71	3.64	3.67
	6th		-			
	Over drive		-			
	Reverse		4.39	4.38	20.46	20.44
B4.8	Hydraulic transmission		NA			
B4.8.1	Type		--			
B5.0	**Number of axles :**		2			
B5.1	Steered		1			
B5.2	Non-steered		1			
B5.3	Driven		1			
B5.4	Non-driven		1			
B6.0	**Front axle**					
B6.1	Type (Single / Multi / Retractable)		Single			
B6.2	Toe-in / Toe out (mm)		1 to 3 mm			
B6.3	Camber angle		0.5º (+1°,-0.5°)			
B6.4	Caster angle		10.55° (+1°/- 3°)			
B6.5	King pin angle		9°±15’			
B7.0	**Rear axle:**					
B7.1	Type (Single/Tandem/Tridem/Multi/Retractable)		Single			
B7.2	Toe-in/ Toe out mm, if applicable		-			
B7.3	Camber angle, if applicable		-			

Manufacturer signature: _____ Test agency: _____

Table 3 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL			
B7.4	Caster angle, if applicable	-	
B7.5	King pin angle, if applicable	-	
B8.0	**Steering system:**		
B8.1	Make	1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd	
B8.1.1	Type (Manual / Power assisted – Hydraulic / Power assisted – Electric / Other)	Power assisted – Hydraulic	
B8.2	**Steering wheel**		
B8.2.1	Identification Mark / Part No./Drawing No.	X0H001XX	
B8.2.2	Position (center/offset)	0	
B8.2.3	Outside dia. (mm)	389mm	
B8.2.4	**Steering column**		
B8.2.4.1	Make	M/s. XXXXXXX Ltd	
B8.2.4.2	Type / Model	Tiltable Steering column with 4 lobe I shaft design	
B8.2.4.3	Detailed drawing with material specifications	FZY0XX00	
B 8.2.4.1	Air Bag – Part No/ID No/Drawing No, Make	NA	
B 8.2.4.2	Air Bag deployment time as per IS 11939/AIS 96/AIS 98 for steering impact	NA	
B8.2.5	**Intermediate shaft**		
B8.2.5.1	Make	M/s. Rane NSK Steering systems Ltd	
B8.2.5.2	Type / Model	4 lobe design	
B8.2.5.3	Detailed drawing with material specifications	FZY0XX00	
B8.3	Maximum No. of rotation of steering wheel from lock to lock	1. XXXX - 4.1 2. YYYY - 4.1	
B8.4	Details of single / multiple combinations to be given in the form of an Annexure with reference to IS: 11939 –1996	NA	
B8.5	Detailed drawing of the mounting arrangement of Steering control assembly showing vertical / tilt / actual angle.	FZY0XX00	
B8.6	Offset of steering column with respect to the seat	0	
B8.7	**Steering Gear**		
B8.7.1	Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others)	Re-circulating ball type	
B8.7.2	Make	1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd	
B8.7.3	Steering gear ratio	1. XXXX – 17.8:1 2. YYYY – 18.2 :1	
B8.8	**Wheel lock angle (deg.)**	**Inner**	**Outer**
B8.8.1	Left	39	25
B8.8.2	Right	39	25
B8.9	**Power Assistance**		
B8.9.1	Type of assistance	Hydraulic	
B8.9.2	Make	XXXX & YYYY	
B8.9.3	Identification No./ Part No.	1. XXXX - FZZ00XXX 2. YYY - FZZ00XXX	
B8.9.4	Pressure setting (kg/cm^2 / bar / kPa)	1. XXXX – 105 Bar Max 2. YYYY – 120 Bar Max	
B8.10	Min turning circle diameter (mm) (as per IS:12222)	**2490 WB**	**2640 WB**
		10.0 m	10.9 m
B8.11	Min. turning circle clearance diameter (mm)	11.0m	11.9 m
B8.12	Coordinates of point defining test turning circle. (Applicable in case of vehicles without complete body which does not cover this point)	NA	
B9.0	**Clearance (Requirement as per AIS-053, If applicable):**	**2490 WB**	**2640 WB**
B9.1	Minimum road clearance (For other than M1)	214/206	193 / 185
B9.2	Road clearance from floor (for buses)	**NA**	

Manufacturer signature: _____ Test agency: _____

Table 3 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL				
B9.3	Approach angle, deg		20.5°	19.4°
B9.4	Departure angle, deg		23.1°	23.5°
B9.5	Ramp-over angle, deg		29.2°	26.7°
B9.6	Minimum Ground Clearance as per IS 9435:2004		214 / 206	193 / 185
B10.0	**Max. stable inclination (For buses as per AIS –031):**		NA	
B10.1	Left		--	
B10.2	Right		--	
B11.0	**Suspension :**		**2490 WB**	**2640 WB**
B11.1	Type and description (Leaf / Coil / Air / Semi-pneumatic/ Torsion bar)		Leaf	
B11.1.1	Front		Parabolic spring- Rigid axle suspension	
B11.1.2	Rear		Semi elliptic spring- Rigid axle suspension	
B11.2	Make			
B11.2.1	Front		Jamna auto industries ltd	
B11.2.2	Rear		Jamna auto industries ltd	
B11.3	Type of spring		Yes	
B11.3.1	Front		parabolic type	
B11.3.2	Rear		Semi elliptic type	
B11.4	If leaf spring		Yes	
B11.4.1	Main spring		Front : parabolic type Rear : Semi elliptic type	
B11.4.1.1	Stack height, at center		Front: 46mm & Rear: 70mm	Front: 51mm & Rear: 91mm
B11.4.1.2	Width at the center point / stack point		Front & Rear: 65mm	Front & Rear: 65mm
B11.4.1.3	Flat length		Front:1055 ± 3mm &Rear: 1140 ± 3.5mm	Front:1055 ± 3mm &Rear: 1180 ± 3.5mm
B11.4.1.4	Free camber		Front: 65mm± 6mm & Rear: 125mm ± 6mm	Front: 74mm± 6mm & Rear: 107mm ± 6mm
B11.4.1.5	No. of leaves	No. of leaves	Front: 3 & Rear: 6	Front: 5 & Rear: 8
		No. of spacers	Front: 2 & Rear: 1	Front: 4 & Rear: 0
B11.4.2	Auxiliary Spring		NA	
B11.4.2.1	Stack height at, at center		--	
B11.4.2.2	Width at the center point / stack point		--	
B11.4.2.3	Flat length		--	
B11.4.2.4	Free camber		--	
B11.4.2.5	No. of leaves		--	
B12.0	**Suspension- Shock absorber:**			
B12.1	Type and Number			
B12.1.1	Front		Double acting hydraulic type – 2 No.s	
B12.1.2	Rear		Double acting hydraulic type – 2 No.s	
B13.0	**Suspension- Stabilizer:**			
B13.1	Front		NA	Yes
B13.2	Rear		NA	NA
B14.0	**Chassis frame:**			
B14.1	Type		Drooped & cranked, variable C section ladder frame	
B15.0	**Displays and tell tales:** (Indicate the type and if the tell tales provided and whether they are symbols or letter. (AIS-071 part 1 and 2 or corresponding Indian Standard))			
B15.1	Head lamp – upper / lower control		0 – 3 Switch provided	
B15.2	Ignition cut-off		Not Provided	
B15.3	Turn signal		Green Color - LH and RH Direction Symbol	
B15.4	Fuel Gauge		LED Bars	
B15.5	Engine coolant temperature		Red Color – Thermometer Symbol	
B15.6	Engine low oil pressure		Red Color – Oil Tank Symbol	
B15.7	High beam indicator		Blue Color – High Beam Symbol	
B15.8	Electrical charge indicator (Battery charge)		Red Color – Battery Symbol	

Manufacturer signature: _____ Test agency: _____

Table 3 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL									
B15.9	Brake failure	Red Color – Parking and Oil symbol together							
B15.10	Front fog light, if provided	Not applicable.							
B15.11	Rear fog light, if provided	Not applicable.							
B15.12	Horn	On steering Wheel							
B15.13	ABS	Red Color – ABS function alert							
B 15.14	Seat Belt Reminder	Not Applicable							
B 15.15	Others (such as, Airbag, HVAC, Content gauge, LPG / CNG changeover switch etc.,)	Not Applicable							
B16.0	**Hood latch:**	**NA**							
B16.1	Make	--							
B16.2	Type	--							
B16.3	Identification No. / Part No.	--							
B17.0	**Wheel guard** (IS 13943 for passenger cars)								
B17.1	Dimension C	X5F00258							
B17.2	Dimension p								
B17.3	Dimension q								
B18.0	**External Projections**								
B18.1	Ornaments	Bumper, Door cladding & body fender.							
B18.2	Projection for head light	NA							
B18.3	Radiator grills (Applicable of on external surface)	NA							
B18.3.1	Gap between individual elements								
B18.3.2	Radius of curvature of individual element	>2.5 mm							
B18.4	Body Panel (In case of radius of curvature of folds in body panels are less than 2.5mm the scaled drawing of folds contour and H value as per IS 13942 is required to be submitted)	NA							
B18.5	Radius of curvature of lateral Rain/Air deflector	NA							
B19.0	**Speedometer:**	7.00 R15 LT				215/75 R15 LT			
B19.1	Type	Electronic							
B19.2	Make, and Identification No. / Part No./ Drawing No.	Pricol Limited	Minda stoneridge	Pricol Limited	Minda stoneridge	Pricol Limited	Minda stoneridge		
		FL102XXX	FL103XXX	FL104XXX	FL103 XXX	FL102XXX	FL103XXX		
B19.3	Range	0 - 130 km/h							
B19.4	Major marking	NA (Digital Indicator)							
B19.5	Minor marking	NA (Digital Indicator)							
B19.6	Speedometer ratio, Pulses / km RAR	2440		2424		2590			
B19.7	Ratio of Speedo drive	0.60		0.60		0.64			
B20.0	**Odometer: (only in case of CNG buses)**	NA							
B20.1	Make	--							
B20.2	Type and Identification No. / Part No.	--							
B21.0	**Safety belt anchorages - specifications**								
B21.1	Design of the type of belts and retractors authorized for fit to the anchorages with which vehicle is equipped:								
	Anchorage Position				**Anchorage on (*)**				
					Vehicle Structure	**Seat Structure**			
	Front	Right hand Seat	Lower anchorage	Outboard Inboard	Anchored on BIW	--			
			Upper anchorage			--			
		Middle Seat	Lower anchorage	Right Left	Anchored on BIW	--			
			Lower anchorage			--			
		Left hand Seat	Lower anchorage	Outboard Inboard	Anchored on BIW	--			

Manufacturer signature: _____ Test agency: _____

<table>
<tr><th colspan="7">Table 3 of AIS 007 (Revision 5)
TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL</th></tr>
<tr><td></td><td></td><td></td><td>Upper anchorage</td><td></td><td></td><td>--</td></tr>
<tr><td></td><td>Rear</td><td>NA</td><td>NA</td><td>NA</td><td>NA</td><td>NA</td></tr>
<tr><td>B21.2</td><td colspan="3">Section views of all the seat belt anchorages.</td><td colspan="3" rowspan="2">BWR04501</td></tr>
<tr><td>B21.3</td><td colspan="3">Reference point on body used for vehicle and seat co-ordinate measurement, X, Y, Z co-ordinates of all the seat belt anchorage points.</td></tr>
<tr><td>B21.4</td><td colspan="3">Weight of seats</td><td colspan="3">Driver seat : 11.5 Kg & Co-driver seat : 11Kg</td></tr>
<tr><td>B21.5</td><td colspan="3">C.G. of seats (Sketch, showing the C.G. location from reference point or from seat anchorage point)</td><td colspan="3">ZZZZZZZZZZ</td></tr>
<tr><td>B22.0</td><td colspan="3">Seats, their anchorages and head restraints</td><td colspan="3"></td></tr>
<tr><td>B22.1</td><td colspan="3">Seats, their anchorages and head restraints – specifications for M1 category.</td><td colspan="3">NA</td></tr>
<tr><td>B22.1.1</td><td colspan="3">Description of seats</td><td colspan="3">--</td></tr>
<tr><td>B22.1.1.1</td><td colspan="3">Make</td><td colspan="3">--</td></tr>
<tr><td>B22.1.1.2</td><td colspan="3">Number of seats fitted or capable of being fitted with head restraints, adjustable or not adjustable.</td><td colspan="3">--</td></tr>
<tr><td>B22.1.1.3</td><td colspan="3">Description of the adjustment, displacement and locking systems of the seat or of its parts and a description of occupant protection system against displacement of luggage.</td><td colspan="3">--</td></tr>
<tr><td>B22.1.2</td><td colspan="3">Description of seat anchorage</td><td colspan="3">--</td></tr>
<tr><td>B22.1.2.1</td><td colspan="3">Longitudinal / Vertical (If applicable) position of the seats during the tests.</td><td colspan="3">--</td></tr>
<tr><td>B22.1.2.2</td><td colspan="3">Drawings, diagrams and plans of the seats, their anchorages on the vehicle, the floor layout, the adjustment and displacement system of the seats and their parts, and their locking devices and of additional occupant protection system against displacement of luggage.</td><td colspan="3">--</td></tr>
<tr><td>B22.1.2.3</td><td colspan="3">In the case of seats fitted with head restraints, the head restraint shall be shown on all drawings, diagrams and photographs.</td><td colspan="3">--</td></tr>
<tr><td>B22.1.3</td><td colspan="3">Seat Drawings showing ‘H point co-ordinates with respect to reference point on body shell</td><td colspan="3">--</td></tr>
<tr><td>B22.1.4</td><td colspan="3">All designed positions i.e. Slider, Height Adjuster, Manikin settings, Torso Angle etc.</td><td colspan="3">--</td></tr>
<tr><td>B22.1.5</td><td colspan="3">Seat anchorage co-ordinates with respect to reference point on body shell to be shown on floor drawing.</td><td colspan="3">--</td></tr>
<tr><td>B22.1.6</td><td colspan="3">Seat Identification No. / Part No/Drawing Number.:
<table><tr><th>Description</th><th>Make</th><th>ID Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer)</th><th>Weight (kg)</th></tr><tr><td></td><td></td><td></td><td></td></tr></table></td><td colspan="3">--</td></tr>
<tr><td>B22.2</td><td colspan="3">Seats, their anchorages and head restrains
(For passenger vehicles of categories other than M1 and goods vehicles of category N).</td><td colspan="3"></td></tr>
</table>

Manufacturer signature: _____ Test agency: _____

Table 3 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL				
B22.2.1	Make	XXXXX YYYY		
B22.2.2	Brief description and drawings of the seat type, its attachment fittings and its adjustment, displacement and locking systems including the minimum distance between fitting points.	D+1		
B22.2.3	Position and arrangement of seats including seat layout.	XXXXXXYY		
B22.2.4	Seats if any which incorporate a safety belt anchorage.	NA		
B22.2.5	Driver Seat and Front Passenger Seat drawings, their anchorages on the vehicle, the floor layout, the adjustment and displacement system of the seats and their parts, and their locking devices, drawings showing 'H point co-ordinates and seat anchorage co-ordinates with respect to reference point on vehicle			
B22.2.6	Seat Identification No. / Part No/Drawing Number.:			
	Description	Make	ID Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer)	Weight (kg)
	Driver seat	XXXX YYY	XXXXXXXX	11.5
	Co-Driver Backrest	XXXX YYY	XXXXXXXX	11
	Co-Driver Cushion	XXXX YYY	XFXXXXXX	
B23.0	**Rear Under run Protective device**	**NA**		
B23.1	Height of lower edge of the device from the ground (mm).	-		
B23.2	Width of the device (mm).	-		
B23.3	Drawing of the rear under-run protective device with dimensions. RUPD cross section, material details etc.	-		
B23.4	Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details~~.~~ tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.	-		
B23.5	RUPD Material with specific grade	-		
B24.0	**Lateral Protection (Side Guards)**	**NA**		
B24.1	Height of the lower edge of the Side Guard (mm).	-		
B24.2	Drawing of the lateral protection device with dimensions. LPD cross section, material details with grade etc.	-		
B24.3	Distance between tyre & LPD front, rear, RH & LH, distance between supports of LPD, distance between device & load body.			
B25.0	**Controls - Specify method of operation, hand operated – left / right, foot operated – left / right Ref. SS12.1 or corresponding Indian Standard).**	-		
B25.1	Ignition	Hand operated [Rotary Switch RHS]		
B25.2	Horn	Hand operated [Horn pad on steering wheel]		
B25.3	Lamps (Head lamp, Tail lamp, Parking lamp and Number plate lamp)	Hand operated [Rotary switch – RHS]		
B25.4	Turn signal	Hand operated [Swivel arm switch – RHS]		
B25.5	Transmission shift lever	Hand operated [LHS]		
B25.6	Wind shield wiper	Hand operated [Swivel arm switch – LHS]		
B25.7	High beam/low beam	Hand operated [Swivel arm switch – RHS]		
B25.8	Parking brake	Hand operated [LHS]		
B25.9	Master switch for electrical	NA		
B25.10	Hazard warning signal	Hand operated [Push button Switch -LHS]		

Manufacturer signature: _____ Test agency: _____

Table 3 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL			
B25.11	Service Brake	Foot operated	
B25.12	Accelerator Pedal (Floor hinged/hanging type)	Foot operated – [Floor hanging Type]	
B25.13	Others	NA	
B26.0	**Safety glass**		
B26.1	**Front wind shield (laminated)**		
B26.1.1	Make	XXXXXX	
B26.1.1.1	BIS License number	CM/L 658XXXX	
B26.1.2	Identification: TAC No. / BIS License No. / E- Marking	CM/L 658XXXX	
B26.1.3	Type (flat/curved, clear/tinted)	Laminated	
B26.1.4	Thickness (mm)	4.66mm	
B26.1.5	Rake Angle	42.06°	
B26.1.6	Method of fixing (for approval of Demisting / Defrosting system)	NA	
B26.2	**Side Windows (Left & Right)**		
B26.2.1	Make	XXXX Ltd.	
B26.2.1.1	BIS License number	CM/L 658XXXX	
B26.2.2.	Identification: TAC No. / BIS License No. / E- Marking	CM/L 658XXXX	
B26.2.3	Type (flat/curved, clear/tinted, toughened/ laminated)	Curved, Tinted, toughened	
B26.2.4	Thickness mm	3.4mm	
B26.2.5	Colour	Clear	
B26.3	**Rear Window**		
B26.3.1	Make	XXXX	YYYY
B26.3.1.1	BIS License number	CM/L 658XXXX	CM/L 620013XXXX
B26.3.2	Identification: TAC No. / BIS License No. / E- Marking	CM/L 658XXXX	CM/L 620013XXXX
B26.3.3	Type (flat/curved, clear/tinted, toughened/ laminated)	Flat	
B26.3.4	Thickness mm	3.5mm	3.5mm
B26.3.5	Radius of curvature (if curved)	FLAT GLASS	
B26.4	**Sun / Moon roof glass**	NA	
B26.4.1	Make	NA	
B26.4.2	Identification: TAC No. / BIS License No. / E- Marking	NA	
B26.4.3	Type (Toughened / Laminated)	NA	
B26.4.4	Thickness mm	NA	
B26.4.5	Colour	NA	
B27.0	**Rear view mirror : [As applicable to the category of vehicle]**		
B27.1	**Interior mirror (Class-I)**	Optional	
B27.1.1	Make	XXXX	
B27.1.2	Identification: TAC No. / BIS License No. / E- Marking	A94506 / E12 01 5480	
B27.1.3	Area of mirror and radius of curvature of reflecting surface	0	
B27.2	**Main mirror (Large) (Class-II)**	NA	
B27.2.1	Make	NA	
B27.2.2	Identification: TAC No. / BIS License No. / E- Marking	NA	
B27.2.3	Area of mirror and radius of curvature of reflecting surface	NA	
B27.3	**Main mirror (Small) (Class-III)**		
B27.3.1	Make	XXXXXYY	

Manufacturer signature: _____ Test agency: _____

Table 3 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL				
B27.3.2	Identification: TAC No. / BIS License No. / E- Marking	RXZ562 (M0XX)		
B27.3.3	Area of mirror and radius of curvature of reflecting surface	Area of mirror 0.021m2 Radius of curvature of reflecting surface R1100+201SPH		
B27.4	**Wide angle mirror (Class-IV)**	NA		
B27.4.1	Make	NA		
B27.4.2	Identification: TAC No. / BIS License No. / E- Marking	NA		
B27.4.3	Area and radius of curvature of the mirror glass reflecting surface	NA		
B27.5	**Close proximity mirror (Class-V)**	NA		
B27.5.1	Make	NA		
B27.5.2	Identification: TAC No. / BIS License No. / E- Marking	NA		
B27.5.3	Area and radius of curvature of the mirror glass reflecting surface	NA		
B27.6	**Front mirror (Class-VI)**	NA		
B27.6.1	Make	NA		
B27.6.2	Identification: TAC No. / BIS License No. / E- Marking	NA		
B27.6.3	Area and radius of curvature of the mirror glass reflecting surface	NA		
B27.7	Devices for indirect vision other than mirrors:	NA		
B27.7.1	Make	NA		
B27.7.2	Identification: TAC No. / BIS License No. / E- Marking	NA		
B27.7.3	Area and radius of curvature of the mirror glass reflecting surface	NA		
B27.8	Brief drawing showing installation dimensions of all mirrors provided on the vehicle and drivers ocular point angle with RH mirror	VFT0XXXX		
B28.0	**Information on safety belt / restraint system:**			
B28.1	Safety belt	Provided		
B28.1.1	Make of seat belt	Samsung		
B28.1.2	Type and configuration	Three-point seat belt with ELR for driver & co-driver outboard, Two point static lap belt for Center seat.		
B28.1.3	Identification No. / Part No.	**Description**	**Make**	**Part number**
		Buckle RH	Samsung	FWS00XXX
		Buckle LH	Samsung	FWS00XXX
		Tounge (ELR & STATIC)	Samsung	FWS00XXX
		Belt anchorage top	Samsung	FWS00XXX
		Buckle CTR	Samsung	FWS01XXX
		Lap Tounge	Samsung	FWS01XXX
B28.2	**Restraint system**			
B28.2.1	Make	Samsong Industries Ltd		
B28.2.2	Type and configuration	Three-point seat belt with ELR for driver & co-driver outboard, Two point static lap belt for Center seat.		
B28.2.3	Identification No. / Part No.	Description	Make	Part number
		Buckle RH	Samsung	FWS00XXX
		Buckle LH	Samsung	FWS00XXX
		Tounge (ELR & STATIC)	Samsung	FWS00XXX
		Belt anchorage top	Samsung	FWS00XXX
		Buckle CTR	Samsung	FWS00XXX
		Lap Tounge	Samsung	FWS00XXX
B28.2.4	Drawings of the relevant parts of the vehicle structure and any seat anchorage reinforcements.	XXXYYYXX		
B28.2.5	Drawings of seat, showing its structure, adjustment system and fixing components, with an indication of the materials used.	XXXXXXXX		

Manufacturer signature: _____ Test agency: _____

Table 3 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL					
B28.2.6	Drawing or photograph of the restraint system as installed.			NA	
B28.2.7	Drawing showing the installation of belts on the vehicle.			XXXXXYYY	
B28.3	**Safety belts and / or other restraint systems :**				
B28.3.1	Number and position of safety belts and restraint systems and seats on which they can be used :				
	Row of Seat	Location*	Type of seat belt	Variant (if applicable)	Belt adjustment device for height (Indicate yes/no/ optional)
	First row of Seats	L	3 pt ELR	No	No
		C	Lap Belt	No	No
		R	3 pt ELR	No	No
	Second row of seats	L	NA	No	No
		C	NA	No	No
		R	NA	No	No
	The table may be extended as necessary for vehicles with more than two rows of seats there are more than three seats across the width of the vehicle. * (L = left-hand side, R= right-hand side, C= Center)				
B28.4	**Emergency exit**			**NA**	
B28.4.1	Position			--	
B28.4.2	Size			--	
B29.0	**Fuel tank :**				
B29.1	Make			M/s. XXXXYYYY	
B29.2	Material (Metallic / Plastic etc.)			Metallic	
B29.3	Nominal thickness mm			1.0mm	
B29.4	Capacity, Litre			40 Lts, 30 Lts	
B29.5	Detailed drawing of the fuel tank assembly with material specifications			FHSXXXXY FHSXXXXY	
B29.6	Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.			A5LXXX00 (50 LTS) A5LXXX00 (40 LTS)	
B29.7	Identification No. / Part No.			FHS00544 FfffXXXcdHXXS00744	
B30.0	**Wheel rim:**				
B30.1	**Front**				
B30.1.1	Make			M/s. XXXXYYYY Limited	
B30.1.1.1	BIS license Number			CM/L-670XXXXXX4 (Chennai)	
B30.1.2	Part number of wheel rim supplier			X5E06XXXX / X5E1XXXX	
B30.1.3	Size			5.5J x 15	
B 30.1.4	Type (Alloy / Sheet metal / spoke)			Sheet metal (Steel)	
B30.2	**Rear**				
B30.1.1	Make			M/s. XXXXYYYY Limited	
B30.1.1.1	BIS license Number			CM/L-6XXXXX14 (Chennai)	
B30.1.2	Part number of wheel rim supplier			X5EXXX00 / X5XXXX00	
B30.1.3	Size			5.4J x 14	
B 30.1.4	Type (Alloy / Sheet metal / spoke)			Sheet metal (Steel)	
B30.3	**Others**				
B30.1.1	Make			M/s. XXXXYYYY Limited	
B30.1.1.1	BIS license Number			CM/L-670XXXXXX4 (Manam)	
B30.1.2	Part number of wheel rim supplier			X5E0XX00 / X5E1XX00	
B30.1.3	Size			5.4J x 14	
B 30.1.4	Type (Alloy / Sheet metal / spoke)			Sheet metal (Steel)	
B31.0	**Door, Door locks and hinges:**				
B31.1	Doors				
B31.1.1	No. of doors			2	
B31.1.2	Position and type of door			Front Side door, hinged	

Manufacturer signature: _____ Test agency: _____

Table 3 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL		
B31.1.3	Detailed drawing of the door including location of the door strengthening bars, Dimensions of door reinforcements from door lower edge, cross section of the bars, material specification of the bar and door sheet metal, number of reinforcements and details of welding / bolting etc., (for side door impact test)	--
B31.2	**Door lock / latch**	
B31.2.1	**Front**	
B31.2.1.1	Make	XXXYYY Pvt ltd.
B31.2.1.2	Identification No. / Part No.	X7K00XXXX
B31.2.2	**Rear**	**NA**
B31.2.2.1	Make	NA
B31.2.2.2	Identification No. / Part No.	NA
B31.3	**Door hinge**	
B31.3.1	**Front**	
B31.3.1.1	Make	XXXYYY Pvt ltd.
B31.3.1.2	Identification No. / Part No.	FDTXXX13
B31.3.2	**Rear**	**NA**
B31.3.2.1	Make	NA
B31.3.2.2	Identification No. / Part No.	NA
B32.0	**Wheel nut, Wheel cap and Hub cap:**	
B32.1	**Wheel Nut (s) / Bolt (s)**	
B32.1.1	Make	M/s. XXXXX Fasteners
B32.1.2	Size	M13 x 1.4
B32.1.3	No. per wheel	5
B32.1.4	Tightening torque on vehicle (recommended by Vehicle Manufacturer)	160 – 170 Nm
B32.1.5	Detailed dimensional drawing along with material specifications	X5M00YYY, X5M0YYYY
B32.2	**Wheel cap (if provided)**	**NA**
B32.2.1	Detailed dimensional drawing along with material specifications	NA
B32.3	**Hub cap**	
B32.3.1	Make	Ashok Leyland Ltd
B32.3.2	Method of fitment (Press/bolted/others)	Press
B32.3.3	Brief dimensional drawing along with press fit diameter as applicable	FKCXXX13
B33.0	**Towing devices:**	NA
B33.1	Type	NA
B33.2	Make	NA
B33.3	Capacity	NA
B34.0	**Coupling devices (for trailers) :**	NA
B34.1	Make	NA
B34.2	Identification mark	NA
B34.3	Type of coupling device for mechanical	NA
B34.4	Type of coupling device for electrical	NA
B34.5	Type of coupling device for brake	NA
B34.6	Dia. Of king pin (mm)	NA
B35.0	**Spray Suppression System**	NA
B35.1	Make	NA
B35.2	Type (Water separator / Pulveriser)	NA
B35.3	Identification No. / Part No.	NA
B35.4	Size	NA
B35.5	Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.	NA

Manufacturer signature: _____ Test agency: _____

Table 3 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL		
B36.0	**Interior Fittings as per IS 15223 or AIS - 047, as applicable**	
B36.1	Instrument Panel (Dash Board)	
B36.2	Make	TXXO IPD
B36.3	Identification No. / Part No./ Drawing No.	BEXX3201
B36.4	Drawing showing the mounting details, overall size and all control switches with dimensions	BWR0XX01
B36.5	Additional details for interior fitting tests to be given (if test is already conducted, this information need not be submitted).	NA
B36.5.1	Instrument Panel Variants with photographs (With / without Airbag, Music system, AC)	NA
B36.5.2	Material used for instrument Panel	PPTYY & PCXXS
B36.5.3	Drawings	
B36.5.3.1	Instrument Panel mounting (With hardware details)	BWRXXYY1
B36.5.3.2	'H' point co-ordinates for each seating position	BWRXXYY1
B36.5.3.3	Cross sectional drawings for each projection more than 3.2	NA
B36.5.3.4	Cross sectional Drawing of Gear shift lever (Knob)	FXA0XXYY (Non AC) / FXA0XXYY (AC)
B36.5.3.5	Drawing of Grab handle with cross section	F0C0XYZ8
B36.5.3.6	Drawing of Sun visor with details of metal wire used	X1E00XYZ (LH) / X1E00XYZ (RH)
B36.5.3.7	Drawing of lamp assembly mounted at roof	FL804XYZ
B36.5.3.8	Drawing of Cigarrete lighter/ charging point	FSC01XYZ
B36.5.3.8	Any other fitment	--
B37.0	**Bumper (for M1 category vehicle):**	NA
B37.1	Make	NA
B37.2	Identification No. / Part No./Drawing No.	NA
B37.3	Installation drawing showing location of Bumper in the front and rear, fitment of the Bumper, dimensions of Bumper, mounting points, details of mounting fasteners and additional fitments on it.	NA
B37.4	Material of Bumper with details (metallic / non-metallic)	NA
B37.5	Test method to be adopted by the test agency (Pendulum impact test, Component level vibration test, or Vehicle level four poster test)	NA
B 38.0	**Flammability requirements of interior materials as per IS: 15061, as applicable.**	
B 38.1	**Seat Upholstery**	
B 38.1.1	Make	M/s. XXXXYYY
B 38.1.2	Material and Composition	Vinyl
B 38.1.3	Identification No. / Part No./Drawing No.	FWR0XYZ0
B 38.2	**Roof lining**	
B 38.2.1	Make	Supreme
B 38.2.2	Material and Composition	PU + Glass fiber + Non woven Fabric FM60XYZ8
B 38.2.3	Identification No. / Part No./Drawing No.	
B 38.3	**Floor lining**	
B 38.3.1	Make	UNXXYY PVT LTD.
B 38.3.2	Material and Composition	Carpet type – Non Woven Fabric
B 38.3.3	Identification No. / Part No./Drawing No.	FGM00XY8 MAIN FLOOR MAT FGM00XY8 CO-DRIVER MAT FGM00XY8 DRIVER SIDE MAT
B 38.4	**Side wall lining**	**NA**
B 38.4.1	Make.	--
B 38.4.2	Material and Composition	--
B 38.4.3	Identification No. / Part No./Drawing No.	--

Manufacturer signature: _____ Test agency: _____

Table 3 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL		
B 38.5	**Rear wall lining**	**NA**
B 38.5.1	Make	--
B 38.5.2	Material and Composition	--
B 38.5.3	Identification No. / Part No./Drawing No.	--
B 38.6	**Interior lining of luggage racks**	**NA**
B 38.6.1	Make	--
B 38.6.2	Material and Composition	--
B 38.6.3	Identification No. / Part No./Drawing No.	--
B 38.7	**Heating and ventilation pipe**	**NA**
B 38.7.1	Make	--
B 38.7.2	Material and Composition	--
B 38.7.3	Identification No. / Part No./Drawing No.	--
B 38.8	**Curtain / Blinds / Hanging material**	**NA**
B 38.8.1	Make	--
B 38.8.2	Material and Composition	--
B 38.8.3	Identification No. / Part No./Drawing No.	--
B 38.9	**Material for Luminaries**	**NA**
B 38.9.1	Make	--
B 38.9.2	Material and Composition	--
B 38.9.3	Identification No. / Part No./Drawing No.	--
B 38.10	**Separation wall**	**NA**
B 38.10.1	Make	--
B 38.10.2	Material and Composition	--
B 38.10.3	Identification No. / Part No./Drawing No.	--
B39.0	**Hand holds (as per AIS - 046)**	
B39.1	No. of hand holds	2
B39.2	Details and dimension of hand hold indicating length, clearance between the hand hold and body and cross sectional area of every hand hold (if required, details may be provided as a separate Annexure)	F0CXYZ68
B39.2.1	Make	XXXX YYYY COMPONENTS LTD.
B39.2.2	Identification No. / Part No./Drawing No.	FGM00XYZ
B39.2.3	Type (Grab handle / Strap / Hand Rail)	GRAB HANDLE
B39.2.4	Material	PP XYZ
B39.2.5	Size	REF DRAWING NO. F0XYZ468
B39.3	Installation drawing of hand hold, showing location of mounting for every seating position, passengers details of mounting fasteners and additional fitments on it.	F0C0XYZ8
B 40.0	**Arrangement of front controls (For M1 category as per AIS – 035)**	**NA**
B 40.1	Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing surfaces, “E” in mm.	--
B 40.2	Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.	--
B 40.3	Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.	--
B 40.4	Drawing showing the parts and arrangement of the foot controls along with dimensions “E”, “H” and “J”	--
B41.0	**Statutory Plates – Vehicle Identification Number (As per AIS - 065)**	

Manufacturer signature: _____ Test agency: _____

Table 3 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL		
B41.1	Drawings and/or photographs of the locations of the statutory plates and inscriptions and of the vehicle identification number	X4W0XYZ4 / VFT0XYZ2
B41.2	Drawings and / or photographs of the official part of the plates and inscriptions (completed example with dimensions)	X4WXYZ44
B41.3	Drawings and / or photographs of the Body Builder's Plates and its location on the vehicle	NA
B41.4	World Manufacturer's Identifier (WMI) code and its location in Vehicle Identification Number (VIN)	1st, 2nd& 3rdCharacters of 17 Characters VIN
B41.5	Location of Vehicle Descriptor Section (VDS) in Vehicle Identification Number (VIN)	4thto 9thCharacter of 17 Characters VIN
B41.6	Location of Vehicle Indicator Section (VIS) in Vehicle Identification Number (VIN)	10thto 17thCharacter of 17 Characters VIN
B41.7	Height of characters in VIN (mm)	4 mm (min in Plate) / 7 mm (min)
B42.0	**Window retention and release for Buses as per IS: 13944**	**NA**
B42.1	**Window safety glass**	**--**
B42.1.1	Method of Window Fixing (sliding or sealed or opening outside etc.)	--
B42.1.2	Brief description of the Provision for emergency exit provided in case of sealed windows	--
B42.1.3	Name of the Window Manufacturer	--
B42.2	Drawing (including Drawing no. & Revision no.) including Plan view, Elevation, LH, RH view of vehicle showing the dimensions of all windows, seating layout and location of emergency exit.	--
B42.3	Detailed brief dimensional drawing of emergency exit indicating its location(s)	--
B42.4	Number of passenger compartment doors provided in addition to driver's door.	--
B42.5	Dimensions Emergency exit identification	--
B42.5.1	Number(s) and Dimensions of Roof Exit (if provided)	--
B42.6	Dimensions of the instructions describing each motion necessary to unlatch and open the exit are provided from the release mechanism.	--
B 43.0	**Front Under run Protective devices as per AIS: 069**	**NA**
B 43.1	Drawing of the vehicle parts relevant to the front under run protection, i.e, drawing of the vehicle and/or chassis with position and mounting and/or fitting of the front under run protective device. Mentioning material details of FUPD & mounting hardware. If the under-run protection is by no special device, the drawing should clearly show as how the required dimensions are met.	--
B 43.2	In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.	--
B43.3	Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.	--
B 44.0	**Coupling Device (for T Category)**	NA
B 44.1	Trade name or mark / approval No(s):	--
B 44.2	Type (Mechanical / Close):	--

Manufacturer signature: _____ Test agency: _____

<table>
<tr><th colspan="3">Table 3 of AIS 007 (Revision 5)
TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL</th></tr>
<tr><td>B 44.3</td><td>Class of coupling device</td><td>--</td></tr>
<tr><td>B 44.4</td><td>Maximum permissible static mass on coupling ball (kg)</td><td>--</td></tr>
<tr><td>B 44.5</td><td>Maximum permissible static vertical load on coupling ball (kgf)</td><td>--</td></tr>
<tr><td>B 44.6</td><td>Drawing showing details of mounting points on the vehicle</td><td>--</td></tr>
<tr><td>B 44.7</td><td>D..... kN
Dc..... kN
S..... kg
U.....tonnes
V..... …kN</td><td>--</td></tr>
<tr><td>B 45.0</td><td>Make of the rear marking plate:</td><td>--</td></tr>
<tr><td>B 45.1</td><td>Rear marking plate type:</td><td>--</td></tr>
<tr><td>B 45.2</td><td>Rear marking plate class:</td><td>--</td></tr>
<tr><td>B 45.3</td><td>Position and nature of the marking:</td><td>--</td></tr>
<tr><td>B46.0</td><td>Child Restraint system (CRS) ;</td><td>--</td></tr>
<tr><td>B46.1</td><td>Seating Positions for Child restraint system (As per AIS-072)</td><td></td></tr>
<tr><td></td><td colspan="2">
<table>
<tr><th rowspan="2">Mass Group</th><th colspan="5">Seating Position (or other site)</th></tr>
<tr><th>Front Passenger</th><th>Rear Outboard</th><th>Rear Centre</th><th>Intermediate Outboard</th><th>Intermediate Centre</th></tr>
<tr><td>Group 0 up to 10 kg</td><td>NA</td><td>NA</td><td>NA</td><td>NA</td><td>NA</td></tr>
<tr><td>Group 0+ up to 13 kg</td><td>NA</td><td>NA</td><td>NA</td><td>NA</td><td>NA</td></tr>
<tr><td>Group I 9 to 18 kg</td><td>NA</td><td>NA</td><td>NA</td><td>NA</td><td>NA</td></tr>
<tr><td>Group II 15 to 25 kg</td><td>NA</td><td>NA</td><td>NA</td><td>NA</td><td>NA</td></tr>
<tr><td>Group III 22 to 36 kg</td><td>NA</td><td>NA</td><td>NA</td><td>NA</td><td>NA</td></tr>
</table>
Key of letters to be inserted in the above table:
U = Suitable for "universal" category restraints approved for use in this mass group.
UF = Suitable for forward-facing "universal" category restraints approved for use in this mass group.
L = Suitable for particular child restraints given on attached list. These restraints may be of the "specific vehicle", "restricted" or "semi-universal" categories.
B = Built-in restraint approved for this mass group.
X = Seat position not suitable for children in this mass group.
</td></tr>
<tr><td>B46.2</td><td>Details of CRS (To be provided if the CRS is not of Universal category (U/UF))</td><td>--</td></tr>
<tr><td>B46.3</td><td>Category of CRS(Semi Universal, Restricted or specific vehicle)</td><td>--</td></tr>
<tr><td>B46.4</td><td>Belt type: (adult) three-point belt (adult) lap belt/ special type belt/retractor;</td><td>--</td></tr>
<tr><td>B46.5</td><td>Trade name or mark</td><td>--</td></tr>
<tr><td>B46.6</td><td>Manufacturer's designation of the child restraint.</td><td>--</td></tr>
<tr><td>B46.7</td><td>Drawings, Diagrams and plans of the child restraint, including any retractor, chair assembly, impact shield fitted;</td><td>--</td></tr>
<tr><td>B46.8</td><td>Drawings, Diagrams and plans of the vehicle structure and the seat structure, as well as of the adjustment system and the attachments, including any energy absorber fitted;</td><td>--</td></tr>
<tr><td>B46.9</td><td>Photographs of the child restraint and/or vehicle structure and seat structure;</td><td>--</td></tr>
<tr><td>B47.0</td><td>Head-On Collision as per AIS :96 and Off-set frontal collision as per AIS:98</td><td>NA</td></tr>
<tr><td>B47.1</td><td>Brief description of the vehicle type as regards its structure, dimensions, lines and constituent materials:</td><td>--</td></tr>
</table>

Manufacturer signature: _____ Test agency: _____

Table 3 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL		
B47.2	Description of the protective system installed in the vehicle:	--
B47.3	Description of interior arrangements or fittings that might affect the tests:	--
B47.4	(Photographs and/or diagrams and drawings permitting the basic identification of the type(s) of vehicle and its possible variants which are covered by the approval)	--
B48.0	**Side-Impact resistance as per AIS: 99**	NA
B48.1	A detailed description, including photographs and/or drawings, of the vehicle type with respect to the structure, the dimensions, the design and the constituent materials, the side walls of the passenger compartment (exterior and interior), including specific details of the protection system, where applicable:	--
B 51.0 **B49.0**	**Pedestrian protection as per AIS-100:** The distance, measured longitudinally on a horizontal plane, between the transverse centerline of the front axle & the R point of the driver's seat is less than 1100 mm (Yes / No).	NA
B49.1	Drawing of Bonnet & wings/ fender including material details	--
B49.2	Drawing / Images of position & arrangement of Engine	--
B49.3	Drawing/ Images of Under bonnet (Engine house) Components.	--
B49.4	Drawing mentioning ground levels for correct ride attitude of the vehicle.	--
B49.5	If lower bumper height is greater than, or equal to 425 mm and less than 500 mm, then test to be conducted with lower leg form or upper leg form	--
B49.6	Identified zones of the bonnet top where the HIC must not exceed 1,000 (HIC1000 Zone) & 1,700 (HIC1700 Zone) respectively with drawing of the complete test impact area with 1000 and 1700 HIC zones mark on it.	--
B50.0	**Speed Governor (As applicable)**	
B50.1	Type of Speed Governor (SLD / SLF)	SLF
B50.2	Speed Limiting Device (SLD) [For vehicles with max. speed above 80 kmph in unladen condition as per AIS-018]	NA
B50.3	Make	M/s. XXXYYY
B50.4	Model	DXM 8.1AP
B50.5	Identification: TAC No. / BIS License No.	AL Development ID : F6NXYZW1 AL Production ID : F6NXYZW0
B50.6	SLD approved for speed/s (Annexure to be provided including information for every Vehicle Model / Variants indicating various SLD approved speeds contributed due to declared transmission options)	--
B51.0	**Temporary Cabin for drive away chassis :**	NA
B51.1	Width of the temporary cabin (mm)	--
B51.2	Height of the temporary cabin (from seat top surface to canopy) (mm)	--
B51.3	Depth or Length of the temporary cabin (Clearance from the rear most back rest when the seat is taken at the rear-most position) (mm)	--
B51.4	Windshield type	--

Manufacturer signature: _____ Test agency: _____

Table 3 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL				
B51.5	Drawing showing the mounting and relevant dimensions	--		
B52.0	**AIS-029 Survival Space for the Protection of the Occupants of the Cab of a Commercial Vehicle**			
B52.1	Drawing showing R-Point height for vehicle w.r.t ground & fiducial point on the vehicle for driver	VFT0XYZ2		
B52.2	**Cabin & Mounts Construction Details :**			
B52.3	Construction: Integrated / Attached with Mounts	4 Mounts, Fixed CABIN		
B52.4	If attached with mounts: Number of cabin mounts: Cabin type: Tiltable / Fixed	Day		
B52.5	Cabin Size: Day / Sleeper / High-Roof / Low Roof / Extended Day Cab / Any combination	B4Z8XYZ1 B0Y7XYZ0		
B52.6	Detailed drawings showing construction details of the cabin in sufficient details alongwith list of materials used in construction of the cabin	B4Z8XYZ1		
B52.7	Drawings of vehicle, showing the position of the cab on the vehicle and the manner of its attachment, and by sufficiently detailed drawings relating to the structure of the cab,	B4Z8XYZ1 B0Y7XYZ0		
B52.8	Layout drawings for different types of cabin mounts available alongwith their position on chassis/frame, material properties of the mounts, attachment details, fastener details etc.			
B52.9	**Interior Space & Standard Fitments**			
B52.10	Layout drawings of the seating layout inside the cabin showing designated position of driver, co-driver, additional passengers, sleeper berths, second row seating positons etc.	BWR0XYZ1		
B52.11	Individual drawings of the driver & other seats showing normal position of use along with adjustment positions clearly identified	--		
B52.12	Layout drawings of the steering mechanism in mounted position along with normal position of use & adjustment positions clearly identified	FZY02XYZ – Upper Column FZY04XYZ – Intermediate column		
B 55.0	**Information for Speed alert**	NA		
B 55.1	Details of audible alert (Alert above 80 km/h)	--		
B 55.2	Details of audible alert Continuous /intermittent.(Alert above 120 km/h)	--		
B 55.3.	if intermittent, Interval between Alerts	--		
B 56.0	**Information for Safety belt reminder**	NA		
B 56.1.	Whether Visual warning for Safety belt reminder provided	--		
B 56.2.	Whether audible warning for Safety belt reminder provided	--		
B 57.0	**Information for manual over-ride (Door, Door locks and hinges) (Yes/No):**	N0		
B 58.0	Information for Reverse Parking Alert System			
B 58.1.	Type of System (Sensor based / Camera based system)	Sensor		
B 58.2.	Sensor based system-(as applicable) information	Ultrasonic Sensor		
B 58.2.1	Make	XXXXX YYYYY Ltd		
B 58.2.2	Identification No. / Part No.	**Description**	**AL Part No.**	**Hella Part No**
		Sensor	FVG00XYZ	015.ABS-00
		ECU	FVG01XYZ	015.ABS-01
		Buzzer	X4Z00XYZ	015.ABS-00

Manufacturer signature: _____ Test agency: _____

Table 3 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART B - VEHICLE OVERALL		
B 58.2.3	Number of sensors and mounting location	Option 1: 4 Sensors mounted on Tail Lamp bracket and load body Option 2: 3 Sensors mounted on Tail Lamp bracket and load body
B 58.3.	Camera based System (as applicable)	NA
B 58.3.1	Make	NA
B 58.3.2	Identification No. / Part No.	NA
B 58.3.3	Camera mounting location	NA
B 58.3.4	Display Make	NA
B 58.3.5	Display Identification No. / Part No.	NA
B 58.3.6	Display location	NA
B 58.3.1	Make	NA

Manufacturer signature: _____ Test agency: _____

Manufacturer signature: _____ Test agency: _____

Table 04E of AIS 007 (Revision 5) TECHNICAL SPECIFICATIONS FOUR WHEELER AND ABOVE WITH GVW≤3500 kg for BS VI Norms		
1.0	Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1)	Diesel
1.1.	Engine Manufacturer:	Ashok Leyland Limited
1.1.1.	Manufacturer's engine model code (as marked on the engine, or other means of identification)	Model Code: ALXYZABC0 Enclosure – 4A
1.1.2.	Location of engine model code and engine serial number on block / crankcase (drawing to be attached)	Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A
1.2.	Internal combustion engine:	
1.2.1.	Specific engine information	
1.2.1.1.	Working principle: positive ignition / compression-ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other)	Compression Ignition, four-stroke, DI, TCIC
1.2.1.2.	Number, arrangement of cylinders:	3, Inline
1.2.1.2.1.	Bore----------(mm)	80
1.2.1.2.2.	Stroke: ----------(mm)	98
1.2.1.2.3.	Firing order:	1-2-3
1.2.1.3.	Engine capacity:----------(cm^3)	1468
1.2.1.4.	Volumetric compression ratio (specify tolerance)	17.5±1.0 : 1
1.2.1.5.	Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:	Refer Enclosure – 4B
1.2.1.5.1	Vehicle with Defeat Device Yes / No	No
1.2.1.5.2	If yes – Details of the Defeat device	NA
1.2.1.6.	Normal engine idling speed (specify tolerance):	700 ± 100 rpm
1.2.1.6.1.	Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only)	NA
1.2.1.7.	High idle engine speed (specify tolerance): -	NA
1.2.1.7.1	High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer)	NA
Vehicles can be fuelled with both petrol and a gaseous fuel but if the petrol system is fitted for emergency purposes or starting only and the petrol tank cannot contain more than 5 litres of petrol, they will be regarded for the test as vehicles which can only run a gaseous fuel.		
1.2.1.8.	Maximum net power:..... kW at min^{-1}	55.1 KW @ 3100 ± 50 rpm
1.2.1.9.	Maximum permitted engine speed as prescribed by the manufacturer- min^{-1}	3500 rpm
1.2.1.10.	Maximum net torque: ……Nm at: -------min^{-1} …(manufacturer's declared value)	170 Nm @ 1600 – 2400 rpm
1.2.1.11	Maximum rated speed (Specify tolerance)	3100 rpm ± 50 rpm
1.2.1.12	Minimum rated speed (Specify tolerance)	1000 rpm ± 100 rpm
1.2.1.13	Engine Performance Declared speed and powers of the engine submitted for type approval) (Speeds to be agreed with the testing agency)	
1.2.1.13.1	Engine Power Table	

Measurement point*	Engine speed rpm	New Power kW**
(1)	1100	15.5
(2)	1600	32.0
(3)	2000	40.5
(4)	2400	48.0

Manufacturer signature: _____ Test agency: _____

Table 04E of AIS 007 (Revision 5) TECHNICAL SPECIFICATIONS FOUR WHEELER AND ABOVE WITH GVW≤3500 kg for BS VI Norms			
	(5)	2800	52.6
	(6)	3000	53.5
	(7)	3100	55.1
	* See reference in AIS 137 ** Net power according to AIS 137		
	Note: In case, if data regarding the Moment of Inertia, is required by the test agency for carrying out the Full Throttle performance test for both the CI / SI engines, the same shall be provided by the manufacturer.		
1.2.2.	Fuel		
1.2.2.1	Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel		Mono Fuel
1.2.2.2	Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H_2 methane / HCNG / Biogas / LNG / Ethanol ((E85) / (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% / Hydrogen.		Diesel
1.2.2.3	Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume		7%
1.2.3.	Fuel feed		
1.2.3.1	By fuel injection (compression-ignition only): (Yes/No)		Yes
1.2.3.1.1	System description:		Unit Pump Common Rail Direct Injection system
1.2.3.1.2	Working principle: direct-injection/pre-chamber/swirl Chamber		Direct Injection
1.2.3.1.3	Injection pump		
1.2.3.1.3.1	Make(s):		M/s. XXXXYYY
1.2.3.1.3.2	Type(s) & Identification Number:		Plunger Type & FHL0XYZ0
1.2.3.1.3.3	Maximum fuel delivery in (mm^3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min^{-1}) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed		13.5 ± 1 kg/hr @ 3300 rpm
1.2.3.1.3.4	Injection advance curve (Specify the tolerances):		ECU Controlled
1.2.3.1.4	Governor		ECU Controlled
1.2.3.1.4.1	Cut-off point:		
1.2.3.1.4.1.1	Cut-off point under load:…(min^{-1})		3300
1.2.3.1.4.1.2	Cut-off point without load (Maximum no-load speed)…(min-1)		3600 ± 100 rpm
1.2.3.1.5	Injector(s):		
1.2.3.1.5.1	Make(s):		M/s. XXXXYYY
1.2.3.1.5.2	Type(s) & Identification Number:		Multi-Hole Injectors (7 holes) FKXXYYZZ
1.2.3.1.5.3	Opening pressure (kPa) (specify the tolerances) or characteristic diagram		15000 to 23000 kPa
1.2.3.1.5.4	**Injection piping**		
1.2.3.1.5.4.1	Length and internal diameter…(mm)		Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm
1.2.3.1.5.4.2	Common rail make & type		M/s. XXXXYYY, Hot Forged Rail Type
1.2.3.1.6	Cold start system		NA
1.2.3.1.6.1	Make(s):		NA
1.2.3.1.6.2	Type(s) & Identification Number:		NA
1.2.3.1.6.3	Description:		NA
1.2.3.1.7	Auxiliary starting aid		NA
1.2.3.1.7.1	Make(s):		NA
1.2.3.1.7.2	Type(s):		NA
1.2.3.1.7.3	System description:		NA
1.2.3.1.8	**Electronic controlled injection: (Yes/No)**		Yes

Manufacturer signature: _____ Test agency: _____

Table 04E of AIS 007 (Revision 5) TECHNICAL SPECIFICATIONS FOUR WHEELER AND ABOVE WITH GVW≤3500 kg for BS VI Norms		
1.2.3.1.8.1	Make(s):	M/s. XXXXYYY
1.2.3.1.8.2	Type(s):	Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU
1.2.3.1.8.3	Description of the system (in the case of systems other than continuous injection, give equivalent details):	NA
1.2.3.1.8.3.1	Make and type of the control unit:	M/s. XYZX and Electronic Control Unit
1.2.3.1.8.3.2	Make and type of the fuel regulator:	M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve
1.2.3.1.8.3.3	Make and type of air-flow sensor:	M/s. Continental and Active type
1.2.3.1.8.3.4	Make and type of fuel distributor:	M/s. Delphi-TVS and Metering type
1.2.3.1.8.3.5	Make and type of throttle housing:	M/s. Padmini and Butterfly Valve type
1.2.3.1.8.3.6	Make and type of water temperature sensor:	M/s. Continental and Thermistor Type
1.2.3.1.8.3.7	Make and type of air temperature sensor:	M/s. Continental and Thermistor Type
1.2.3.1.8.3.8	Make and type of air pressure sensor:	M/s. Continental and Piezo Resistance Type
1.2.3.2	**By fuel injection (positive ignition only): (Yes/No)**	No
1.2.3.2.1	Working principle: intake manifold (single/multi-point)/direct injection/other (specify)	NA
1.2.3.2.2	Make(s) of Injection Pump :	NA
1.2.3.2.3	Type(s) & Identification Number of Injection Pump:	NA
1.2.3.2.4	System description (in the case of systems other than continuous injection give equivalent details):	NA
1.2.3.2.4.1	Make and type of the control unit:	NA
1.2.3.2.4.2	Make and type of the fuel regulator:	NA
1.2.3.2.4.3	Make and type of the air-flow sensor:	NA
1.2.3.2.4.4	Make and type of the micro-switch:	NA
1.2.3.2.4.5	Make and type of the throttle housing:	NA
1.2.3.2.4.6	Make and type of the water temperature sensor:	NA
1.2.3.2.4.7	Make and type of the air temperature sensor:	NA
1.2.3.2.5	Injectors: Opening pressure (Specify the tolerances):..... (kPa) or characteristic diagram:	NA
1.2.3.2.5.1	Make(s):	NA
1.2.3.2.5.2	Type(s):	NA
1.2.3.2.5.3	Identification Number	NA
1.2.3.2.5.4	Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed	NA
1.2.3.2.5.5	Cut-off point under load (min-1)	NA
1.2.3.2.6	Injection timing:	NA
1.2.3.2.7	Cold start system:	NA
1.2.3.2.7.1	Operating principle(s):	NA
1.2.3.2.7.2	Operating limits/settings(Specify the tolerances)	NA
1.2.3.3	Feed pump	
1.2.3.3.1	Pressure (Specify the tolerances).....(kPa) or characteristic diagram:	320 ± 50 kPa
1.2.4	Electrical system	
1.2.4.1	Rated voltage:…V, positive/negative ground	12V, Negative Ground
1.2.4.2	Generator/ Alternator	Alternator
1.2.4.2.1	Type:	DC – Alternator with Voltage regulator
1.2.4.2.2	Nominal output: …VA	1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps
1.2.5	Ignition	NA
1.2.5.1	Make(s):	NA
1.2.5.2	Type(s):	NA
1.2.5.3	Working principle:	NA

Manufacturer signature: _____ Test agency: _____

Table 04E of AIS 007 (Revision 5) TECHNICAL SPECIFICATIONS FOUR WHEELER AND ABOVE WITH GVW≤3500 kg for BS VI Norms			
1.2.5.4	Ignition advance curve or map(Specify the tolerances):	NA	
1.2.5.5	Static ignition timing (Specify the tolerances):.....degrees before TDC	NA	
1.2.5.6	Spark plugs	NA	
1.2.5.6.1	Make	NA	
1.2.5.6.2	Type	NA	
1.2.5.6.3	Identification number	NA	
1.2.5.6.4	Gap setting----------(mm)	NA	
1.2.5.6.5	Nominal resistance (kilo ohm) (if resistive type)	NA	
1.2.5.7	Ignition coil(s)	NA	
1.2.5.7.1	Make	NA	
1.2.5.7.2	Type	NA	
1.2.5.7.3	Identification number	NA	
1.2.5.8	**Ignition condenser**	NA	
1.2.5.8.1	Make	NA	
1.2.5.8.2	Type	NA	
1.2.5.8.3	Identification number	NA	
1.2.5.9	**EMI suppressor cap / Device / Electronic unit**	NA	
1.2.5.9.1	Make	NA	
1.2.5.9.2	Type (Resistive/Capacitive)	NA	
1.2.5.9.3	Identification number	NA	
1.2.5.9.4	Nominal resistance (kilo ohm)	NA	
1.2.5.9.5	Terminology and Drawing of interference Suppression equipment	NA	
1.2.5.10	**H.T. Cable**	NA	
1.2.5.10.1	Make and Place	NA	
1.2.5.10.2	Type (Resistive/Non-resistive)	NA	
1.2.5.10.3	Length (mm) (if resistive type)	NA	
1.2.5.10.4	Outside dia. (mm) (if resistive type)	NA	
1.2.5.10.5	Nominal resistance kilo ohm, (if resistive type)	NA	
1.2.5.11	**Radiator**		
1.2.5.11.1	Radiator drawing(s)	Refer enclosure 4C2 & 4C4	Refer enclosurev 4C1 & 4C3
1.2.5.11.2	Make	Mahle Behr	Tata Toyo
1.2.5.11.3	Type (s)	Cross flow	
1.2.5.11.4	Relief valve pressure setting	0.9 kg/cm2	
1.2.5.12	Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).	Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC)	
1.2.5.12.1	Make (s)	M/s. XXXX	
1.2.5.12.1.1	Number of blades and Identification/Part No	9	
1.2.5.12.1.2	Material of blades (metal / plastic)	Plastic	
1.2.5.12.2	Type(s) [Fixed / Viscous / Electrical driven]	Electrical driven	
1.2.5.12.3	Drive ratio	NA	
1.2.5.12.4	Fan diameter (mm)	360	
1.2.5.12.5	Max. Speed of fan (in rev/min)	3200 ± 150 (Non-AC) 3400 ± 150 (AC)	
1.2.5.13	Radiator core open area (cm²)	1906 cm2	
1.2.6	Cooling system: liquid/air	Liquid	
1.2.6.1	Nominal setting of the engine temperature control mechanism (^{O}C)::	85 deg C	
1.2.6.2	Liquid	Yes	
1.2.6.2.1	Nature of liquid:	Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L	

Manufacturer signature: _____ Test agency: _____

Table 04E of AIS 007 (Revision 5) TECHNICAL SPECIFICATIONS FOUR WHEELER AND ABOVE WITH GVW≤3500 kg for BS VI Norms			
		Option 3 : Water + 50 % Antifreeze coolant capacity : approx..8 L	
1.2.6.2.2	Circulating pump(s): (Yes/No)	Yes	
1.2.6.2.3	Characteristics of Circulating pump	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm	
1.2.6.2.3.1	Make(s):	XXXXX Fasteners Ltd	
1.2.6.2.3.2	Type(s):	Impeller Type	
1.2.6.2.4	Drive ratio(s):	1: 1.12	
1.2.6.2.5	Description of the fan and its drive mechanism:	Electric Fan , Motor Driven	
1.2.6.3	Air		
1.2.6.3.1	Blower: (Yes/No)	No	
1.2.6.3.2	Characteristics of blower	NA	
1.2.6.3.2.1	Make(s):	NA	
1.2.6.3.2.2	Type(s):	NA	
1.2.6.3.3	Drive ratio(s):	NA	
1.2.7	Intake system:		
1.2.7.1	Pressure charger: (Yes/No) Turbocharger / Supercharger	Yes, Turbo charger	
1.2.7.1.1	Make(s):	M/s. XYZ, India	
1.2.7.1.2	Type(s) & Identification Number:	Waste Gate with Vacuum Actuator, X3M1XXXX	
1.2.7.1.3	Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable)	Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa	
1.2.7.2	Inter-cooler: (Yes/No)	Yes	
1.2.7.2.1	Type: air-air / air-water	Air – Air	
1.2.7.2.1.1	Make :	Mahle Behr	Tata Toyo
1.2.7.2.2	Identification Number:	B1V42XXX	B1V42YYY
1.2.7.2.2.1	Air pressure drop across the intercooler	15 kPa	
1.2.7.3	Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:… (kPa) Maximum allowable:… (kPa)	Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet)	
1.2.7.4	Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):	Intake manifold with Electronic throttle valve Refer Enclosure 4D	
1.2.7.4.1	Intake manifold description including dimensional drawings and/or photographs):	Aluminium Intake manifold with pressure sensor Refer Enclosure 4D	
1.2.7.4.2	Air filter, Dimensional drawings	Refer Enclosure 4E	
1.2.7.4.2.1	Make(s):	M/s. Fleet guard	
1.2.7.4.2.2	Type(s) & Identification Number:	Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0	
1.2.7.4.3	Intake silencer, Dimensional drawing	NA	
1.2.7.4.3.1	Make(s):	NA	
1.2.7.4.3.2	Type(s) & Identification Number:	NA	
1.2.8	Exhaust system		
1.2.8.1	Description and Dimensional drawing of the exhaust manifold:	Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F	
1.2.8.2	Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust pipe, tail pipe and	Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly	

Manufacturer signature: _____ Test agency: _____

Table 04E of AIS 007 (Revision 5) TECHNICAL SPECIFICATIONS FOUR WHEELER AND ABOVE WITH GVW≤3500 kg for BS VI Norms		
	exhaust outlet location, indicated in a Schematic dimensional drawing.)	Refer Enclosure 4G
1.2.8.2.1	Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly	13.05 ± 1 litre Refer Enclosure 4G
1.2.8.2.2.	Description and drawing of the exhaust Muffler/Silencer:	NA
1.2.8.2.2.1	Type(s) & Identification Number:	NA
1.2.8.2.2.1.1	Make	NA
1.2.8.2.2.2	Number:……	NA
1.2.8.2.3	Internal diameter of exhaust pipe (mm)(min)	diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G
1.2.8.2.4	Minimum distance between exhaust pipe(s) and the fuel line	>150 mm
1.2.8.2.5	Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided)	NA
1.2.8.2.5.1	Material	NA
1.2.8.2.5.2	Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.	NA
1.2.8.3	Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): … (kPa)	30.0 kpa (150 mm after Turbocharger)
1.2.9	Minimum cross-sectional areas of inlet and outlet ports:	Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm
1.2.10	**Valve timing or equivalent data:**	
1.2.10.1	Max. lift of valves	
1.2.10.1.1	Inlet (mm)	8.00
1.2.10.1.2	Exhaust (mm)	8.00
1.2.10.2	Angle of valves / port (w.r.t. top dead center)	
1.2.10.2.1	Inlet	
1.2.10.2.1.1	Opening	17° bTDC
1.2.10.2.1.2	Closing	51° aBDC
1.2.10.2.2	Exhaust	
1.2.10.2.2.1	Opening	66° bBDC
1.2.10.2.2.2	Closing	25° aTDC
1.2.10.2.3	Transfer	NA
1.2.10.2.3.1	Opening	NA
1.2.10.2.3.2	Closing	NA
1.2.10.3	Reference and/or setting ranges (Specify the tolerances):	On Cold Engine
1.2.10.4	Valve gap (Hot or Cold as applicable)	
1.2.10.4.1	Inlet	0.3 to 0.4 mm
1.2.10.4.2	Exhaust	0.3 to 0.4 mm
1.2.10.5	Distribution by ports	NA
1.2.10.5.1	Volume of crank-case cavity with piston at TDC	NA
1.2.10.5.2	Reed valve fitted (Yes / No)	No
1.2.10.5.3	Description of inlet ports, scavenging and exhaust ports with corresponding timing.	NA
1.2.11	Measures taken against air pollution:	
1.2.11.1	Device for recycling crankcase gases (description and drawings):	Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J
1.2.11.2	Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:	Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR)
1.2.11.2.1	Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other	Yes, LNT+DPF

Manufacturer signature: _____ Test agency: _____

Table 04E of AIS 007 (Revision 5) TECHNICAL SPECIFICATIONS FOUR WHEELER AND ABOVE WITH GVW≤3500 kg for BS VI Norms		
1.2.11.2.1.1	Catalytic converter Make Canner / Loader	Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF
1.2.11.2.1.2	Catalytic converter Identification Number	FZL1XXYY
1.2.11.2.1.2.1	Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification)	XXX / DDMMYY / X (Serial No. / Date / Shift)
1.2.11.2.1.3	Number of catalytic converters and elements (provide the information below for each separate unit:	Catalytic Converter – 1No Element – 1 No.
1.2.11.2.1.4	Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):	Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G
1.2.11.2.1.5	Type of catalytic action:	Adsorption, desorption and regeneration
1.2.11.2.1.5.1	Regeneration type for LNT (periodic / continuous):	Continuous
1.2.11.2.1.6	Total charge of precious metal: (g/vehicle)	6.68 g/Vehicle
1.2.11.2.1.7	Relative concentration (%): Pt: Rh:Pd	90% : 5% : 5%
1.2.11.2.1.8	Substrate (structure and material):	Structure: Oval shape, Honey Comb type Material: Cordierite.
1.2.11.2.1.9	Cell density:.....cells per sq. inch / cm	300 CPSI
1.2.11.2.1.10	Type of casing for catalytic converter(s):	Gap Bulk Density (GBD) Type
1.2.11.2.1.11	Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):	Refer Enclosure 4G
1.2.11.2.1.12	Heat shield: (Yes/No)	Yes
1.2.11.2.1.13	**Selective Catalytic Reduction (SCR) and its components**:.....	
1.2.11.2.1.13.1	Principle and Characteristics:	Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation.
1.2.11.2.1.13.2	Make:	Canner - M/s. XXXX Loader - M/s. YYYY
1.2.11.2.1.13.3	Type and Identification number:	Type - Passive NOx Reduction and FWH0YYYY
1.2.11.2.1.13.3.1	Total charge of precious metal:	211 g/vehicle (Cu Wash coat 90%)
1.2.11.2.1.13.3.2	Substrate (structure and material):	Structure: Oval shape, Honey Comb type Material: Cordierite
1.2.11.2.1.13.3.3	Relative Concentration (%):	0%:0%:0% (Pt:Pd:Rh)
1.2.11.2.1.13.3.4	Dimensions and shape of the SCR (volume, etc.):	Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical
1.2.11.2.1.13.3.5	Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification)	XXX / DDMMYY / X (Serial No. / Date / Shift)
1.2.11.2.1.13.4	Dosing ECU:	No
1.2.11.2.1.13.4.1	Make:	NA
1.2.11.2.1.13.4.2	Identification number:	NA
1.2.11.2.1.13.4.3	Calibration Identification number:	NA

Manufacturer signature: _____ Test agency: _____

Table 04E of AIS 007 (Revision 5) TECHNICAL SPECIFICATIONS FOUR WHEELER AND ABOVE WITH GVW≤3500 kg for BS VI Norms		
1.2.11.2.1.13.4.4	Calibration Verification Number:	NA
1.2.11.2.1.13.4.5	**Urea Dosing Unit**:	No
1.2.11.2.1.13.4.5.1	Dosing Unit:	NA
1.2.11.2.1.13.4.5.1.1	Make:	NA
1.2.11.2.1.13.4.5.1.2	Type and Identification number:	NA
1.2.11.2.1.13.4.5.2	Supply Unit:	NA
1.2.11.2.1.13.4.5.2.1	Make:	NA
1.2.11.2.1.13.4.5.2.3	Type and Identification number	NA
1.2.11.2.1.13.4.5.3	Dosing Injector:	NA
1.2.11.2.1.13.4.5.3.1	Make:	NA
1.2.11.2.1.13.4.5.3.2	Type and Idetification number :	NA
1.2.11.2.1.13.4.6	NOx Sensor (Before /After of SCR):	No
1.2.11.2.1.13.4.6.1	Make:	NA
1.2.11.2.1.13.4.6.2	Type and Identification number:	NA
1.2.11.2.1.13.4.7	Ad Blue Tank:	No
1.2.11.2.1.13.4.7.1	Tank Capacity (l) with minimum-maximum :	NA
1.2.11.2.1.13.4.7.2	Location :	NA
1.2.11.2.1.14	Regeneration systems/method of exhaust after-treatment systems, description:	Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system
1.2.11.2.1.14.1	The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:	65 cycles
1.2.11.2.1.14.1.1	Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137)	2 cycles
1.2.11.2.1.14.2	Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:	Soot load determined by pressure difference across diesel particulate filter (DPF)
1.2.11.2.1.14.3	Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):	Engine Soot Model
1.2.11.2.1.14.4	Description of method used to load system in the test procedure described in AIS 137	Type I Operating cycle on chassis dynamometer
1.2.11.2.1.14.5	Normal operating temperature range (°C):	400 to 500 °C
1.2.11.2.1.14.6	Consumable reagents (where appropriate)	NA
1.2.11.2.1.14.7	Type and concentration of reagent needed for catalytic action (where appropriate):	NA

Manufacturer signature: _____ Test agency: _____

Table 04E of AIS 007 (Revision 5) TECHNICAL SPECIFICATIONS FOUR WHEELER AND ABOVE WITH GVW≤3500 kg for BS VI Norms		
1.2.11.2.1.14.8	Normal operational temperature range of reagent (where appropriate):	NA
1.2.11.2.1.14.9	International standard (where appropriate):	NA
1.2.11.2.1.14.10	Frequency of reagent refill: continuous/maintenance (where appropriate):	NA
1.2.11.2.1.14.11	Normal operating temperature range (°C):	NA
1.2.11.2.2	**Oxygen sensor: (Yes/No)**	Yes
1.2.11.2.2.1	Type	Wide band lambda
1.2.11.2.2.2	Location of oxygen sensor:	Upstream of LNT and downstream of DPF
1.2.11.2.2.3	Control range of oxygen sensor (Specify the tolerances):	0.8 to 9 λ
1.2.11.2.2.4	Make of oxygen sensor:	XXXXXXXX
1.2.11.2.2.5	Identification / Part number	YYYYYYY
1.2.11.2.3	Air injection: (Yes/No)	No
1.2.11.2.3.1	Type (pulse air, air pump, etc.) & Identification Number:	NA
1.2.11.2.4	Exhaust gas recirculation (EGR): (Yes/No)	Yes (Integrated with EGR Cooler)
1.2.11.2.4.1	Make and number of EGR's :	M/s. Paxxxx and One
1.2.11.2.4.2	Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other)	Cooled, Progressive
1.2.11.2.4.3	Identification Number:	FEW0XXYY
1.2.11.2.4.4	Characteristics (flow rate, etc.):	Flow rate 101 kg/hr (Maximum)
1.2.11.2.4.5	Water cooled system: (Yes/No)	Yes
1.2.11.2.4.5.1	EGR Cooler Make and Identification Number	M/s. XXXXXX and FEW0YYYY
1.2.11.2.5	Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:	No
1.2.11.2.5.1	Detailed description of the devices and their state of tune:	NA
1.2.11.2.5.2	Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):	NA
1.2.11.2.5.3	Drawing with dimensions of the carbon canister:	NA
1.2.11.2.5.3.1	Canister Make :	NA
1.2.11.2.5.3.2	Canister Identification/ Part number :	NA
1.2.11.2.5.3.3	Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification)	NA
1.2.11.2.5.4	Mass of dry charcoal:……g	NA
1.2.11.2.5.5	Canister working capacity:……..g	NA
1.2.11.2.5.6	Canister bed volume:……litre	NA
1.2.11.2.5.7	Canister Purge and vent strategy description:	NA
1.2.11.2.5.8	Purge valve make and Identification Number	NA
1.2.11.2.5.9	Fuel tank pressure control valve : (Yes/No)	NA
1.2.11.2.5.9.1	Make and Identification Number	NA
1.2.11.2.5.9.2	Operating pressure (kPa)	NA
1.2.11.2.5.10	Liquid fuel hoses material :	NA
1.2.11.2.5.11	Schematic drawing with dimensions of the fuel tank with indication of capacity and material:	NA
1.2.11.2.5.12	Drawing of the heat shield between tank and exhaust system:	NA
1.2.11.2.5.13	Regeneration system/method. Description and/or drawing (Periodic / Continuous)	NA
1.2.11.2.6	Particulate trap: (Yes/No)	Yes

Manufacturer signature: _____ Test agency: _____

Table 04E of AIS 007 (Revision 5) TECHNICAL SPECIFICATIONS FOUR WHEELER AND ABOVE WITH GVW≤3500 kg for BS VI Norms				
1.2.11.2.6.1	Number of Particulate trap and elements (provide the information below for each separate unit:	Particulate Trap - 1No. Element - 1No.		
1.2.11.2.6.2	Make and Identification Number Canner & Loader	Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY		
1.2.11.2.6.2.1	Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification)	XXX / DDMMYY / X (Serial No. / Date / Shift)		
1.2.11.2.6.3	Dimensions and shape of the particulate trap (capacity):	Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre		
1.2.11.2.6.4	Type and design of particulate trap: Volume in litres	Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres		
1.2.11.2.6.5	Location of the particulate trap (reference distances in the exhaust line):	Refer Enclosure 4G		
1.2.11.2.6.6	Total charge of precious metal:	0.81 g/vehicle		
1.2.11.2.6.7	Relative concentration (%): Pt:Pd:Rh	90% : 10% : 0		
1.2.11.2.6.8	Substrate (structure and material):	Structure - Oval shape, Honey Comb type Material - Silicon Carbide		
1.2.11.2.6.8.1	Cell density (CPSI)	300 CPSI		
1.2.11.2.7	Electronic Control Unit (ECU)	Yes		
1.2.11.2.7.1	Make & number of ECU's	M/s. XXXX & One		
1.2.11.2.7.2	Identification number	Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ		
1.2.11.2.7.3	Calibration Identification Number	P15D4BS6UP80LNT1		
1.2.11.2.7.4	Calibration Verification Number	**Version**	**7.00 R15**	**215/75 R15 LT**
		Version 00	A4 9D CF BZ	9D 56 DA 0Y
1.2.11.2.8	On-board-diagnostic (OBD) system: ((Yes/No))	Yes		
1.2.11.2.8.1	Written description and/or drawing of the Malfunction Indicator (MI):	Refer Enclosure 4K		
1.2.11.2.8.2	List and purpose of all components monitored by the OBD system:	Refer Enclosure 4K		
1.2.11.2.8.3	Written description (general working principles) for:	Refer Enclosure 4K		
1.2.11.2.8.3.1	Positive ignition engines	NA		
1.2.11.2.8.3.1.1	Catalyst monitoring:	NA		
1.2.11.2.8.3.1.2	Misfire detection:	NA		
1.2.11.2.8.3.1.3	Oxygen sensor monitoring:	NA		
1.2.11.2.8.3.1.4	Other components monitored by the OBD system:	NA		
1.2.11.2.8.3.	Compression-ignition engines	Yes		
1.2.11.2.8.3.2.1	Catalyst monitoring	Refer Enclosure 4K		
1.2.11.2.8.3.2.2	Particulate traps monitoring:	Refer Enclosure 4K		
1.2.11.2.8.3.2.3	Electronic fueling system monitoring:	Refer Enclosure 4K		
1.2.11.2.8.3.2.4	NOx control system monitoring (SCR/LNT/NOx absorber):	Refer Enclosure 4K		
1.2.11.2.8.3.2.5	Other components monitored by the OBD system:	Refer Enclosure 4K		

Manufacturer signature: _____ Test agency: _____

Table 04E of AIS 007 (Revision 5) TECHNICAL SPECIFICATIONS FOUR WHEELER AND ABOVE WITH GVW≤3500 kg for BS VI Norms							
1.2.11.2.8.4	Criteria for MI activation (fixed number of driving cycles or statistical method):				Refer Enclosure 4K		
1.2.11.2.8.5	List of all OBD output codes and formats used (with explanation of each):				Refer Enclosure 4K		
1.2.11.2.8.5.1	OBD Communication protocol standard				Refer Enclosure 4K		
1.2.11.2.8.6	The following additional information shall be provided by the vehicle manufacturer for the purposes of enabling the manufacture of OBD-compatible replacement or service parts and diagnostic tools and test equipment, unless such information is covered by intellectual property rights or constitutes specific know-how of the manufacturer or or its supplier(s).						
1.2.11.2.8.6.1	A description of the type and number of the pre-conditioning cycles used for the original type approval of the vehicle.				3 EUDC		
1.2.11.2.8.6.2	A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.				Refer Enclosure 4K		
1.2.11.2.8.6.3	A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions-related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.				Refer Enclosure 4K		
1.2.11.2.8.6.4	The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:				Refer Enclosure 4K		
Component	**Fault code**	**Monitoring strategy**	**Fault detection criteria**	**MI activation criteria**	**Secondary parameters**	**Preconditioning**	**Demonstration test**
Catalyst	P0420	Oxygen sensor 1 and 2 signals	Difference between sensor 1 and sensor 2 signals	3rd cycle	Engine speed, engine load, A/F mode, catalyst temperature	Two Type I cycles	Type I
1.2.11.2.8.6.5	OBD vehicle family if any				AL XYZ engine family		
1.2.11.2.8.6.5.1	Parameters defining the OBD family (to be attached) (in reference with AIS 137)				Refer OBD Family - Annexure		
1.2.11.2.9	Torque limiter (Yes/No) (Applicable for OBD)				No		
1.2.11.2.9.1	Description of the torque limiter activation:				NA		
1.2.11.2.9.2	Description of the full load curve limitation:				NA		
1.2.11.2.10	Other systems (description and operation):				NA		
1.2.11.2.10.1	Systems to ensure the correct operation of NOx control measures				NA		

Manufacturer signature: _____ Test agency: _____

Table 04E of AIS 007 (Revision 5) TECHNICAL SPECIFICATIONS FOUR WHEELER AND ABOVE WITH GVW≤3500 kg for BS VI Norms		
1.2.11.2.10.2	Driver inducement system if applicable	NA
1.2.11.2.10.2.1	Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross-reference to be determined]. : (Yes/No)	NA
1.2.11.2.10.2.2	Activation of the creep mode disable after restart /disable after fueling /disable after parking	NA
1.2.11.2.102.3	Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol)	NA
1.2.11.2.10.2.4	When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures	NA
1.2.12	Details of any devices designed to influence fuel economy (if not covered by other items):	NA
1.2.13	LPG fuelling system: (Yes/No)	No
1.2.13.1	Electronic engine management control unit for LPG fuelling	NA
1.2.13.1.1	Make(s):	NA
1.2.13.1.2	Type and Identification Number :	NA
1.2.13.1.3	Calibration Identification Number	NA
1.2.13.1.4	Calibration Verification Number (CVN)	NA
1.2.13.2.1	Description of the safeguarding of the catalyst at switch-over from petrol to LPG or back:	NA
1.2.13.2.2	System layout (electrical connections, vacuum connections, compensation hoses, etc.)	NA
1.2.14	NG fuelling system: (Yes/No)	No
1.2.14.1	Approval number	NA
1.2.14.2	Electronic engine management control unit for NG fuelling	NA
1.2.14.2.1	Make(s):	NA
1.2.14.2.2	Type and Identification Number :	NA
1.2.14.2.3	Calibration Identification Number	NA
1.2.14.2.4	Calibration Verification Number (CVN)	NA
1.2.14.2.5	Emission-related adjustment possibilities:	NA
1.2.14.3	Further documentation:	NA
1.2.14.3.1	Description of the safeguarding of the catalyst at switch-over from petrol to NG or back	NA
1.2.14.3.2	System layout (electrical connections, vacuum connections, compensation hoses, etc.):	NA
1.2.15	Hydrogen fuelling system: (Yes/No)	No
1.2.15.1	Electronic engine management control unit for hydrogen fuelling	NA
1.2.15.1.1	Make(s):	NA
1.2.15.1.2	Identification Number :	NA
1.2.15.1.3	Calibration Identification Number	NA
1.2.15.1.4	Calibration Verification Number (CVN)	NA
1.2.15.1.5	Emission-related adjustment possibilities:	NA
1.2.15.2	Further documentation	NA
1.2.15.2.1	Description of the safeguarding of the catalyst at switch-over from petrol to hydrogen or back:	NA
1.2.15.2.	System lay-out (electrical connections, vacuum connections compensation hoses, etc.):	NA
1.2.16	H2NG fuelling system: (Yes/No)	No
1.2.16.1	Percentage of hydrogen in the fuel (maximum specified by the manufacturer)	NA
1.2.16.2	Electronic engine management control unit for H2NG fuelling	NA
1.2.16.2.1	Make(s):	NA

Manufacturer signature: _____ Test agency: _____

Table 04E of AIS 007 (Revision 5) TECHNICAL SPECIFICATIONS FOUR WHEELER AND ABOVE WITH GVW≤3500 kg for BS VI Norms		
1.2.16.2.2	Identification Number :	NA
1.2.16.2.3	Calibration Identification Number	NA
1.2.16.2.4	Calibration Verification Number (CVN)	NA
1.2.16.2.5	Emission-related adjustment possibilities:	NA
1.2.16.3	Further documentation	NA
1.2.16.3.1	Description of the safeguarding of the catalyst at switch-over from petrol to H2NG or back:	NA
1.2.16.3.2	System lay-out (electrical connections, vacuum connections compensation hoses, etc.):	NA
1.3.	Electric motor(Traction)	No
1.3.1.	Type (winding, excitation):	NA
1.3.1.1.	Maximum hourly output: kW(manufacturer's declared value)	NA
1.3.1.1.1.	Maximum net power:kW(manufacturer's declared value)	NA
1.3.1.1.2.	Maximum 30 minutes power: kW (manufacturer's declared value)	NA
1.3.1.2.	Operating voltage: V	NA
1.3.1.3	Make and Identification Number:	
1.3.2.	Battery (Traction)	NA
1.3.2.1.	Number of cells:	NA
1.3.2.2.	Mass: ……(kg)	NA
1.3.2.3.	Capacity: Ah (Amp-hours)	NA
1.3.2.4.	Position:	NA
1.4.	Engines or motor combinations	NA
1.4.1.	Hybrid Electric Vehicle: (Yes/No)	No
1.4.2.	Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging[9]	NA
1.4.3.	Operating mode switch: with/without	NA
1.4.3.1.	Selectable modes	NA
1.4.3.1.1.	Pure electric: (Yes/No)	NA
1.4.3.1.2.	Pure fuel consuming: (Yes/No)	NA
1.4.3.1.3.	Hybrid modes: (Yes/No) (if yes, short description)	NA
1.4.4.	Description of the energy storage device: (traction battery, capacitor, flywheel/generator...)	NA
1.4.4.1.	Make(s):	NA
1.4.4.2.	Type(s) :	NA
1.4.4.3.	Identification number:	NA
1.4.4.4.	Kind of electrochemical couple:	NA
1.4.4.5.	Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J)	NA
1.4.4.6.	Charger: on board/external/without	NA
1.4.5.	Electric machines (describe each type of electric machine separately)	NA
1.4.5.1.	Make:	NA
1.4.5.2.	Type:	NA
1.4.5.3.	Primary use: traction motor/generator	NA
1.4.5.3.1.	When used as traction motor: mono motor / multi motors (number):	NA
1.4.5.4.	Maximum power: kW	NA
1.4.5.5.	Working principle:	NA
1.4.5.5.1.	Direct current/alternating current/number of phases:	NA
1.4.5.5.2.	Separate excitation/series/compound	NA
1.4.5.5.3.	Synchronous/asynchronous	NA
1.4.6.	Control unit	NA
1.4.6.1.	Make:	NA
1.4.6.2.	Type :	NA
1.4.6.3.	Identification number:	NA

Manufacturer signature: _____ Test agency: _____

Table 04E of AIS 007 (Revision 5) TECHNICAL SPECIFICATIONS FOUR WHEELER AND ABOVE WITH GVW≤3500 kg for BS VI Norms		
1.4.7.	Power controller	NA
1.4.7.1.	Make:	NA
1.4.7.2.	Type:	NA
1.4.7.3.	Identification number:	NA
1.4.8.	Vehicle electric range..... km (according to AIS 137):	NA
1.4.9.	Manufacturer's recommendation for preconditioning:	NA
1.5.	**CO2 (g/km) (Applicable for category M1 with GVW<3.5 T)**	NA
1.5.1	Declared : **(rounded to 3 decimal places)**	NA
1.5.2	Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*	NA
1.5.2.1	Fuel Equivalent Fuel Consumption (Actual Fuel)	NA
1.5.2.2	Petrol Equivalent Fuel Consumption (Petrol Equivalent)	NA
1.5.2.3	*calculated on the declared CO2	NA
1.5.2.4	CO2 Reducing technologies Available (Yes/No)	No
1.5.2.4.1	Regenerative braking	NA
1.5.2.4.2	Start-Stop System	NA
1.5.2.4.3	Tyre pressure monitoring system	NA
1.5.2.4.4	6 or more Speed Transmission	NA
1.5.2.4.5	Any other technology (manufacture to enclose detail)	NA
1.5.3	Electric energy consumption for electric vehicles Wh/km	NA
1.5.3.1	Electric energy consumption for pure electric vehicles Wh/km	NA
1.5.3.2	Electric energy consumption for externally chargeable hybrid electric vehicles	NA
1.5.3.2.1	Electric energy consumption (condition A, combined) Wh/km	NA
1.5.3.2.2	Electric energy consumption (condition B, combined) Wh/km	NA
1.5.3.2.3	Electric energy consumption (weighted combined) ….Wh/km	NA
1.6	Temperatures permitted by the manufacturer	NA
1.6.1.	Cooling system	Yes
1.6.1.1.	Liquid cooling	Yes
1.6.1.1.1.	Maximum temperature at outlet: …….(°C)	100 °C
1.6.1.2.	Air cooling	NA
1.6.1.2.1.	Reference point:	NA
1.6.1.2.2.	Maximum temperature at reference point: …….(°C)	NA
1.6.2.	Maximum outlet temperature of the inlet intercooler: .. (°C)	45 °C
1.6.3.	Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: …(°C)	740 °C
1.6.4.	Fuel temperature	
1.6.4.1.	Minimum: .. (°C)	- 8
1.6.4.2.	Maximum: .. (°C)	42
1.6.5.	Lubricant temperature	
1.6.5.1.	Minimum: …(°C)	- 5
1.6.5.2.	Maximum: …(°C)	120
1.7	Lubrication system	Yes

Manufacturer signature: _____ Test agency: _____

Table 04E of AIS 007 (Revision 5) TECHNICAL SPECIFICATIONS FOUR WHEELER AND ABOVE WITH GVW≤3500 kg for BS VI Norms		
1.7.1	Description of the system	Forced Lubrication system
1.7.1.1	Position of the lubricant reservoir:	Sump beneath crankcase
1.7.1.2	Feed system (by pump/injection into intake/mixing with fuel, etc.)	Pump
1.7.2	Lubricating pump	Yes
1.7.2.1	Make(s):	M/s. ZXWA
1.7.2.2	Type(s):	XXYY Type
1.7.3	Mixture with fuel	No
1.7.3.1	Percentage:	
1.7.4	Oil cooler: (Yes/No)	Yes
1.7.4.1	Drawing(s):…, or	Enclosure 4L
1.7.4.1.1	Make(s):	M/s. XXYY or Any other equivalent
1.7.4.1.2	Type(s):	DONUT
2.0	Transmission	
2.1	Moment of inertia of engine flywheel:	0.1212 kg/m2
2.1.1	Additional moment of inertia with no gear engaged:	0.4565 kg/m2
2.2	Clutch (type):	Diaphragm
2.2.1	Maximum torque conversion:	1.5 times
2.3	Transmission Control Unit (ECU) (Yes/No):	No
2.3.1	Type and Identification Number :	NA
2.3.2	Calibration Identification Number	NA
2.3.3	Calibration Verification Number (CVN)	NA
3.0	Vehicle Family	
3.1	OBD IUPR Family if any	XX+XYY
3.2	In-Service family if any	XXYYS OBD II

Manufacturer signature: _____ Test agency: _____

Table 5 of AIS -007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART D – BRAKES		
D1.0	**Service brakes:**	
D1.1	Make	M/s Brakes India Private Limited
D1.2	Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others)	Vacuum assisted hydraulic
D1.3	Control system & braking wheel	Hydraulic acting on all wheels
D1.4	Schematic layout indicating method of split of brake system, location of valves, reservoirs, ABS components etc. (Attach drawing and indicate the drawing number)	AJ1XXX00
D1.5	**Anti-Lock braking system Provided (Yes / No / Optional)**	NA
D1.5.1	If yes, details of ABS	-
D1.5.2	Make	-
D1.5.3	Category of ABS	-
D1.5.4	Nos. of directly controlled wheel(s)	-
D1.5.5	Brief description of failure warning tell–tale	-
D1.5.6	**Wheel Speed Sensors**	-
D1.5.6.1	No. of sensors	-
D1.5.6.2	Make of sensors	-
D1.5.6.3	Type of sensors	-
D1.5.7	**Modulator**	-
D1.5.7.1	Nos. of Modulators	-
D1.5.7.2	Make of Modulators	-
D1.5.7.3	Identification No. / Part No. of Modulator	-
D1.5.7.4	Brief description and features	-
D1.5.8	**Controller**	-
D1.5.8.1	Nos. of Controller	-
D1.5.8.2	Make of Controller	-
D1.5.8.3	Identification No. / Part No. of Controller	-
D1.5.8.4	Brief description and features	-
D1.5.9	**Height of Center of Gravity (mm)**	-
D1.5.9.1	Un laden condition	-
D1.5.9.2	Laden condition	-
D 1.5.10	**Slack adjuster (**only applicable for M2, M3, N and T category of vehicles.)	-
D 1.5.10.1	Front (Automatic / Manual)	-
D 1.5.10.1.1	Make of slack adjuster	-
D 1.5.10.1.2	Model and Type	-
D 1.5.10.2	Rear (Automatic / Manual)	-
D 1.5.10.2.1	Make of slack adjuster	-
D 1.5.10.2.2	Model and Type	-
D2.0	**Brake lining or pad**	
D2.1	Nominal Dimensions, (mm) (Length x Width x thickness)	
D2.1.1	Front wheel	132x47.3x10.35
D2.1.2	Rear wheel	LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6
D2.1.3	Others	NA
D2.1.4	Type of liner wear indicator (window/ acoustic/ optical/ any other)	Window
D2.2	**Effective area per axle (cm^2)**	
D2.2.1	Front	250 sq.cm
D2.2.2	Rear	570 sq.cm
D2.2.3	Others	NA
D2.3	**Make** and material designation (only applicable for M2, M3, N and T category of vehicles.)	
D2.3.1	Front wheel / axle	Option 1: SYY 6XXXYA

Manufacturer signature: _____ Test agency: _____

			Option 2: XXXXXX 5009	
D2.3.2	Rear wheel / axle		APACHE 4059M	
D2.3.3	Others		NA	
D2.3.4	Whether asbestos or asbestos-free		Asbestos Free	
D3.0	**Brake drum or disc**			
D3.1	Front axle (Disc / Drum)		DISC	
D3.1.1	Effective Diameter (mm)		212	
D3.2	Rear axle (Disc / drum)		DRUM	
D3.2.1	Effective Diameter (mm)		282	
D3.3	Other axle (Disc / Drum)		NA	
D3.3.1	Effective diameter (mm)		NA	
D 3.5	Material (if the braking surface is not ferrous)		NA	
D 3.5.1	Front wheel		--	
D 3.5.1	Rear wheel		--	
D 3.6	Others		--	
D4.0	Master cylinder or brake valve			
D4.1	Make		M/s XXXX YYYY	
D4.2	Type (only applicable for M2, M3, N and T category of vehicles.)		N Category	
D4.3	Inner diameter of the master cylinder (mm)		22.81	
D 4.4	Operating stroke (mm)		41	
D 4.5	Type of supply tank (only applicable for M2, M3, N and T category of vehicles.)			
D5.0	**Wheel cylinder / Wheel Chamber**			
D5.1	Diameter (mm)			
D5.1.1	Front		41 mm	
D5.1.2	Rear		23.4 mm	
D5.1.3	Others		NA	
D5.2	Type (single acting/double acting)			
D5.2.1	Front		SINGLE ACTING	
D5.2.2	Rear		DOUBLE ACTING	
D5.2.3	Others		NA	
D5.2.4	Make of wheel cylinder / slave cylinder		M/s XXXXXYYY Limited	
D6.0	**Booster :**			
D6.1	Make		SINGLE ACTING	
D6.2	Type		DOUBLE ACTING	
D6.3	Boost ratio		NA	
D6.4	Size of the booster (mm) (diameter)		M/s XXXX YYYY	
D7.0	**Vacuum or air assistance**			
D7.1	**Pressure**			
D7.1.1	Nominal (P2 as per IS:11852)		0.8 ~ 0.9	
D7.1.2	Cut in		NA	
D7.1.3	Cut out		NA	
D7.2	Type of vacuum pump or air compressor		ROTARY VANE TYPE – VACUUM PUMP	
D7.2.1	Air compressor cubic capacity (cc)		**NA**	
D7.2.2	Free air delivery, l/min		--	
D7.3	Type of pressure regulator		--	
D7.4	No. of tanks		--	
D7.5	Tank Capacity (l)		NA	
			Description	Capacity
D7.5.1	Tank 1		Front	NA
D7.5.2	Tank 2		Rear	NA
D7.5.3	Tank 3		Auxiliary	NA
D7.5.4	Tank 4		Parking	
D7.6	**Brake Chamber**	**Front**	**Rear**	**Parking**
D7.6.1	Make and type	NA	NA	NA
D7.6.2	Size mm	NA	NA	NA
D7.6.3	Inner diameter mm	NA	NA	NA
D7.6.4	Stroke mm	NA	NA	NA

Manufacturer signature: _____ Test agency: _____

D8.0	**Brake hose (if Hydraulic)**			
D8.1	Make and Identification No.		**MAKE:** XXX GIRLING **TAC No:** AB2XX1. **CoP No:** SYL/29/2020-2021/300000YY60/ADDM/XXXX	
D8.1.1	Type of expansion (HL / HR)		HL	
D8.2	Free Length of hoses		FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR : L1 – 405, L2 – 360.9	
D8.3	Thickness of lining (mm)		0.6	
D8.4	Nominal bore dia. (mm)		3.1	
D9.0	**Failure Warning device for braking**			
D9.1	Type (Visual display/ audible/others)		Visual display	
D9.2	Operation pressure (kg/cm2 / bar / kPa)		NA	
D9.3	Type of safety device		NA	
D10.0	**Parking brake**			
D10.1	Make		M/s XXXXXYY	
D10.2	Type (mechanical/spring brake)		MECHANICAL	
D10.3	Acting on Transmission/wheel		WHEEL	
D10.4	Control System & Braking wheel		Hand operated & Rear wheel	
D10.5	Lining / pad		Lining	
D10.6	Name of producer		M/s XXXXYYY	
D10.7	Dimension (mm) (D x W x T)		LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6	
D10.8	Area (cm ^{2})		279 sq.cm	
D10.9	Material		XYZ 420	
D10.10	Diameter of brake drum/disc (mm)		272	
D11.0	**Secondary brake**			
D11.1	Type			
D11.2	Description		UNFAILED PART OF THE SERVICE BRAKE	
D12.0	**Additional retarding devices**		NA	
D12.1	Type		--	
D12.2	Description		--	
D12.3	Deceleration at 30 km/h, m/s2		--	
D13.0	**Brake fluid**			
D13.1	Make		Ms. GULF Oil	
D13.2	Trade name		DOT 4	
D13.3	Specification/ grade as per Indian standard		DOT 4	
D 13.4	Brake fluid level indicator (Yes/No)		Yes	
D 13.5	Brake adjustment			
D 13.5.1	Type (Manual/Auto) Front and Rear		Manual	
D 13.5.2	Lever length			
D 13.6	Pneumatic Brake Actuator		NA	
D 13.6.1	Type (as per IS)		NA	
D14.0	**Load distribution:**			
		Laden kg	Unladen kg	F/R Ratio (simulated for fully built)
	Front axle	Refer Annexure T7 - B		
	Rear axle			
	Other axle			
	Total			
D15.0	**Proportioning valve**		NA	
D15.1	Make		-	
D15.2	Characteristics		-	
D15.3	Identification		-	
D16.0	**Apportioning valve**		NA	
D16.1	Make		-	
D16.2	Characteristics		-	

Manufacturer signature: _____ Test agency: _____

D16.3	Identification	-
D17.0	**Load sensing valve**	LCRV – BYPASS ASSY
D17.1	Make	M/s XXXXYYY
D17.2	Characteristics	0.2 Slope
D17.3	Identification	Coil spring mounted in Rear circuit
D18.0	**Valve**	NA
D18.1	Make	-
D18.2	Characteristics	-
D18.3	Identification	-
D19.0	**Other valves**	NA
D19.1	Function	-
D19.2	Make	-
D19.3	Characteristics	-
D19.4	Identification	-
D 20.0	**TECHNICAL SPECIFICATION – ADDITIONAL INFORMATION FOR M1 AND N1 CATEGORY OF VEHICLES**	
D 20.1	Tyre dimensions	NA
D 20.1.1	Temporary-use spare wheel/tyre dimensions:	NA
D 20.1.2	Vehicle meets the technical requirements of AIS 110: Yes/No	No
D 20.2	Brake Assist System (Fitted : Yes/No)	No
D 20.2.1	Category of Brake Assist System A / B	--
D 20.2.1.1	For category A systems, define the force threshold at which the ratio between pedal force and brake pressure increases	--
D 20.2.1.2	For category B systems, define the brake pedal speed which must be achieved in order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval)	--
D 20.3	Adequate documentation according to annex G (Complex electronics vehicle control systems) provided in respect of the following system(s): ..... Yes / No / Not applicable	--
D 20.3.1	List of complex functions covered:	--
D 20.3.2	Manufacturer's documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.	--
D 20.3.3	Manufacturer's documents explain the functionality and safety concept of the system and describe how the operational status may be checked.	--
D 20.3.4	A list of input and output variables, including their working range, is provided.	--
D 20.3.5	Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.	--
D 20.3.6	Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.	--
D 20.3.7	Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non-fault conditions.	--
D 20.3.8	System architecture, design methods and tools identified.	--
D 20.3.9	Information regarding design provisions in case of failure is provided including any error messages, warning signals, partial performance conditions, back-up modes and/or removal of high-level functions.	--
D 20.3.10	Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.	--
D 20.3.11	Document reference and date of inspection:	--
D 20.3.12	Vehicle complies with all other performance requirements of IS 15986 : 2015 and meets manufacturer's specifications under non-fault conditions.	--

Manufacturer signature: _____ Test agency: _____

D 20.3.13	Under fault conditions, vehicle response corresponds to that described in the manufacturer's documents / failure analysis and safety concept is verified.	--
	(see annex 4 of this report for test data)	--
D 20.4	Electronic Stability Control system as per AIS:133 (Fitted : Yes/No)	NO
D 20.4.1	If yes, details of ESC	--
D 20.4.2	ESC System Technical Documentation. To ensure a vehicle is equipped with an ESC system that meets the definition of "ESC System" in paragraph 2.0, the vehicle manufacturer shall make available to the test agency, upon request, the documentation specified below.	--
D 20.4.3	System diagram identifying all ESC system hardware.	--
D 20.4.4	A brief written explanation sufficient to describe the ESC system basic operational characteristics.	--
D 20.4.4.1	Logic diagram	--
D 20.4.4.2	Understeer information.	--
D 20.4.4.3	Static Stability Factor	--
D 20.4.4.4	Make and Country of manufacturer(if imported)	--
D 20.4.5	Software Id / version	--
D 20.4.6	Hardware Id	--
D 20.4.7	Brief description of failure warning tell–tale	--
D 20.4.8	Control function (Directional / Roll / Directional and Roll)	--
D 20.4.9	Steering Angle Sensor	--
D 20.4.9.1	Make and Country of manufacturer(if imported)	--
D 20.4.9.2	Identification No. / Part No.	--
D 20.4.9.3	Brief description and features	--
D 20.4.10	Yaw Rate Sensor	--
D 20.4.10.1	Make and Country of manufacturer(if imported)	--
D 20.4.10.2	Identification No. / Part No.	--
D 20.4.10.3	Brief description and features	--
D 20.5	Additional Component details, if any	--
D 20.5.1	Component 1	--
D 20.5.1.1	Make and Country of manufacturer(if imported)	--
D 20.5.1.2	Identification No. / Part No.	--
D 20.5.2	Component 2	--
D 20.5.2.1	Make and Country of manufacturer(if imported)	--
D 20.5.2.2	Identification No. / Part No.	--
D 20.5.3	Component 3	--
D 20.5.3.1	Make and Country of manufacturer(if imported)	--
D 20.5.3.2	Identification No. / Part No.	--
D 21.0	**TECHNICAL SPECIFICATION – ADDITIONAL INFORMATION FOR OTHER THAN M1 CATEGORY OF VEHICLES**	NA
D 21.1	Motor Vehicle stability function information document	--
D 21.1.1	System	--
D 21.1.2	System variants	--
D 21.1.3	System options	--
D 21.1.3.1	Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control	--
D 21.1.4	System configurations (where appropriate)	--
D 21.1.5	System identification including software level identifier	--
D 21.2	Applications:	--
D 21.2.1	List of motor vehicles by description and configuration that are covered by the information document	--
D 21.2.2	Schematic diagrams of the respective configurations installed on the motor vehicles defined in item 2.1. above with consideration given to the following:	--
D 21.2.2.1	Lift axles	--
D 21.2.2.2	Steering axles	--

Manufacturer signature: _____ Test agency: _____

D 21.2.2.3	Anti-lock braking configurations	--
D 21.2.3	Scope of application with respect to suspension:	--
D 21.2.3.1	Air	--
D 21.2.3.2	Mechanical	--
D 21.2.3.3	Rubber	--
D 21.2.3.4	Mixed	--
D 21.2.3.4	Anti-roll bar	--
D 21.2.4	Additional information (if applicable) to the application of the directional control and roll-over control functions, for example:	--
D 21.2.4.1	Wheelbase, track, centre of gravity height	--
D 21.2.4.2	Wheel type (single or twin) and tyre type (e.g. structure, category of use, size)	--
D 21.2.4.3	Gearbox type (e.g. manual, automated manual, semi-automatic, automatic)	--
D 21.2.4.4	Drive train options (e.g. retarder)	--
D 21.2.4.5	Differential type/differential lock(s) (e.g. standard or self-locking, automatic or driver selected)	--
D 21.2.4.1	Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.	--
D 21.3	Component description:	--
D 21.3.1	Sensor(s) External to the Controller	--
D 21.3.1.1	Function	--
D 21.3.1.2	Limitations on the location of the sensors	--
D 21.3.1.3	Identification (e.g. part numbers)	--
D 21.3.2	Controller(s)	--
D 21.3.2.1	General description and function	--
D 21.3.2.2	Functionality of internal sensors (if applicable)	--
D 21.3.2.3	Hardware identification (e.g. part numbers)	--
D 21.3.3.4	Software identification	--
D 21.3.3.5	Limitations on the location of the controller(s)	--
D 21.3.3.6	Additional features	--
D 21.3.3	Modulators	--
D 21.3.3.1	General description and function	--
D 21.3.3.2	Hardware identification (e.g. part numbers)	--
D 21.3.3.3	Software identification (if applicable)	--
D 21.3.3.4	Limitations	--
D 21.3.4	Electrical Equipment	--
D 21.3.4.1	Circuit diagrams	--
D 21.3.4.2	Powering methods	--
D 21.3.5	Pneumatic circuits	--
D 21.3.6	Safety aspects of the electronic system in accordance with Annex S of IS 11852	--
D 21.3.6.1	List of complex functions covered:	--
D 21.3.6.2	Manufacturer's documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.	--
D 21.3.6.3	Manufacturer's documents explain the functionality and safety concept of the system and describe how the operational status may be checked.	--
D 21.3.6.4	A list of input and output variables, including their working range, is provided.	--
D 21.3.6.5	Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.	--
D 21.3.6.6	Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.	--
D 21.3.6.7	Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non-fault conditions.	--

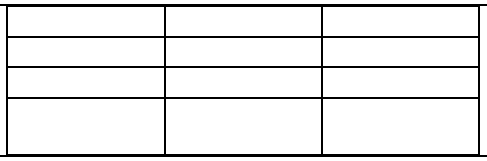
Manufacturer signature: _____ Test agency: _____

D 21.3.6.8	System architecture, design methods and tools identified.	--
D 21.3.6.9	Information regarding design provisions in case of failure is provided including any error messages, warning signals, partial performance conditions, back-up modes and/or removal of high-level functions.	--
D 21.3.6.10	Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.	--
D 21.3.6.11	Document reference and date of inspection:	--
D 21.3.6.12	Vehicle complies with all other performance requirements of IS 15986:2015 and meets manufacturer's specifications under non-fault conditions.	--
D 21.3.6.13	Under fault conditions, vehicle response corresponds to that described in the manufacturer's documents / failure analysis and safety concept is verified.	--
	(see annex 4 of this report for test data)	--
D 21.3.7	Electro-magnetic compatibility	--
D 21.3.7.1	Documentation demonstrating compliance with AIS-004(Part 3).	--
D 22.0	**TECHNICAL SPECIFICATION – ADDITIONAL INFORMATION (For other than M1 and N1)**	NA
D 22.1	Name of manufacturer	--
D 22.2	System name	--
D 22.3	System variations	--
D 22.4	Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control	--
D 22.5	System configurations (where appropriate)	--
D 22.6	System identification	--
D 22.7	Additional information (if applicable) to the application of the directional control and/or the roll-over control function(s)	--
D 22.8	Component description	--
D 22.9	Sensors external to the controller (a) Function; (b) Limitations on the location of the sensors; (c) Identification, e.g. part numbers.	--
D 22.10	Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional features.	--
D 22.11	Modulators (a) General description and function; (b) Identification; (c) Limitations.	--
D 22.12	Electrical equipment (a) Circuit diagrams; (b) Powering methods.	--
D 22.13	Pneumatic circuits System schematics including anti-lock braking configurations associated with the trailer types defined in paragraph 6.2.1. of this annex.	--
D 22.14	Safety aspects of the electronic system in accordance with Annex S of IS 11852	--
D 22.14.1	List of complex functions covered:	--
D 22.14.2	Manufacturer's documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.	--
D 22.14.3	Manufacturer's documents explain the functionality and safety concept of the system and describe how the operational status may be checked.	--

Manufacturer signature: _____ Test agency: _____

D 22.14.4	A list of input and output variables, including their working range, is provided.	--
D 22.14.5	Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.	--
D 22.14.6	Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.	--
D 22.14.7	Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non-fault conditions.	--
D 22.14.8	System architecture, design methods and tools identified.	--
D 22.14.9	Information regarding design provisions in case of failure is provided including any error messages, warning signals, partial performance conditions, back-up modes and/or removal of high-level functions.	--
D 22.14.10	Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.	--
D 22.14.11	Document reference and date of inspection:	--
D 22.14.12	Vehicle complies with all other performance requirements of IS 15986 : 2015 and meets manufacturer's specifications under non-fault conditions.	--
D 22.14.13	Under fault conditions, vehicle response corresponds to that described in the manufacturer's documents / failure analysis and safety concept is verified.	--
D 22.15	Electro-magnetic compatibility	--
D 22.16	Documentation demonstrating compliance with AIS-004(Part 3).	--
D 23.0	**Vehicle is/ is not equipped to tow a trailer with electric braking system (Yes/No)**	No

Manufacturer signature: _____ Test agency: _____

Table 6 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART E – ELECTRICAL		
E1.0	**Battery**	
E1.1	Type & number	Lead Acid Battery & 1 Number
E1.2	Voltage	12V
E1.3	Rating (Ah)	50 Ah
E2.0	**Wind Screen Wiper**	
E2.1	Type (Manual / power)	Manual
E2.2	No. of wipers	Two
E2.3	**Wiper motor**	
E2.3.1	Make	XXYY System
E2.3.2	Type	12V
E2.3.3	Identification mark	X5XYZ100
E2.3.4	Rated voltage	12V
E2.3.5	Number of sweep Frequencies	2
E2.3.6	Highest sweep frequency (Cycles / min)	62±8 rpm
E2.3.7	Lowest sweep frequency (Cycles / min)	40±5 rpm
E2.4	**Wiper arm**	Provided
E2.4.1	Length	500mm
E2.4.2	Make	M/s XYZ
E2.5	**Wiper blade**	
E2.5.1	Length	450 mm
E2.5.2	Make	BCS (SYNDICATE WIPER SYSTEM)
E2.5.3	Identification	X5XY0258
E2.6	**Washer tank**	BWXY4501
E2.6.1	Type	Plastic
E2.6.2	Make	XYZ
E2.6.3	Identification No. / Part No.	X5BXYZ168
E2.6.4	Capacity l	1.5Ltr
E2.6.5	Material	H.D.P.E
E2.6.6	Washer tank motor / Washer pump	ELECTRIC MOTOR WITH PUMP
E2.6.6.1	Make	XYZ
E2.6.6.2	Model	PEAD64XYZS00
E2.6.7	Nozzle(s)	XYZ
E2.6.7.1	No. Of Nozzles	2
E2.7	**Defroster**	NA
E2.7.1	Type	-
E2.7.2	Make	-
E2.7.2.1	Details of Demisting and / or Defrosting system	-
E2.7.2.2	Schematic Diagram of Engine cooling system	-
E2.7.2.3	Schematic Diagram of Heating system	-
E2.7.2.4	Cross sectional view of complete air handling system including position of louvers (vents) on dash board	-
E2.7.2.4.1	Blower identification	-
E2.7.2.5	Blower Voltage and capacity (Watts)	-
E2.8	If defrost system is fitted, specify defrost test temperature (-8 +/- 2 deg C / -18 +/- 3 deg C)	
E2.7	Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver's vision points, Angle obstruction of 'A' pillar & related dimensions as per related standards.	
E3.0	**Horn**	
E3.1	Make	M/s. XXYY, India
E3.2	Type (As per IS: 1884 – 1993)	2B
E3.3	Operating voltage	12V

Manufacturer signature: _____ Test agency: _____

Table 6 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART E – ELECTRICAL		
E3.4	Identification: TAC No. / BIS License No. / E-Marking	CXU105
E3.5	Number	One
E3.6	Sketch showing mounting of horn	VFTXYZ05
E3.7	Brief Dimensional Drawing indicating the shape and material of the body work at the front of the horn, which might affect the level of the sound, emitted by the horn and have a masking effect	
E3.8	Maximum vehicle speed for continuous operation (km/h) (Only for AC horns)	80 Kmph
E4.0	**Lighting Installation requirements**	
E4.1	Brief Dimensional Drawing indicating installation details of all light & light signaling devices of the vehicle as per AIS-008	VFXYZ1501
E4.2	Head lamp leveling system (manual / automatic)	Manual
E4.2.1	Stop position (if manual)	0
E4.2.2	Initial inclination	-1.70%
E4.2.3	Drawing showing initial inclination angle on the head lamp	VFTXYZ506
E4.2.4	Drawing showing type and controls of dipped beam leveling device with table showing positions of switch for various vehicle loading condition	VFTXYZ507
E4.3	Tell-Tale for leveling switch	Yes
E4.3.1	Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles)	NA
E5.0	**Head lamp**	
E5.1	**Main beam**	
E5.1.1	Make	M/s. XYZ Limited
E5.1.2	Type of lens (Glass / Plastic)	Plastic
E5.1.3	Identification: TAC No. / BIS License No. / E-Marking	ALSS4 / E9-2ZXYZ8
E5.1.4	Number and Colour of Lens	Two, Clear
E5.2	**Dipped beam**	
E5.2.1	Make	M/s. XYZ Limited
E5.2.2	Type of lens (Glass / Plastic)	Plastic
E5.2.3	Identification: TAC No. / BIS License No. / E-Marking	AL0XYZ4
E5.2.4	Number and Colour of Lens	Two, Clear
E5.3.0	Head Lamp cleaning device provided (Yes / No) (For Headlamps having intensity more than 2000 lumen)	NA
E5.3.1	Cleaner Type as per AIS-083	--
E5.3.1.1	Make	--
E5.3.2	A list, specifying the parts which constitute the headlamp cleaner and drawings thereof, (e.g. pumps, nozzles, valves, motors and wipers);	--
E5.3.3	A brief technical description indicating the length of the cleaning period, the consumption of cleaning fluid during the cleaning period and the minimum capacity of the container provided;	--
E5.3.4	Capacity class of the fluid container: 25/50	--
E5.3.5	Drawings showing the installation to a vehicle	--
E5.3.6	Drawings showing the relative attachment between the headlamp(s) & the wiper(s), nozzle(s), or corresponding parts,	--
E5.3.7	Drawings showing the cleaning principle employed	--

Manufacturer signature: _____ Test agency: _____

Table 6 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART E – ELECTRICAL		
E5.3.8	Where appropriate, the part of the illuminating surface of the headlamp relevant to the cleaner shall also be shown	--
E5.4	**Bend Lighting , provided (Yes / No)**	NO
E5.4.1	Cornering Lamp (if provided)	--
E5.4.2	Make	--
E5.4.3	Identification: TAC No. / BIS License No. / E-Marking	--
E5.4.4	Type of lens (Glass / Plastic)	--
E5.4.5	Number and Colour of Lens	--
E5.5	**Day Time Running Lamp (if provided)**	NO
E5.5.1	Make	--
E5.5.2	Identification: TAC No. / BIS License No. / E-Marking	--
E5.5.3	Type of lens (Glass / Plastic)	--
E5.5.4	Number and Colour of Lens	--
E6.0	**Front Fog Lamp**	NA
E6.1	Make	--
E6.2	Type of lens (Glass / Plastic)	--
E6.3	Identification: TAC No. / BIS License No. / E-Marking	--
E6.4	Number and Colour of Lens	--
E7.0	**Rear Fog Lamp**	NA
E7.1	Make	--
E7.2	Identification: TAC No. / BIS License No. / E-Marking	--
E7.3	Number and Colour of Lens	--
E8.0	**Side Marker lamps**	NA
E8.1	Make	-
E8.2	Identification: TAC No. / BIS License No. / E-Marking	-
E8.3	Number and colour of Lens	-
E9.0	**Registration Plate lamp**	-
E9.1	Make	M/s. XYZ Limited
E9.2	Identification: TAC No. / BIS License No. / E-Marking	AA2XY4
E9.3	Number and colour of Lens	Two & White
E10.0	**Position lamp / Parking Lamp - Front**	--
E10.1	**Front Position Lamp**	--
E10.1.1	Make	M/s. XYZLimited
E10.1.2	Identification: TAC No. / BIS License No. / E-Marking	ALXY44
E10.1.3	Number and colour of Lens	Two, Amber
E10.2	**Front Parking Lamp**	NA
E10.2.1	Make	--
E10.2.2	Identification: TAC No. / BIS License No. / E-Marking	--
E10.2.3	Number and colour of Lens	--
E11.0	**Position lamp / Parking Lamp - Rear**	**Integrated in Rear Combination Lamp**
E11.1	**Rear Position Lamp**	Provided
E11.1.1	Make	M/s. XYZ Limited
E11.1.2	Identification: TAC No. / BIS License No. / E-Marking	AAXYZ4
E11.1.3	Number and colour of Lens	Two & Red
E11.2	Rear Parking Lamp	NA
E11.2.1	Make	-
E11.2.2	Identification: TAC No. / BIS License No. / E-Marking	-
E11.2.3	Number and colour of Lens	-

Manufacturer signature: _____ Test agency: _____

Table 6 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART E – ELECTRICAL			
E12.0	**Stop lamp (S1 / S2)**	**Integrated in Rear Combination Lamp**	
E12.1	Make	M/s. XYZ Limited	
E12.2	Identification: TAC No. / BIS License No. / E-Marking	AAXY64	
E12.3	Number and colour of Lens	Two & Red	
E13.0	**Stop lamp (S3) for M1 category & N category as applicable**	NA	
E13.1	Make	-	
E10.2	Identification: TAC No. / BIS License No. / E-Marking	-	
E10.3	Number and colour of lens	-	
E14.0	**Reversing lamp**	**Integrated in Rear Combination Lamp**	
E14.1	Make	M/s. XYZ Limited	
E14.2	Identification: TAC No. / BIS License No. / E-Marking	AA2364	
E14.3	Number and colour of Lens	One & White	
E15.0	**Direction indicator Lamp**	-	
E15.1	**Front**	Integrated in Head Lamp	
E15.1.1	Make	M/s. XYZ Limited	
E15.1.2	Identification: TAC No. / BIS License No. / E-Marking	AL0344 / E9-23468	
E15.1.3	Number and colour of Lens	Two, Amber	
E15.2	**Rear**	Integrated in Rear Combination Lamp	
E15.2.1	Make	M/s. XYZ Limited	
E15.2.2	Identification: TAC No. / BIS License No. / E-Marking	AA2364	
E15.2.3	Number and colour of Lens	Two & Amber	
E15.3	**Side**	Provided Category - 5	Provided Category - 6
E15.3.1	Make	M/s. XYZZ Industries Limited	M/s. XYZ Limited
E15.3.2	Identification: TAC No. / BIS License No. / E-Marking	C90756	E4 61712
E15.3.3	Number and colour of Lens	Two, Amber	Two, Amber
E15.4	**Flasher for Direction Indicators**	Electronic Flasher	
E15.4.1	Flashing Frequency (No of flashes / minute)	85	
E16.0	**Hazard warning signal**		
E16.1	**Front**		
E16.1.1	Make	M/s. XYZ limited	
E16.1.2	Identification: TAC No. / BIS License No. / E-Marking	ALXY44 / E9-23468	
E16.1.3	Number and colour of Lens	Two, Amber	
E16.2	**Rear**		
E16.2.1	Make	M/s. XYZ Limited	
E16.2.2	Identification: TAC No. / BIS License No. / E-Marking	AA2364	
E16.2.3	Number and colour of Lens	Two & Amber	
E16.3	**Side**	Provided Category - 5	Provided Category - 6
E16.3.1	Make	M/s. XYZZ Industries Limited	M/s. XYZ Limited
E16.3.2	Identification: TAC No. / BIS License No. / E-Marking	C90756	E4 61712
E16.3.3	Number and colour of Lens	Two, Amber	Two, Amber
E17.0	**Reflector**		
E17.1	Front		
E17.1.1	Make	M/s. XYZ limited Industries Limited	
E17.1.2	Type	CATEGORY 1A	
E17.1.3	Identification: TAC No. / BIS License No. / E-Marking	TAC NO- CXXYY80	
E17.1.4	Number and colour of Lens	Two & White	
E17.1.5	Reflective surface Area	34 cm^2	

Manufacturer signature: _____ Test agency: _____

Table 6 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART E – ELECTRICAL		
E17.1.6	Shape (Square / rectangular / circular / elliptical /other)	Round
E17.2	**Rear**	Integrated in Rear Combination Lamp
E17.2.1	Make	M/s. XXYYXX
E17.2.2	Type	CATEGORY 1B
E17.2.3	Identification: TAC No. / BIS License No. / E-Marking	AAXXY64
E17.2.4	Number and colour of Lens	Two & Red
E17.2.5	Reflective surface Area	24.3 cm^2
E17.2.6	Shape (Square / rectangular / circular / elliptical /other)	Rectangular
E17.3	**Side**	NA
E17.3.1	Make	-
E17.3.2	Type	-
E17.3.3	Identification: TAC No. / BIS License No. / E-Marking	-
E17.3.4	Number and colour of Lens	-
E17.3.5	Reflective surface Area	-
E17.3.6	Shape (Square / rectangular / circular / elliptical /other)	-
E18.0	**End – outline marker lamp (Top light)**	NA
E18.1	**Front**	-
E18.1.1	Make	-
E18.1.2	Type of lens (Glass / Plastic)	-
E18.1.3	Identification: TAC No. / BIS License No. / E-Marking	-
E18.1.4	Number and colour of Lens	-
E18.2	**Rear**	-
E18.2.1	Make	-
E18.2.2	Type of lens (Glass / Plastic)	-
E18.2.3	Identification: TAC No. / BIS License No. / E-Marking	-
E18.2.4	Number and colour of Lens	-
E 5.0 to E 18.0 - Installation Details.	Diagram of vehicle indicating location, reference axis, mark of apparent surface, contour of vehicle parts limiting geometric visibility of all lights and light signaling devices, location of extreme outer edges and longitudinal median plane of vehicle including following dimensions in mm.	
	Along width of vehicle-horizontal distance between inner illuminating surfaces, distance between inner illuminating surfaces and outer most part of vehicle and distance between nearest point of illuminating surfaces of indicators and dipped-beam head lamp.	
	Along length of vehicle (where applicable) - distance between transverse plane corresponding to the longitudinal rearmost extremity to center of reference of rear indicators.	
	Heights of highest and lowest point of illuminating surfaces.	
E19.0	**Automotive bulbs**	
E19.1	**Head lamp bulb (main beam**)	
E19.1.1	Make	
E19.1.2	Category as per AIS –034	
E19.1.3	Identification: TAC No. / BIS License No. / E-Marking	
E19.1.4	**Head lamp bulb (Dipped beam)**	
E19.1.5	Make	
E19.1.6	Category as per AIS –034	

Manufacturer signature: _____ Test agency: _____

Table 6 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART E – ELECTRICAL			
E19.1.7	Identification: TAC No. / BIS License No. / E-Marking		
E19.2	**Parking Lamp bulb – Front**		
E19.2.1	Make		
E19.2.2	Category as per AIS –034		
E19.2.3	Identification: TAC No. / BIS License No. / E-Marking		
E19.3	**Parking Lamp bulb – Rear**		
E19.3.1	Make		
E19.3.2	Category as per AIS –034		
E19.3.3	Identification: TAC No. / BIS License No. / E-Marking		
E19.4	**Direction indicator lamp bulb - front**		
E19.4.1	Make		
E19.4.2	Category as per AIS –034		
E19.4.3	Identification: TAC No. / BIS License No. / E-Marking		
E19.5	**Direction indicator lamp bulb - rear**		
E19.5.1	Make		
E19.5.2	Category as per AIS –034		
E19.5.3	Identification: TAC No. / BIS License No. / E-Marking		
E19.6	**Direction indicator lamp bulb - side**		
E19.6.1	Make		
E19.6.2	Category as per AIS –034		
E19.6.3	Identification: TAC No. / BIS License No. / E-Marking		
E19.7	**Front Position Lamp bulb**		
E19.7.1	Make		
E19.7.2	Category as per AIS –034		
E19.7.3	Identification: TAC No. / BIS License No. / E-Marking		
E19.8	**Rear Position Lamp (tail lamp) Bulb**		
E19.8.1	Make		
E19.8.2	Category as per AIS –034		
E19.8.3	Identification: TAC No. / BIS License No. / E-Marking		
E19.9	**Stop lamp bulb**		
E19.9.1	Make		
E19.9.2	Category as per AIS –034		
E19.9.3	Identification: TAC No. / BIS License No. / E-Marking		
E19.10	**Number plate lamp bulb**		
E19.10.1	Make		
E19.10.2	Category as per AIS –034		
E19.10.3	Identification: TAC No. / BIS License No. / E-Marking		
E19.11	**End out Marker bulb**		
E19.11.1	Make		
E19.11.2	Category as per AIS –034		
E19.11.3	Identification: TAC No. / BIS License No. / E-Marking		
E19.12	**Reversing lamp bulb**		
E19.12.1	Make		
E19.12.2	Category as per AIS –034		
E19.12.3	Identification: TAC No. / BIS License No. / E-Marking		
E19.13	**Stop Lamp Bulb (S3)**	**NA**	
E19.13.1	Make	-	

Manufacturer signature: _____ Test agency: _____

Table 6 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART E – ELECTRICAL		
E19.13.2	Category as per AIS –034	-
E19.13.3	Identification: TAC No. / BIS License No. / E-Marking	-
E19.14	**Front Fog Lamp Bulb**	**NA**
E19.14.1	Make	-
E19.14.2	Category as per AIS –034	-
E19.14.3	Identification: TAC No. / BIS License No. / E-Marking	-
E19.15	**Rear Fog Lamp Bulb**	NA
E19.15.1	Make	-
E19.15.2	Category as per AIS –034	-
E19.15.3	Identification: TAC No. / BIS License No. / E-Marking	-
E19.16	**Side Marker Lamp Bulb**	NA
E19.16.1	Make	-
E19.16.2	Category as per AIS –034	-
E19.16.3	Identification: TAC No. / BIS License No. / E-Marking	-
E19.17	**Cornering lamp bulb (if provided)**	NA
E19.17.1	Make	-
E19.17.2	Category as per AIS –034	-
E19.17.3	Identification: TAC No. / BIS License No. / E-Marking	-
E19.18	**Day time Running lamp bulb (if provided)**	NA
E19.18.1	Make	-
E19.18.2	Category as per AIS –034	-
E19.18.3	Identification: TAC No. / BIS License No. / E-Marking	-
E19.19	**Bending lamp bulb (if provided)**	NA
E19.19.1	Make	-
E19.19.2	Category as per AIS –034	-
E19.19.3	Identification: TAC No. / BIS License No. / E-Marking	-
E20.0	**Warning Triangle**	
E20.1	Make	M/s. XXYYZZ Pvt Ltd
E20.2	Identification: TAC No. / BIS License No. / E-Marking	V9XYZ1
E21.0	**Reflective tape (AIS: 090)**	
E 21.1	Make	M/s. XXXYYYY
E 21.2	Width of tape(s) in mm	20 mm
E 21.3	Identification: TAC No. / BIS License No. / E-Marking	A9XXX4
E 21.4	Dimensional Drawing indicating installation details of reflective Tapes at front, Rear & side of the vehicle / load body / container / Tanker etc., as per AIS-090	VFT01XYZ
E22.0	**Electromagnetic Compatibility as per AIS:004 (Part 3)**	Refer Annexure 6A
E22.1	Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:	--
E22.2	Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):	--
E22.3	List of all subassemblies, which includes an electronic oscillator or switching frequency greater than 9 kHz (like ECU, steering, suspension, Body	--

Manufacturer signature: _____ Test agency: _____

Table 6 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART E – ELECTRICAL			
	Control Module, etc.). The list shall include Device name, Make/s and Part No. /Type ID/Drawing No Please fill information about Vehicle Alarm System (VAS)/ Immobilizer as per Annexure A to Table 6, if fitted. List of Electrical/Electronic Systems which are not previously listed		
E 22.3.1	List of all Electrical components, which include Broadband EMI sources (like HAVC Motor, Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No	--	
E 22.4	Table of installation and use of RF transmitters in the vehicle(s), if applicable (see paragraph 3.1.8. of this standard):		
	Frequency bands (Hz)	Max Output Power (W)	Antenna Position at Vehicle, specific conditions for installation and/or use
E22.5	Vehicle equipped with 24 GHz short-range radar equipment: yes/no/optional		
E 23.0	**Protective Devices as per AIS 075**		
E 23.1	Name of the protective device manufacturer	Jay Switches	
E23.1.1	Type/Part no./Identification number(ID)/Drawing No.	FKXYZS00	
E 23.2	Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions	XXYY Steering System	
E 23.2.1	Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions	FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly	
E 23.3	System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems	Locks the Steering	
E 23.4	Type of Protective Device (Electronic / Mechanical)	Mechanical	
E 23.5	Mechanism of Locking (for Steering) a) Normal blocking type b) Yield or Slipping Mechanism c) Freely rotating Steering Wheel	Normal Blocking Type	
E 23.6	Whether driver warning feature present in the vehicle in case if driver opens the side door with the device deactivated?	NO	
E 24.0	VLT SYSTEM DETAILS (If fitted)	NA	
E 24.1	Make:	--	
E 24.2	Type	--	
E 24.3	Model No	--	
E 24.4	Part No	--	
E 24.5	Installation layout: Attach drawing showing location in vehicle.	--	
E 25.0	**EMERGENCY BUTTONS (If fitted)**	NA	
E 25.1	Make	--	
E 25.2	Model No.	--	
E 25.3	Part No.	--	
E 25.4	Connector used	--	
E 25.5	Connector used for antennas	--	
E 25.5.1	Main GSM antenna	--	
E 25.5.2	GPS antenna	--	

Manufacturer signature: _____ Test agency: _____

Table 6 of AIS 007 (Revision 5) TECHNICAL SPECIFICATION – FOUR WHEELERS AND ABOVE PART E – ELECTRICAL				
E 25.5.3	WLAN antenna	--		
E 26.0	**SYSTEM SOFTWARE (If fitted)**	NA		
E 26.1	Make	--		
E 26.2	Version	--		
E 26.3	Operating System Details with Version	--		
E 27.0	**COMMUNICATION PROTOCOL USED (If fitted)**	NA		
E 27.1	Vehicle to Centre	--		
E 27.1.1	VLT to Control Centre	--		
E 27.1.2	Command Set for Configurations	--		
E 27.2	**DESCRIPTION OF DEVICE (If fitted)**	NA		
E 27.6	**DRAWINGS(If fitted)**	--		
E 27.7	Device/Emergency Button/ System Drawing.	--		
E 27.8	**INSTRUCTIONS MANUAL (If fitted)**	NA		
		Blower	**Low-Cost Heater**	**AC**
E28.0	**Air Conditioning System for cabin of N2 and N3 categories of vehicle as per AIS-056(Rev.1) (Provided -Yes/No)**	Optional		
E28.1	Make(s):	M/S. XYZS		
E28.2	Part No/Identification No:	B0CXYZ02	B0C12XYZ1	B0XYZ202
E28.3	Schematic diagram of AC system			
E28.4	Cooling Capacity (kW)	NA	NA	4 kW
E29.0	**Ventilation System for cabin of N2 and N3 categories of vehicle as per AIS-056(Rev.1) (Provided - Yes/No)**	Optional		
E29.1	Information for Ventilation system	--		
E29.1.1	Total area (cm^2) of the unblocked openings and their locations excluding the area of Side Door Windows (Drawing to be provided)	F0Y8XYZ5		
E29.1.2	Cabin volume (m^3)	1.89 m_3		
E29.2	Name of manufacturer of the installed blower(s)	M/S. XYZS		
E29.3	Total flow rate (m^3 /hr) of all the blowers installed	400 ± 10% m^3/Hr		
E29.4	No. of speeds offered in blower (s)	4	4	4
E29.5	Whether air vents have been provided for direction control, If yes, their locations	Yes (4) LH/CLH/CRH/RH	Yes (4) LH/CLH/CRH/RH	Yes (4) LH/CLH/CRH/RH
E29.6	Schematic diagram of the Cabin Ventilation System	B0CXYZ02	B0C12XYZ1	B0XYZ202

E 30.0	Rear marking plate for Long and Heavy Vehicles	-		
E 30.1	Make	-		
E 30.2	Identification	-		
E 30.3	Class	-		
E 30.4	Installation drawing with mounting dimensions of rear marking plate on vehicle	-		
E 31.0	**Any other features (As declared by the vehicle manufacturer)**			
E 31.1	Information for Reverse Parking Alert System			
E 31.2	Type of System (Sensor based / Camera based system)	Sensor		
E 31.3	Sensor based system (as applicable) information	Ultrasonic Sensor		
E 31.4	Make	XYZ Automotive Ltd		
E 31.6	Identification No. / Part No.	Description	AL Part No.	Hella Part No
		Sensor	FXXX0700	015.XYZ-00
		ECU	FXX01000	015.XYZ-01
		Buzzer	X4XX0400	015.XYZ-00
E 31.7	Number of sensors and mounting location	Option 1: 4 Sensors mounted on Tail Lamp bracket and Frame load body Option 2: 3 Sensors mounted on Tail Lamp bracket and load body		
E 31.8	Camera based System (as applicable)	NA		
E 31.9	Make	--		

Manufacturer signature: _____ Test agency: _____

E 31.10	Identification No. / Part No.	--
E 31.11	Camera mounting location	--
E 31.12	Display Make	--
E 31.13	Display Identification No. / Part No.	--
E 31.14	Display location	--
Information to be submitted for Type approval of VAS and/or Immobilizer Information sheet for AIS-076		
1.0	**Description of the VAS**	-
a.	Make	-
b.	Type	-
c.	Serial Number	-
d.	Name and Address of manufacture	-
1.1	A detailed description of the alarm system and of the vehicle parts related to alarm system installed with Block Diagram	
1.2	A list of main components comprising the alarm system	-
1.2.1	Components placed in Engine compartment	-
1.2.2	Components placed in Passenger compartment	-
1.3	Availability of radio/wireless transmission Yes/No	-
1.3.1	Details of radio/wireless transmission if any	-
1.3.2	frequency	-
1.3.3	Availability of WPC approval for wireless technology Yes/No	-
1.4	The measures taken against false alarm	-
1.5	Range of protection offered by the device. Details about the detection mechanism (opening of Door, Bonnet, luggage compartment, intrusion etc.)	-
1.6	Details & specification of audible alarm	-
1.7	Details & specification of optical alarm	-
1.8	Details & specification of radio alarm if fitted	-
1.9	Method of setting/ unsetting the device	-
1.10	Type of key used for unsetting (Mechanical Key / electronic device)	-
1.11	Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code)	-
1.12	Exit delay in Sec	-
1.13	Entry Delay in sec	-
1.14	Details of status display (0.5cd)	-
1.15	Details of self check , automatic failure indication	-
1.6	Details & specification of audible alarm	-
1.7	Details & specification of optical alarm	-
1.8	Details & specification of radio alarm if fitted	-
1.9	Method of setting/ unsetting the device	-
1.10	Type of key used for unsetting (Mechanical Key / electronic device)	-
1.11	Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code)	-
1.12	Exit delay in Sec	-
1.13	Entry Delay in sec	-
1.14	Details of status display (0.5cd)	-
1.15	Details of self check , automatic failure indication	-
1.16	Details about the panic alarm , if provided	-
2.0	**Description of the Immobilizer**	-
a.	Make	-
b.	Type	-
c.	Serial Number	-
d.	Name and Address of manufacture	-
2.1	A detailed description of the vehicle immobilizer and of the vehicle and the measures taken against inadvertent activation with block diagram.	
2.2	The vehicle system(s) on which the immobilizer acts.	-
2.3	Details of radio/wireless transmission if any	-

Manufacturer signature: _____ Test agency: _____

2.4	Method of setting/ unsetting the device.	-
2.5	Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code)	-
2.6	A list of main components comprising the device	-
3.0	**Instructions (Applicable for both VAS & Immobilizer)**	-
3.1	Instructions for use	-
3.2	Instructions for maintenance (if any)	-

Manufacturer signature: _____ Test agency: _____

Annexure T6 - A List of Electricals & Electronics components					
S.No.	**Name**	**Make and Type**		**Part No / Id No**	**Test Report Number**
1	ECU	XXXX		YYYY	YYYY
2	Instrument Cluster	XXXX		YYYY	YYYY
		XXXX		YYYY	YYYY
3	Speedometer & Odo	XXXX		YYYY	YYYY
		XXXX		YYYY	YYYY
4	ABS ECU	XXXX		YYYY	YYYY
5	AC unit	XXXX		YYYY	YYYY
6	Auto AC controller	XXXX		YYYY	YYYY
7	Front Wiper Motor	XXXX		YYYY	YYYY
8	Hazard/ Flasher Controller	XXXX		YYYY	YYY
9	LED Interior Lamps	XXXX		YYYY	YYY
10	Alternator	XXXX		YYYY	YYY
					YYY
11	Radiator assembly (Cooling fan)	XXXX		YYYY	YYY
					YYY
12	Radiator assembly (Cooling fan)	XXXX		YYYY	YYY
					YYY
13	RPAS	XXXX	YYYY	YYYY	YYY
			YYY	YYY	
			YYY	YYY	
14	Remote Central Locking system	XXXX		YYYY	YYY
15	Telematics Control Unit	XXXX		YYYY	YYY
16	Low Cost Heater	XXXX		YYYY	YYY
17	Blower	XXXX		YYYY	YYY

Manufacturer signature: _____ Test agency: _____

ASHOK LEYLAND

Table 07 of AIS 007 (Revision 5) BRIEF TECHNICAL SPECIFICATIONS FOR MOTOR VEHICLES				
A. Manufacturer's name and address	**Ashok Leyland Limited,** No:1, Sardar Patel, Road, Guindy, Chennai – 600 032.			
Importer's name and address (in case of CBU)	NA			
Vehicle data				
Basic model	**BADA DOST i9 TNZX**			
Type / Description	Fully built vehicle, D+2, Non AC, 3.39 Ton GVW			
Category of the vehicle	N1			
Variant(s)	BADA DOST i8 TNZX			
Type / Description				
Category of variant(s) as per IS 14272:2011	N1			
Engine				
Make	**Ashok Leyland Limited** Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126.			
Model name / identification	ALXYZABC0			
Type	Compression Ignition			
Bore x stroke (mm)	80 X 98			
No. of cylinders	3			
Displacement (cm^3)	1468 cc			
Compression ratio	16.5±1.0 : 1			
Max. Engine output (KW @ rpm)	55.1 Kw @ 3300 rpm			
Max. Torque (Nm @ rpm)	170Nm @ 1600 to 2400 rpm			
Air cleaner type	Dry type			
Type of Fuel	Diesel			
Clutch				
Type	Dry / Single plate/Mechanical actuation			
Gear box				
Make and model	**RSB Transmission (T210R)**		**AVTEC (A5M210)**	
Type	Manual gearbox with fully synchro mesh & Rev with sliding mesh			
No. of gears	5 Forward and 1 Reverse			
Gear ratio	**Gear Ratio**	**Overall Ratio**	**Gear Ratio**	**Overall Ratio**
1st	4.62	4.60	21.53	21.44
2nd	2.68	2.69	12.67	12.71
3rd	1.51	1.49	7.34	7.25
4th	1.0	1.0	4.56	4.56
5th	0.70	0.71	3.64	3.67
Reverse	4.39	20.36	4.38	20.34
Drive Axle (Front / Rear / All)	Rear			
Front axle ratio	NA			
Rear axle ratio	5.13 : 1			
Steering				
Type / Description	Re-circulating ball type			
Steering Gear Ratio	1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd			
Steering wheel diameter mm	400mm			
Frame				
Long member size (mm)	**2490 WB**		**2640 WB**	
	4435 mm		4735 mm	
Number of cross members	7 - Total (Bolted & Rivited)			
Suspension				
Type				
Front	Front and Rear rigid leaf suspension			
Rear				
Spring – Front	Leaf spring			
Spring – Rear				
Anti-roll bar				

Manufacturer signature: _____ Test agency: _____

Table 07 of AIS 007 (Revision 5) BRIEF TECHNICAL SPECIFICATIONS FOR MOTOR VEHICLES					
Front		Yes			
Rear		NA			
Shock absorbers					
Front		Double acting hydraulic 2 No's			
Rear					
Brake					
Service brake Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others)		Vacuum assisted hydraulic			
Auto Slack Adjuster Fitted (Yes / No / Optional)		Yes (Automatic wear adjustment)			
Anti-Lock braking system Provided(Yes/No/Optional)		Optional			
Front (Disc / Drum)		Disc			
Rear (Disc / Drum)		Drum			
Total braking area (cm^2)		Front axle-254 sq.cm Rear axle-580 sq.cm			
Parking brake type		Mechanical			
Secondary brake type		Un failed part of the service brake			
Wheels and tyres		**Apollo**	**Ceat**	**Apollo**	**Ceat**
Wheel rim size					
Front		5.4J x 14			
Rear					
Tyre size designation including ply rating		7.00 R15 LT 12 PR		215/75 R15 LT	
Speed index					
Front		Q	L	S	S
Rear					
Load index / Load rating					
Front		114/109	114/109	115/113	115/113
Rear					
Tyre Type (Radial / Cross / Tube / Tubeless)					
Front		Radial, Tube Type		Radial, Tubeless	
Rear					
Laden Tyre pressure (kg/cm2 /kPa/ psi)					
Front		3.61 kg/cm2		2.56 kg/cm2	
Rear		6.77 kg/cm2		3.26 kg/cm2	
Electrical system					
System voltage (V)		12 V			
Battery rating (Ah)		50 Ah			
Wiper motor		XYZ Ltd.			
Wiping system (Brief description)		Twin blade, Tandem Pattern			
Fuel tank					
Material		Metallic			
Capacity (l)		50 lts , 40 lts			
Dimensions		**2490WB**		**2640WB**	
Wheel base (mm)		2490		2640	
Overall width (mm)		1842			
Overall Length (mm)	CC	4750		4988	
	FSD / HSD	4925		5225	
Overall height (mm)		1961		1948	
Wheel Track (mm)					
Front track (mm)		1480			
Rear track (mm)		1395			
Minimum Ground Clearance as per IS 9435:2004 (mm) (M1)		NA			
		2490WB		**2640WB**	
Min. ground clearance (mm) (other than M1)		214 / 193		193 / 185	
Maximum gradeability in 1st gear		GVW 2890 kg		GVW 3390 kg	
		10 Deg		8.5 Deg	

Manufacturer signature: _____ Test agency: _____

Table 07 of AIS 007 (Revision 5) BRIEF TECHNICAL SPECIFICATIONS FOR MOTOR VEHICLES											
Dimensions			**L**	**W**	**H**	**L**	**W**	**H**			
		FSD	2951	1750	490	3250	1750	490			
		HSD	2983	1819	1092	3283	1819	1092			
	Container (As per Annexure T7 – B)	Type 1	2951	1750	1850	3250	1750	1850			
		Type 2			1950			1950			
	Tipper (As per Annexure T7 – B)	Tipper	3180	1760	668	3060	1750	709			
Load body platform area (l x b) (mm)		FSD	4.8 m^2			5.3 m^2					
		HSD	5.0 m^2			5.5 m^2					
Body overhang (mm)			**2490 WB**			**2640 WB**					
Front end			1297								
Rear end		CC	863			951					
		FSD / HSD	938			1188					
Frame overhang mm (in case of vehicles without complete body)											
Front end			937								
Rear end			808			958					
Weights			Base	Variant 1	Variant 2	Variant 3	Variant 4				
Maximum GVW kg (for rigid vehicles)			Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B								
Maximum Permissible FAW											
Maximum Permissible RAW											
Other axle			-	-		-	-				
Maximum GCW kg (for articulated / combination Vehicles)			-	-		-	-				
Maximum Permissible FAW			-	-		-	-				
Maximum Permissible RAW			-	-		-	-				
Other axle			-	-		-	-				
Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg)			-	-		-	-				
Front axle			-	-		-	-				
Rear axle			-	-		-	-				
Other axle			-	-		-	-				
Total			-	-		-	-				
CO2 (g/km) (Applicable for category M1 with GVW<3.5 T)											
Declared (Rounded to 3 decimal places)			NA	NA		NA	NA				
Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles (Applicable for category M1 with GVW<3.5 T) (Rounded to 3 decimal places)*											
Fuel Equivalent Fuel Consumption (Actual Fuel)			NA	NA		NA	NA				
Petrol Equivalent Fuel Consumption (Petrol Equivalent)			NA	NA		NA	NA				
*calculated on the declared CO2											
Vehicle Max Speed in unladen condition (km/h)			Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF								
Vehicle Max Speed in laden condition (km/h)			Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF								
Seating											
Seating capacity			D+2								
Sketch showing seating layout with vehicle dimensions (mm) (All category of vehicles)			BWXY4501								

Manufacturer signature: _____ Test agency: _____

Manufacturer signature: _____ Test agency: _____

ANNEXURE T7 - A																	
S. No	Vehicle Model	FSD						HSD						FAW (Kg)	RAW (Kg)	GVW (Kg)	
		FAW (Kg)	RAW (Kg)	ULW (Kg)	F/R Ratio	Reference Mass (Kg) (ULW+150Kg)	Equivalent Inertia (Kg) as per AIS 137 - Part 3	FAW (Kg)	RAW (Kg)	ULW (Kg)	F/R Ratio	Reference Mass (Kg) (ULW+150Kg)	Equivalent Inertia (Kg) as per AIS 137 - Part 3				
Base Model																	
1	BADA DOST i9 TNZX	1011	613	1624	1.65	1774	1810	1021	671	1692	1.52	1842	1810	1493	1897	3390	
2	BADA DOST i8 TNZX	1011	613	1624	1.65	1774	1810	1021	671	1692	1.52	1842	1810	1425	1465	2890	

ANNEXURE T7 – A																			
S. No	Vehicle Model	CAB CHASSIS VARIANTS				FSD		HSD		Recommended Load Body						FAW (Kg)	RAW (Kg)	GVW (Kg)	
										Type 1		Type 2		Tipper					
		FAW (Kg)	RAW (Kg)	ULW (Kg)	F/R Ratio	Reference Mass (Kg) (ULW+150Kg)	Equivalent Inertia (Kg) as per AIS 137 - Part 3	Reference Mass (Kg) (ULW+150Kg)	Equivalent Inertia (Kg) as per AIS 137 - Part 3	Reference Mass (Kg) (ULW+150Kg)	Equivalent Inertia (Kg) as per AIS 137 - Part 3	Reference Mass (Kg) (ULW+150Kg)	Equivalent Inertia (Kg) as per AIS 137 - Part 3	Reference Mass (Kg) (ULW+150Kg)	Equivalent Inertia (Kg) as per AIS 137 - Part 3				
3	BADA DOST i9 TNZX CC	989	479	1468	2.06	1774	1810	1842	1810	1843	1810	1848	1810	1908	1930	1493	1997	3490	
4	BADA DOST i8 TNZX CC	989	479	1468	2.06	1774	1810	1842	1810	1843	1810	1848	1810	1908	1930	1425	1465	2890	

Manufacturer signature: _____ Test agency: _____

Manufacturer signature: _____ Test agency: _____

Table 8 of AIS-007 (Revision 5)

Rule No.	Subject	Name of the Manufacturer (Please give information for every supplier / vendor under the same para, separate lines)	TAC No. / BIS License No / Test Report No. as applicable. (indicate validity date) (Application Ref No. allotted by concerned Test Agency, If approval is in the process)	Possible date of submission of required approval, if the same is in process	CoP Cert No. with validity date (wherever applicable)
95	**Tyres** (Compliance to IS 15633 / IS 15627/IS 15636)				
	Front				
	Rear				
	Spare wheel (as applicable)				
100	**Safety Glass** a) Windscreen b) Side c) Rear (For 3 & 4 Wheeler)				
101	**Windscreen Wiping System** a) Wiping System b) Washing System c) Wiper Blade (For 3 & 4 Wheelers)				
104	**Reflex Reflector** a) Front, White b) Rear, Red c) Side, Amber				
	CNG / LPG Kit Components				
	Cylinder (as per Gas Cylinder Rule, 2004)				
	Cylinder Valve / Multi Function Valve (as per Gas Cylinder Rule, 2004)				

Manufacturer signature: _____ Test agency: _____

Table 8 of AIS-007 (Revision 5)

	Petrol Solenoid valve				
	CNG/LPG Rigid Pipe				
	CNG/ LPG High Pressure Flexible Hose				
	CNG/ LPG Low Pressure Flexible Hose				
	Electrical Fuses				
	Ventilation Hose/ Conduit				
	Electrical Wiring Harness				
	Seat Upholstery, Roof, Side linings				
	Non-moisture retaining Hard rubber for cylinder mounting (as applicable)				
118	**Speed Limiter Installation Test Report as per AIS-018 (SLD / SLF)**				
119	**Horns(s)** Horn Installation (For all vehicles)				
123	**Pillion Hand Holds** (For all vehicles)				
124/ 1	**Automotive Bulbs** (Mention category of bulb/s as per AIS-034)				
	Main Beam head Lamp				
	Dipped Beam Head Lamp				
	Daytime Running Lamp				
	Cornering Lamp				
	Front Position / parking lamp				
	Front Fog Lamp				
	Front Direction Indicator Lamp				
	Front End-out Marker Lamp				

Manufacturer signature: _____ Test agency: _____

	Rear Direction indicator Lamp				
	Rear Position / Parking Lamp				

Table 8 of AIS-007 (Revision 5)

	Reversing Lamp				
	Rear Fog Lamp				
	Rear Registration Plate Lamp				
	Rear End-out Marker Lamp				
	High Mounted Stop Lamp				
124/2	Hydraulic Brake Hose (For all vehicles – as applicable)				
124/3	Hydraulic Brake Fluid (For all vehicles – as applicable)				
124/5	Steering Impact a) Head Form Test b) Body Block Test c) Crash Test (For M1 category having GVW not more than 1500 kg)				
124/6	Side Door Impact Test (For passenger cars)				
124/7	Fuel Tank (Provide details in case of multiple capacities / suppliers)				
	i) Fuel Tank (metallic) or ii) Fuel Tank (plastic) (For Four Wheelers)				
124/8	Wheel Rims (For Four wheelers)				
124/9	Control Cables (For two wheelers below 50 CC)				
124/10	Pneumatic Coupling (For N category of vehicles)				

Manufacturer signature: _____ Test agency: _____

124/15	Accelerator Control Systems (Only for Four Wheelers)				

Table 8 of AIS-007 (Revision 5)

124/16	Door Locks & Hinges (Only for Four Wheelers)				
	Door Hinges				
	Front Door Hinges				
	Rear Door Hinges				
	Door Locks				
	Front Door Lock				
	Rear Door Lock				
124/17	Hood Latch (For passenger cars)				
124/20	**i)** Lighting Signalling & Indicating Systems (For 4 Wheelers)				
	Head Lamp (Main Beam)				
	Head Lamp (Dipped Beam)				
	Front Position / Parking Lamp				
	Cornering Lamp (if provided)				
	Front Direction Indicator				
	Front Fog lamp				
	Day-Time Running Lamp (if provided)				
	Front end-out marker Lamp / Top Lights				
	Rear end-out marker Lamp / Top Lights				
	Stop Lamp				
	Rear Position / Parking Lamp				
	Rear Direction Indicator				

Manufacturer signature: _____ Test agency: _____

	Rear Registration Plate Lamp				
	Side Direction Indicator Lamp				
	Side Marker lamp				
	Head Lam Cleaning Device				

Table 8 of AIS-007 (Revision 5)

	ii) Lighting and Signalling Installation Requirements (for 4 wheelers)				
	Report No(s). for Base Model / Variants (if already issued)				
124/21	Electromagnetic Radiation (EMI) (for all combinations of spark plug, ignition coil, HT cable, Ignition System, ECU and suppress cap) (For all vehicles)				
124/22	Towing Devices (For 4 wheelers) as applicable				
124/24	Lighting and Signaling installation requirements for 2 & 3wheelers, including Trailers, semi-Trailers				
	Report No(s). for Base Model / Variants (if already issued)				
124/25	Fuel Tank for 2 & 3wheelers (metallic or Non-Metallic) (Indicate Nominal capacity)				
124/32	Lighting and light signaling devices for 2 wheelers, 3 wheelers and their trailers and semi trailers.				

Manufacturer signature: _____ Test agency: _____

	Rear Position / Parking Lamp				
	Rear Direction Indicator				
	Reversing lamp for 3 Wheeler				
	Rear Registration Plate Lamp				
	Side Direction Indicator Lamp				

Table 8 of AIS-007 (Revision 5)

124/33	Spray Suppression System Installation test report as per AIS-013				
124/34	Drivers field of vision for M1 category of vehicles.				
124/35	Survival space for protection of occupants in a cab.				
124/36	Strength of superstructure of passenger vehicles.				
124/37	Flammability requirements for M3 category vehicles with more than 22 passengers.				
124/38	Interior fittings for M1 category				
124/39	Windscreen wiping system requirements for 3 wheelers				
124/42	Handholds for L5, M & N category vehicles				
124/43	Wheel Rims for L category vehicles				
124/44	Protective Devices for L category vehicles				
124/46	Defrost & Demist Systems for M1 category vehicles				
124/48	Spray Suppression test for				

Manufacturer signature: _____ Test agency: _____

124/1 A	Vehicle Rear Under run Protection And Lateral Protection (For four wheelers)				
125/1 A	Safety Belt and Safety Belt Anchorages (For four wheelers)				

Table 8 of AIS-007 (Revision 5)

125/ (2)	Rear View Mirror and Rear View Mirror Installation Requirements as per AIS-002 (For all vehicles as referred in AIS-001)				
	Interior Mirror (Class-I)				
	Main Mirror large (Class-II)				
	Main Mirror small (Class-III)				
	Wide Angle Mirror (Class-IV)				
	Close proximity Mirror (Class-V)				
	Front Mirror (Class-VI)				
	Mirrors for L category vehicle with bodywork (Class-VII)				
125/1 C	Seat Size, Anchorages and Head Restraints (For four wheelers)				
138	Warning Triangles				

Note :

1) Please enclose copies for TAC / CoP / BIS License / ECE Certificate / Test Reports wherever required by the testing agency.
2) Fill all the columns. If any clause is not applicable, mention “NA” in corresponding column. Do not keep it blank.
3) In case samples are submitted to testing agency, please provide Reference No. if the approval is in process.)

Manufacturer signature: _____ Test agency: _____

Table 8 of AIS-007 (Revision 5)

Component test reports / certificates

Continuation document ALP-829383b7-T8-R3-C · TEST – Bada Dost rear axle 5.13

Engineering draft. Manufacturer and test-agency acceptance / signatures remain pending.

Rule / standard / subject	Component / source model	Report / issuer / edition	Applicability justification
AIS-010 / AIS-012 / AIS-034; selected device standard Lamp / light-source component approval	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-001 (Part 1) Mirror reflecting-surface dimensions	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-001 (Part 1) Mirror radius of curvature	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-001 (Part 1) Mirror reflectivity	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-001 (Part 1) Mirror impact / mounting verification	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
IS 15140; safety-belt assemblies and restraint systems Seat-belt assembly approval and installation	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	Component / source model	Report / issuer / edition	Applicability justification
IS 2553 Part 2; CMVR 100 Safety glazing approval and installation	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
IS 15633 / IS 15636; confirm tyre class Tyre component approval and markings	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
IS 1884 Horn component approval	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-057; device class / edition review Retro-reflector component approval	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-010 / selected notified headlamp standard Headlamp component approval and photometry	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-004 Part 3 Electrical / electronic subassembly EMC approval	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-037; applicable notified component standards Safety-critical component type approval and CoP	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Table 9 of AIS-007 (Revision 5)

INFORMATION ON TEST REPORTS FOR VEHICLE LEVEL AND SYSTEM LEVEL COMPLIANCE

Rule No.	Subject	Model	No. of test report / certificate *	Issued by	Justification for applicability as per CEA for the model under consideration

* Xerox copies of the certificates to be submitted in case if it is from another testing agency or whenever necessary.

Manufacturer signature: _____ Test agency: _____

Table 9 of AIS-007 (Revision 5)

Vehicle and system test reports / other-model evidence

Continuation document ALP-829383b7-T9-R3-C · TEST – Bada Dost rear axle 5.13

Engineering draft. Manufacturer and test-agency acceptance / signatures remain pending.

Rule / standard / subject	Component / source model	Report / issuer / edition	Applicability justification
CMVR 96; applicable IS 11852 / IS 15986 route Service braking performance	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
CMVR 96; applicable IS 11852 / IS 15986 route Parking braking performance	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
CMVR 96; applicable braking standard; N1 fitment / mandate review ABS / electronic braking applicability and performance	Evidence not established	No accepted report linked	Confirm ABS fitted / applicable. Unknown fitment / scope is retained for review.
AIS-008 Lighting installation and geometric visibility	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-008 Dipped-beam inclination / levelling verification	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-010 / AIS-012 / AIS-034; selected device standard Lamp / light-source component approval	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	Component / source model	Report / issuer / edition	Applicability justification
AIS-002 (Part 1) Indirect-vision installation and field of view	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-001 (Part 1) Mirror reflecting-surface dimensions	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-001 (Part 1) Mirror radius of curvature	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-001 (Part 1) Mirror reflectivity	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-001 (Part 1) Mirror impact / mounting verification	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
IS 15139; safety-belt anchorages Seat-belt anchorage strength	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
IS 15140; safety-belt assemblies and restraint systems Seat-belt assembly approval and installation	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-023; N-category seats, anchorages and head restraints; IS 15546 where applicable Seats, seat anchorages and head restraints	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	Component / source model	Report / issuer / edition	Applicability justification
AIS-029; applicable N-category cab scope / notification Cab structural strength applicability	Evidence not established	No accepted report linked	Confirm Cab-strength standard applies. Unknown fitment / scope is retained for review.
AIS-098; category / mass / edition review Offset frontal collision applicability	Evidence not established	No accepted report linked	Confirm Frontal-impact scope applies. Unknown fitment / scope is retained for review.
IS 2553 Part 2; CMVR 100 Safety glazing approval and installation	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
IS 15802 / IS 15804 / AIS-045; allocation and scope review Wiping, washing and demisting applicability	Evidence not established	No accepted report linked	Confirm Windscreen fitted. Unknown fitment / scope is retained for review.
IS 15633 / IS 15636; confirm tyre class Tyre component approval and markings	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Applicable tyre / vehicle installation provisions Tyre load/speed rating and installation suitability	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-110 Temporary spare / run-flat configuration	Evidence not established	No accepted report linked	Confirm Temporary spare / run-flat fitted. Unknown fitment / scope is retained for review.

Rule / standard / subject	Component / source model	Report / issuer / edition	Applicability justification
CMVR 98; IS 12222 / IS 11948 Steering and manoeuvrability verification	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
IS 13942; applicable N-category external projection scope External projections / construction review	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
IS 12056 / applicable notified fuel-tank standard Fuel tank / fire-risk and installation verification	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
IS 1884 Horn component approval	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
IS 15796 Audible warning installation / sound level	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
CMVR 117; IS 11827; current edition review Speedometer indication and installation	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-137 (Part 3); route / fuel review Type I tailpipe emissions	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-137 (Part 3) Type III crankcase emissions applicability	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	Component / source model	Report / issuer / edition	Applicability justification
AIS-137 (Part 3) Type V durability / deterioration factors	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-137 (Part 3) OBD monitoring and diagnostic demonstration	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-137 (Part 3), applicable amendments Real-driving emissions / route applicability	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
IS 14599; CMVR 124; current edition / route review Engine net-power measurement	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
CMVR 120; IS 3028; current edition review Exterior / pass-by noise	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-004 (Part 3) Vehicle / ESA electromagnetic emissions	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-004 (Part 3) Vehicle / ESA electromagnetic immunity	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-065 VIN, manufacturer plate and mandatory markings	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	Component / source model	Report / issuer / edition	Applicability justification
AIS-017 (Part 6) Whole-vehicle safety conformity of production	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-017 / CMVR; confirm current notified procedure Whole-vehicle type-approval dossier / extension	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
CMVR / VAHAN; administrative work item Registration / approval identifiers	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Applicable notified CMVR / marking provisions Mandatory labels and statutory markings applicability	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
CMVR 93 Overall dimensions and overhang	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
CMVR 95; IS 11825; applicable axle-weight notifications Permissible GVW, payload and axle loading	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
CMVR 98; IS 12222 Turning circle and manoeuvrability	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
CMVR 96; applicable IS 11852 / IS 15986 route Secondary / residual braking	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	Component / source model	Report / issuer / edition	Applicability justification
CMVR 96; applicable IS 11852 / IS 15986 route Brake fade, recovery and laden/unladen performance	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
CMVR 96; applicable braking standard Brake circuit, warning, control and installation checks	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
CMVR 96; IS 11852:2019; GSR 834(E), 11 Nov 2025 2027 braking standard transition	Evidence not established	No accepted report linked	Confirm manufacture date and whether this is a new or existing model. Final notified route and edition still need confirmation.
AIS-057; device class / edition review Retro-reflector component approval	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-090; CMVR marking provisions Conspicuity tape and rear marking applicability	Evidence not established	No accepted report linked	Confirm Transport vehicle. Unknown fitment / scope is retained for review.
AIS-089; heavy/long category and scope review Heavy / long vehicle rear marking plate scope check	Evidence not established	No accepted report linked	Heavy-category scope is not assumed for N1. Enable the relevant scope fact if the actual classification / special route brings it into scope.
AIS-012; selected signalling device standard Direction, stop, position and registration lamps	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	Component / source model	Report / issuer / edition	Applicability justification
AIS-010 / selected notified headlamp standard Headlamp component approval and photometry	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-012 / device-specific standard; AIS-008 Front / rear fog lamp approval and installation	Evidence not established	No accepted report linked	Confirm Fog lamps fitted. Unknown fitment / scope is retained for review.
Selected notified lighting device standard; AIS-008 Additional driving lamp approval and installation	Evidence not established	No accepted report linked	Confirm Additional driving lamps fitted. Unknown fitment / scope is retained for review.
AIS-034 / selected device and light-source standard Replaceable filament / LED light-source approval	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
IS 16694; CMVR restraint provisions Seat-belt and restraint installation	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-099; applicable N1 scope / notified revision Lateral collision scope and occupant protection	Evidence not established	No accepted report linked	Confirm Side-impact scope applies. Unknown fitment / scope is retained for review.
IS 14225 / applicable notified door-lock standard Door latches and retention systems	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	Component / source model	Report / issuer / edition	Applicability justification
AIS-023; IS 15546 where applicable; N-category scope review Seat head-restraint installation and strength	Evidence not established	No accepted report linked	Confirm Head restraints fitted / required. Unknown fitment / scope is retained for review.
IS 15223 / applicable notified scope Cab interior fittings and injury protection scope	Evidence not established	No accepted report linked	Confirm Interior-fittings scope applies. Unknown fitment / scope is retained for review.
AIS-084; applicable category and scope Windscreen defrosting and demisting	Evidence not established	No accepted report linked	Confirm Defrost/demist scope applies. Unknown fitment / scope is retained for review.
AIS-021 / IS 15802 / IS 15804; scope review Forward field of vision and wiping coverage	Evidence not established	No accepted report linked	Confirm Forward-vision scope applies. Unknown fitment / scope is retained for review.
Applicable wheel/rim and tyre installation standard Wheel / rim specifications and load suitability	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
CMVR 118; AIS-018 Speed limiting function / device and tamper protection	Evidence not established	No accepted report linked	Confirm Transport vehicle. Unknown fitment / scope is retained for review.
AIS-071; applicable category scope Identification of controls, tell-tales and indicators	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	Component / source model	Report / issuer / edition	Applicability justification
CMVR 124; notified fuel-system standard Fuel leakage and installation safety	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-024 / AIS-025 / AIS-028; retrofit route Gas-fuel conversion kit type approval and installation	Evidence not established	No accepted report linked	Confirm Gas-fuel retrofit / conversion. Unknown fitment / scope is retained for review.
Applicable notified LNG requirements and PESO approval LNG fuel-system and cylinder approval route	Evidence not established	No accepted report linked	Confirm LNG fuel route. Unknown fitment / scope is retained for review.
AIS-137 Part 3; CMVR 115 Emission family, road-load and worst-case selection	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-137 Part 3; CMVR 115 Mass-emission conformity of production	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
CMVR 115; AIS-137 applicable in-service provisions In-service conformity / PUC applicability	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
CMVR 115; AIS-137 applicable diesel provisions Diesel free-acceleration smoke / opacity route	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	Component / source model	Report / issuer / edition	Applicability justification
Applicable AIS-137 / notified consumption measurement procedure Fuel consumption and CO2 declaration	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-137 Part 3 OBD interface, fault detection and calibration identification	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-137 Part 3; applicable diesel provisions SCR reagent / inducement and dosing system	Evidence not established	No accepted report linked	Confirm SCR fitted. Unknown fitment / scope is retained for review.
AIS-004 Part 3 Electrical / electronic subassembly EMC approval	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-037; applicable notified component standards Safety-critical component type approval and CoP	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-017 Part 6; CMVR 126A Vehicle safety CoP plan, test records and surveillance	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
AIS-017; CMVR 126 / 126A Manufacturer / production-plant extension review	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	Component / source model	Report / issuer / edition	Applicability justification
AIS-017; CMVR 126; category-specific body scope Incomplete / completed vehicle and body-builder approval	Evidence not established	No accepted report linked	Confirm Multi-stage / body-builder route. Unknown fitment / scope is retained for review.
Applicable CMVR / notified load-body requirements Goods-body, load restraint and body mounting review	Evidence not established	No accepted report linked	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.
Applicable CMVR / notified coupling standard Trailer coupling approval and installation scope	Evidence not established	No accepted report linked	Confirm Trailer coupling fitted. Unknown fitment / scope is retained for review.
CMVR 129–137; applicable dangerous-goods standards Dangerous-goods vehicle special requirements	Evidence not established	No accepted report linked	Confirm Dangerous-goods use. Unknown fitment / scope is retained for review.
AIS-125; applicable CMVR notification Road ambulance conversion and equipment scope	Evidence not established	No accepted report linked	Confirm Ambulance conversion. Unknown fitment / scope is retained for review.
AIS-169; current final notification required Adapted-vehicle approval and controls	Evidence not established	No accepted report linked	Confirm Adapted vehicle. Unknown fitment / scope is retained for review.
Applicable category-specific AIS / CMVR requirements Other special-purpose / conversion route review	Evidence not established	No accepted report linked	Confirm Other special-purpose conversion. Unknown fitment / scope is retained for review.

Rule / standard / subject	Component / source model	Report / issuer / edition	Applicability justification
IS 14812; heavy-vehicle scope Rear underrun protection category scope check	Evidence not established	No accepted report linked	Heavy-category scope is not assumed for N1. Enable the relevant scope fact if the actual classification / special route brings it into scope.
IS 14682; heavy-vehicle scope Lateral protection category scope check	Evidence not established	No accepted report linked	Heavy-category scope is not assumed for N1. Enable the relevant scope fact if the actual classification / special route brings it into scope.
AIS-069; heavy-vehicle scope Front underrun protection category scope check	Evidence not established	No accepted report linked	Heavy-category scope is not assumed for N1. Enable the relevant scope fact if the actual classification / special route brings it into scope.
CMVR 125H; AIS-140; transport-use and exemption review Vehicle location tracking / emergency button applicability	Evidence not established	No accepted report linked	Confirm Location-tracking / emergency-button requirement applies. Unknown fitment / scope is retained for review.
CMVR fitness / permit / registration provisions Transport fitness, permit and registration handover	Evidence not established	No accepted report linked	Confirm Transport vehicle. Unknown fitment / scope is retained for review.

Table 10 of AIS-007 (Revision 5)

INFORMATION NEEDED FOR APPLYING CRITERIA FOR EXTENSION OF APPROVAL

Rule No:	Subject:	Notified Standard	CEA as per doc. no.

Value for each variant

Variants / CEA Parameters	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1)									
(2)									
(3)									
(4)									
(5)									
(6)									
(7)									
(8)									
(9)									
(10)									

Manufacturer signature: _____ Test agency: _____

Table 10 of AIS-007 (Revision 5)

Criteria for extension of approval

Continuation document ALP-829383b7-T10-R3-C · TEST – Bada Dost rear axle 5.13

Engineering draft. Manufacturer and test-agency acceptance / signatures remain pending.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 96; applicable IS 11852 / IS 15986 route Service braking performance	Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Axle spacing in case of multi axle vehicles.: NA → NA Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA → NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rating, Speed rating, Load rating or Load		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	& 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Final Drive (4x2 or 4x4 or 6x2 or 6x4 or others) Front / Rear / Both: (4 x 2) Rear → (4 x 2) Rear Rear axle make: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Rear axle ratio: 4.56 → 5.13 Differential type: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2.		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	YYYY – 18.2 :1 Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tank Symbol → Red Color – Oil Tank Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Parking brake: Hand operated [LHS] → Hand operated [LHS] Service Brake: Foot operated → Foot operated Service Brake: → Type of coupling device for brake: NA → NA Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing surfaces, “E” in		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	· Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Make: M/s Brakes India Private Limited → M/s Brakes India Private Limited Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others):		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vacuum assisted hydraulic → Vacuum assisted hydraulic Control system & braking wheel: Hydraulic acting on all wheels → Hydraulic acting on all wheels Schematic layout indicating method of split of brake system, location of valves, reservoirs, ABS components etc. (Attach	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	drawing and indicate the drawing number): AJ1XXX00 → AJ1XXX00 Anti-Lock braking system Provided (Yes / No / Optional): NA → NA If yes, details of ABS: - → - Make: - → - Category of ABS: - → - Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Modulator: - → - Nos. of Modulators: - → - Make of Modulators: - → - Identification No. / Part No. of Modulator: - → - Brief description and features: - → - Controller: - → - Nos. of Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Height of Center of Gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(mm): - → - Un laden condition: - → - Laden condition: - → - Slack adjuster (only applicable for M2, M3, N and T category of vehicles.): - → - Front (Automatic / Manual): - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make of slack adjuster: - → - Model and Type: - → - Rear (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Front wheel: 132x47.3x10.35 → 132x47.3x10.35 Rear wheel: LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 → LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 Others: NA → NA Type of liner wear indicator (window/ acoustic/ optical/ any other): Window → Window Front: 250 sq.cm → 250 sq.cm Rear: 570 sq.cm → 570 sq.cm Others: NA → NA Front wheel / axle: Option 1: SYY 6XXXYA → Option 1: SYY 6XXXYA Rear wheel / axle: APACHE 4059M → APACHE 4059M Others: NA → NA Whether asbestos or asbestos-free: Asbestos Free → Asbestos Free Front axle (Disc / Drum): DISC → DISC Effective Diameter (mm): 212 → 212 Rear axle (Disc / drum): DRUM → DRUM Effective Diameter (mm): 282 → 282 Other axle (		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Disc / Drum): NA → NA Effective diameter (mm): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material (if the braking surface is not ferrous): NA → NA Front wheel: -- → -- Rear wheel: -- → -- Others: -- → -- Make: M/s XXXX YYYY → M/s XXXX YYYY Type (only applicable for M2, M3, N and T category of vehicles.): N Category → N Category Inner diameter of the master cylinder (mm): 22.81 → 22.81 Operating stroke (mm): 41 → 41 Type of supply tank (only applicable for M2, M3, N and T category of vehicles.): → Brake Chamber: Rear → Rear Brake Chamber: Parking → Parking Make and type: NA → NA Make and type: NA → NA Size mm: NA → NA Size mm: NA → NA Inner diameter mm: NA → NA Inner diameter mm: NA → NA Stroke mm: NA → NA Stroke mm: NA → NA Make and Identification No.: MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/XXXX → MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/XXXX		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type of expansion (HL / HR): HL → HL Free Length of hoses: FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	L1 – 405, L2 – 360.9 → FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR : L1 – 405, L2 – 360.9 Thickness of lining (mm): 0.6 → 0.6 Nominal bore dia. (mm): 3.1 → 3.1 Type (Visual display/ audible/others): Visual display → Visual display Operation pressure (kg/cm2 / bar / kPa): NA → NA Type of safety device: NA → NA Make: M/s XXXXXYY → M/s XXXXXYY Type (mechanical/spring brake): MECHANICAL → MECHANICAL Acting on Transmission/wheel: WHEEL → WHEEL Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Lining / pad: Lining → Lining Name of producer: M/s XXXXYYY → M/s XXXXYYY Dimension (mm) (D x W x T): LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 → LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 Area (cm 2): 279 sq.cm → 279 sq.cm Material: XYZ 420 → XYZ 420	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diameter of brake drum/disc (mm): 272 → 272 Type: UNFAILED PART OF THE SERVICE BRAKE → UNFAILED PART OF THE SERVICE BRAKE Make: Ms. GULF Oil → Ms. GULF Oil Trade name: DOT 4 → DOT 4 Specification/ grade as		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per Indian standard: DOT 4 → DOT 4 Brake fluid level indicator (Yes/No): Yes → Yes Type (Manual/Auto) Front and Rear: Manual → Manual Lever length: → Pneumatic Brake Actuator: NA → NA Type (as per IS): NA → NA Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Brake Assist System (Fitted : Yes/No): No → No Category of Brake Assist System A / B: -- → -- For category A systems, define the force threshold at which the ratio between pedal force and brake pressure increases: -- → -- For category B systems, define the brake pedal speed which must be achieved in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Motor Vehicle stability function information document: -- → -- System: -- → -- System variants: -- → -- System options: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- System configurations (where appropriate): -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Schematic diagrams of the respective configurations installed on the motor vehicles defined in item 2.1. above with consideration given to the following:: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lift axles: -- → -- Steering axles: -- → -- Anti-lock braking configurations: -- → -- Wheelbase, track, centre of gravity height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → --		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Pneumatic circuits System schematics including anti-lock braking configurations associated with the trailer types defined in paragraph 6.2.1. of this annex.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Vehicle is/ is not equipped to		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tow a trailer with electric braking system (Yes/No): No → No Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm^3): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2. YYYY Pvt.Ltd Service brake Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others): Vacuum assisted hydraulic → Vacuum assisted		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	hydraulic Auto Slack Adjuster Fitted (Yes / No / Optional): Yes (Automatic wear adjustment) → Yes (Automatic wear adjustment) Anti-Lock braking system Provided(Yes/No/Optional): Optional → Optional Front (Disc / Drum): Disc → Disc Rear (Disc / Drum): Drum → Drum Total braking area (cm²): Front axle-254 sq.cm Rear axle-580 sq.cm → Front axle-254 sq.cm Rear axle-580 sq.cm Parking brake type: Mechanical → Mechanical Secondary brake type: Un failed part of the service brake → Un failed part of the service brake Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 3.26 kg/cm2 Wiper motor: XYZ Ltd. → XYZ Ltd. Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Engine Manufacturer::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	no-load speed)...(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter...(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) -		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66°	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method)::	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles –		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Make: XXYY System → XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model: PEAD64XYZS00 → PEAD64XYZS00 Schematic Diagram of Engine cooling system: - → - Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph A list, specifying the parts which constitute the headlamp cleaner and drawings thereof, (e.g. pumps, nozzles, valves, motors and wipers);: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor, Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	only in case of steering column locking systems: Locks the Steering → Locks the Steering Components placed in Engine compartment: - → - Frequency bands (Hz) · Max Output Power (W): Antenna Position at Vehicle, specific conditions for installation and/or use → Antenna Position at Vehicle, specific conditions for installation and/or use Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 96; applicable IS 11852 / IS 15986 route Parking braking performance	Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Axle spacing in case of multi axle vehicles.: NA → NA Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA → NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer - TC7719XX000 003XXXF → Refer - TC7719XX000	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L-756XXXX → CM/L-756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Final Drive (4x2 or 4x4 or 6x2 or 6x4 or others) Front / Rear / Both: (4 x 2) Rear → (4 x 2) Rear Rear axle make: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Rear axle ratio: 4.56 → 5.13 Differential type: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2.	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	YYYY – 18.2 :1 Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Parking brake: Hand operated [LHS] → Hand operated [LHS] Service Brake: Foot operated → Foot operated Service Brake: → Type of coupling device for brake: NA → NA Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing surfaces, “E” in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → -- Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	· Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Make: M/s Brakes India Private Limited → M/s Brakes India Private Limited Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others):		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vacuum assisted hydraulic → Vacuum assisted hydraulic Control system & braking wheel: Hydraulic acting on all wheels →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydraulic acting on all wheels Schematic layout indicating method of split of brake system, location of valves, reservoirs, ABS components etc. (Attach drawing and indicate the drawing number): AJ1XXX00 → AJ1XXX00 Anti-Lock braking system Provided (Yes / No / Optional): NA → NA If yes, details of ABS: - → - Make: - → - Category of ABS: - → - Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Modulator: - → - Nos. of Modulators: - → - Make of Modulators: - → - Identification No. / Part No. of Modulator: - → - Brief description and features: - → - Controller: - → - Nos. of Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Height of Center of Gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(mm): - → - Un laden condition: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Laden condition: - → - Slack adjuster (only applicable for M2, M3, N and T category of vehicles.): - → - Front (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Rear (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Front wheel: 132x47.3x10.35 → 132x47.3x10.35 Rear wheel: LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 → LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 Others: NA → NA Type of liner wear indicator (window/ acoustic/ optical/ any other): Window → Window Front: 250 sq.cm → 250 sq.cm Rear: 570 sq.cm → 570 sq.cm Others: NA → NA Front wheel / axle: Option 1: SYY 6XXXYA → Option 1: SYY 6XXXYA Rear wheel / axle: APACHE 4059M → APACHE 4059M Others: NA → NA Whether asbestos or asbestos-free: Asbestos Free → Asbestos Free Front axle (Disc / Drum): DISC → DISC Effective Diameter (mm): 212 → 212 Rear axle (Disc / drum): DRUM → DRUM Effective Diameter (mm): 282 → 282 Other axle (		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Disc / Drum): NA → NA Effective diameter (mm): NA → NA Material (if the braking surface is not ferrous): NA → NA Front wheel: -- → -- Rear wheel: -- → -- Others: -- → -- Make: M/s XXXX YYYY → M/s XXXX YYYY Type (only applicable for M2, M3, N and T category of vehicles.): N Category → N Category Inner diameter of the master cylinder (mm): 22.81 → 22.81 Operating stroke (mm): 41 → 41 Type of supply tank (only applicable for M2, M3, N and T category of vehicles.): → Brake Chamber: Rear → Rear Brake Chamber: Parking → Parking Make and type: NA → NA Make and type: NA → NA Size mm: NA → NA Size mm: NA → NA Inner diameter mm: NA → NA Inner diameter mm: NA → NA Stroke mm: NA → NA Stroke mm: NA → NA Make and Identification No.: MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/XXXX → MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/X	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	XXX Type of expansion (HL / HR): HL → HL Free Length of hoses: FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	L1 – 405, L2 – 360.9 → FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR : L1 – 405, L2 – 360.9 Thickness of lining (mm): 0.6 → 0.6 Nominal bore dia. (mm): 3.1 → 3.1 Type (Visual display/ audible/others): Visual display → Visual display Operation pressure (kg/cm2 / bar / kPa): NA → NA Type of safety device: NA → NA Make: M/s XXXXXYY → M/s XXXXXYY Type (mechanical/spring brake): MECHANICAL → MECHANICAL Acting on Transmission/wheel: WHEEL → WHEEL Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Lining / pad: Lining → Lining Name of producer: M/s XXXXYYY → M/s XXXXYYY Dimension (mm) (D x W x T): LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 → LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 Area (cm 2): 279 sq.cm → 279 sq.cm Material: XYZ 420 → XYZ 420	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diameter of brake drum/disc (mm): 272 → 272 Type: UNFAILED PART OF THE SERVICE BRAKE → UNFAILED PART OF THE SERVICE BRAKE Make: Ms. GULF Oil → Ms. GULF Oil Trade name: DOT 4 → DOT 4 Specification/ grade as		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per Indian standard: DOT 4 → DOT 4 Brake fluid level indicator (Yes/No): Yes → Yes Type (Manual/Auto) Front and Rear: Manual → Manual Lever length: → Pneumatic Brake Actuator: NA → NA Type (as per IS): NA → NA Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Brake Assist System (Fitted : Yes/No): No → No Category of Brake Assist System A / B: -- → -- For category A systems, define the force threshold at which the ratio between pedal force and brake pressure increases: -- → -- For category B systems, define the brake pedal speed which must be achieved in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Motor Vehicle stability function information document: -- → -- System: -- → -- System variants: -- → -- System options: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- System configurations (where appropriate): -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Schematic diagrams of the respective configurations installed on the motor vehicles defined in item 2.1. above with consideration given to the following:: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lift axles: -- → -- Steering axles: -- → -- Anti-lock braking configurations: -- → -- Wheelbase, track, centre of gravity height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → --		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Pneumatic circuits System schematics including anti-lock braking configurations associated with the trailer types defined in paragraph 6.2.1. of this annex.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Vehicle is/ is not equipped to		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tow a trailer with electric braking system (Yes/No): No → No Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm^3): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2. YYYY Pvt.Ltd Service brake Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others): Vacuum assisted hydraulic → Vacuum assisted		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	hydraulic Auto Slack Adjuster Fitted (Yes / No / Optional): Yes (Automatic wear adjustment) → Yes (Automatic wear adjustment) Anti-Lock braking system Provided(Yes/No/Optional): Optional → Optional Front (Disc / Drum): Disc → Disc Rear (Disc / Drum): Drum → Drum Total braking area (cm²): Front axle-254 sq.cm Rear axle-580 sq.cm → Front axle-254 sq.cm Rear axle-580 sq.cm Parking brake type: Mechanical → Mechanical Secondary brake type: Un failed part of the service brake → Un failed part of the service brake Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 3.26 kg/cm2 Wiper motor: XYZ Ltd. → XYZ Ltd. Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Engine Manufacturer::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:...(min-1): 3300 → 3300 Cut-off point without load (Maximum	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) -		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number:: NA → NA Make: NA → NA Number:……..: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): … (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66°	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method)::	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles –		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Make: XXYY System → XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model: PEAD64XYZS00 → PEAD64XYZS00 Schematic Diagram of Engine cooling system: - → - Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph A list, specifying the parts which constitute the headlamp cleaner and drawings thereof, (e.g. pumps, nozzles, valves, motors and wipers);: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor, Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	only in case of steering column locking systems: Locks the Steering → Locks the Steering Components placed in Engine compartment: - → - Frequency bands (Hz) · Max Output Power (W): Antenna Position at Vehicle, specific conditions for installation and/or use → Antenna Position at Vehicle, specific conditions for installation and/or use Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 96; applicable braking standard; N1 fitment / mandate review ABS / electronic braking applicability and performance	Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Axle spacing in case of multi axle vehicles.: NA → NA Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA	CEA / report needs review	Confirm ABS fitted / applicable. Unknown fitment / scope is retained for review.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA → NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer - TC7719XX000 003XXXF → Refer - TC7719XX000	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L-756XXXX → CM/L-756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Final Drive (4x2 or 4x4 or 6x2 or 6x4 or others) Front / Rear / Both: (4 x 2) Rear → (4 x 2) Rear Rear axle make: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Rear axle ratio: 4.56 → 5.13 Differential type: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2.	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	YYYY – 18.2 :1 Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Parking brake: Hand operated [LHS] → Hand operated [LHS] Service Brake: Foot operated → Foot operated Service Brake: → Type of coupling device for brake: NA → NA Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing surfaces, “E” in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → -- Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	· Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Make: M/s Brakes India Private Limited → M/s Brakes India Private Limited Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others):		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vacuum assisted hydraulic → Vacuum assisted hydraulic Control system & braking wheel: Hydraulic acting on all wheels →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydraulic acting on all wheels Schematic layout indicating method of split of brake system, location of valves, reservoirs, ABS components etc. (Attach drawing and indicate the drawing number): AJ1XXX00 → AJ1XXX00 Anti-Lock braking system Provided (Yes / No / Optional): NA → NA If yes, details of ABS: - → - Make: - → - Category of ABS: - → - Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Modulator: - → - Nos. of Modulators: - → - Make of Modulators: - → - Identification No. / Part No. of Modulator: - → - Brief description and features: - → - Controller: - → - Nos. of Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Height of Center of Gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(mm): - → - Un laden condition: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Laden condition: - → - Slack adjuster (only applicable for M2, M3, N and T category of vehicles.): - → - Front (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Rear (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Front wheel: 132x47.3x10.35 → 132x47.3x10.35 Rear wheel: LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 → LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 Others: NA → NA Type of liner wear indicator (window/ acoustic/ optical/ any other): Window → Window Front: 250 sq.cm → 250 sq.cm Rear: 570 sq.cm → 570 sq.cm Others: NA → NA Front wheel / axle: Option 1: SYY 6XXXYA → Option 1: SYY 6XXXYA Rear wheel / axle: APACHE 4059M → APACHE 4059M Others: NA → NA Whether asbestos or asbestos-free: Asbestos Free → Asbestos Free Front axle (Disc / Drum): DISC → DISC Effective Diameter (mm): 212 → 212 Rear axle (Disc / drum): DRUM → DRUM Effective Diameter (mm): 282 → 282 Other axle (		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Disc / Drum): NA → NA Effective diameter (mm): NA → NA Material (if the braking surface is not ferrous): NA → NA Front wheel: -- → -- Rear wheel: -- → -- Others: -- → -- Make: M/s XXXX YYYY → M/s XXXX YYYY Type (only applicable for M2, M3, N and T category of vehicles.): N Category → N Category Inner diameter of the master cylinder (mm): 22.81 → 22.81 Operating stroke (mm): 41 → 41 Type of supply tank (only applicable for M2, M3, N and T category of vehicles.): → Brake Chamber: Rear → Rear Brake Chamber: Parking → Parking Make and type: NA → NA Make and type: NA → NA Size mm: NA → NA Size mm: NA → NA Inner diameter mm: NA → NA Inner diameter mm: NA → NA Stroke mm: NA → NA Stroke mm: NA → NA Make and Identification No.: MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/XXXX → MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/X	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	XXX Type of expansion (HL / HR): HL → HL Free Length of hoses: FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	L1 – 405, L2 – 360.9 → FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR : L1 – 405, L2 – 360.9 Thickness of lining (mm): 0.6 → 0.6 Nominal bore dia. (mm): 3.1 → 3.1 Type (Visual display/ audible/others): Visual display → Visual display Operation pressure (kg/cm2 / bar / kPa): NA → NA Type of safety device: NA → NA Make: M/s XXXXXYY → M/s XXXXXYY Type (mechanical/spring brake): MECHANICAL → MECHANICAL Acting on Transmission/wheel: WHEEL → WHEEL Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Lining / pad: Lining → Lining Name of producer: M/s XXXXYYY → M/s XXXXYYY Dimension (mm) (D x W x T): LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 → LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 Area (cm 2): 279 sq.cm → 279 sq.cm Material: XYZ 420 → XYZ 420	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diameter of brake drum/disc (mm): 272 → 272 Type: UNFAILED PART OF THE SERVICE BRAKE → UNFAILED PART OF THE SERVICE BRAKE Make: Ms. GULF Oil → Ms. GULF Oil Trade name: DOT 4 → DOT 4 Specification/ grade as		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per Indian standard: DOT 4 → DOT 4 Brake fluid level indicator (Yes/No): Yes → Yes Type (Manual/Auto) Front and Rear: Manual → Manual Lever length: → Pneumatic Brake Actuator: NA → NA Type (as per IS): NA → NA Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Brake Assist System (Fitted : Yes/No): No → No Category of Brake Assist System A / B: -- → -- For category A systems, define the force threshold at which the ratio between pedal force and brake pressure increases: -- → -- For category B systems, define the brake pedal speed which must be achieved in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Motor Vehicle stability function information document: -- → -- System: -- → -- System variants: -- → -- System options: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- System configurations (where appropriate): -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Schematic diagrams of the respective configurations installed on the motor vehicles defined in item 2.1. above with consideration given to the following:: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lift axles: -- → -- Steering axles: -- → -- Anti-lock braking configurations: -- → -- Wheelbase, track, centre of gravity height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → --		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Pneumatic circuits System schematics including anti-lock braking configurations associated with the trailer types defined in paragraph 6.2.1. of this annex.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Vehicle is/ is not equipped to		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tow a trailer with electric braking system (Yes/No): No → No Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm^3): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2. YYYY Pvt.Ltd Service brake Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others): Vacuum assisted hydraulic → Vacuum assisted		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	hydraulic Auto Slack Adjuster Fitted (Yes / No / Optional): Yes (Automatic wear adjustment) → Yes (Automatic wear adjustment) Anti-Lock braking system Provided(Yes/No/Optional): Optional → Optional Front (Disc / Drum): Disc → Disc Rear (Disc / Drum): Drum → Drum Total braking area (cm²): Front axle-254 sq.cm Rear axle-580 sq.cm → Front axle-254 sq.cm Rear axle-580 sq.cm Parking brake type: Mechanical → Mechanical Secondary brake type: Un failed part of the service brake → Un failed part of the service brake Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 3.26 kg/cm2 Wiper motor: XYZ Ltd. → XYZ Ltd. Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Engine Manufacturer::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:...(min-1): 3300 → 3300 Cut-off point without load (Maximum	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) -		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66°	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method)::	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles –		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Make: XXYY System → XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model: PEAD64XYZS00 → PEAD64XYZS00 Schematic Diagram of Engine cooling system: - → - Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph A list, specifying the parts which constitute the headlamp cleaner and drawings thereof, (e.g. pumps, nozzles, valves, motors and wipers);: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor, Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	only in case of steering column locking systems: Locks the Steering → Locks the Steering Components placed in Engine compartment: - → - Frequency bands (Hz) · Max Output Power (W): Antenna Position at Vehicle, specific conditions for installation and/or use → Antenna Position at Vehicle, specific conditions for installation and/or use Example of Engine No: HJEZXYZ891 → HJEZXYZ891		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 117; IS 11827; current edition review Speedometer indication and installation	Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	– SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L-756XXXX → CM/L-756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube /		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 Speedometer:: 7.00 R15 LT → 7.00 R15 LT Speedometer:: 215/75 R15 LT → 215/75 R15 LT Type: Electronic →		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Electronic Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneridge → Minda stoneridge Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneri dge → Minda stoneri dge Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneridge → Minda stoneridge Make, and Identification No. / Part No./ Drawing No.: FL102XXX → FL102XXX Make, and Identification No. / Part No./ Drawing No.: FL103XXX → FL103XXX Make, and Identification No. / Part No./ Drawing No.: FL104XXX → FL104XXX Make, and Identification No. / Part No./ Drawing No.: FL103 XXX → FL103 XXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make, and Identification No. / Part No./ Drawing No.: FL102XXX → FL102XXX Make, and Identification No. / Part No./ Drawing No.: FL103XXX → FL103XXX Range: 0 -		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	130 km/h → 0 - 130 km/h Major marking: NA (Digital Indicator) → NA (Digital Indicator) Minor marking: NA (Digital Indicator) → NA (Digital Indicator) Speedometer ratio, Pulses / km RAR: 2440 → 2440 Speedometer ratio, Pulses / km RAR: 2424 → 2424 Speedometer ratio, Pulses / km RAR: 2590 → 2590 Ratio of Speedo drive: 0.60 → 0.60 Ratio of Speedo drive: 0.60 → 0.60 Ratio of Speedo drive: 0.64 → 0.64 Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Tyre pressure monitoring system: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
AIS-137 (Part 3); route / fuel review Type I tailpipe emissions	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E-		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	:1 Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio /	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Wheel type		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39 Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC Overall Length (mm): 4750 → 4750	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961 Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Declared (Rounded to 3 decimal places): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel → Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA Operating limits/settings(Specify the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type):		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump: Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/ Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa)		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum allowable:... (kPa): Minimum : 0 kPa	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s)::		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure at rated engine speed and at 100 per cent load (location of measurement): … (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:……..g: NA → NA Canister working capacity:……..g: NA → NA Canister bed volume:……..litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output		
(continued)	codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	operation of NOx control measures: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45 °C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: ...(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Minimum: ...(°C): - 5 → - 5 Maximum: ...(°C): 120 →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump → Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Schematic Diagram of Engine cooling system: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → --		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly → FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Steering Components placed in Engine compartment: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
AIS-137 (Part 3), applicable amendments Real-driving emissions / route applicability	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R →	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E-		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L-	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	:1 Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Installation drawing showing rear	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Wheel type		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Management of the engine or any other source(s) of motive power e.g.Torque	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Management, control of fuel supply etc.: -- → -- Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39 Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Wheels and tyres: Apollo → Apollo Wheels and tyres:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC Overall Length (mm): 4750 → 4750 Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel → Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: …VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type):		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(AC) Radiator core open area (cm^2): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 %	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump: Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa)		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s)::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification Number:: NA → NA Make: NA → NA Number:……: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure at rated engine speed and at 100 per cent load (location of measurement): … (kPa): 30.0 kpa (150	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawings):: Closed Loop Crankcase ventilation with external oil separator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	means of identification): XXX / DDMMYY / X	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:……..g: NA → NA Canister working capacity:……..g: NA → NA Canister bed volume:……..litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output		
(continued)	codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	operation of NOx control measures: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45 °C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: ...(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Minimum: ...(°C): - 5 → - 5 Maximum: ...(°C): 120 →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump → Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Schematic Diagram of Engine cooling system: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → --		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly → FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Steering Components placed in Engine compartment: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 120; IS 3028; current edition review Exterior / pass-by noise	Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Vehicle kerb weight kg: Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C → Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA → NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Maximum (Including driver) for completely built vehicles: D+2 → D+2 Maximum (Including driver) for completely built vehicles: → Make: APOLLO → APOLLO Make: CEAT → CEAT Make:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L-756XXXX → CM/L-756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E-		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev →		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Rear axle ratio: 4.56 → 5.13 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Weight of seats: Driver seat : 11.5 Kg & Co-driver seat : 11Kg → Driver seat : 11.5 Kg & Co-driver seat : 11Kg Seat Identification No. / Part No/Drawing Number.: ID Number/ Part Number/D rawing number Weight Description Make (Seat (kg) Manufactu rer/ Vehicle Manufactu rer): -- → -- Installation drawing showing rear extremity of vehicle, chassis rear		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	dimensions.: NA → NA Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	20.44 → 20.44 Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ -- Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39 Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Weights: Base → Base Weights: Variant 1 → Variant 1 Weights: Variant 2 → Variant 2 Weights: Variant 3 → Variant 3 Weights: Variant 4 → Variant 4 Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% / Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter...(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	4C1 & 4C3 →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::::	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump: Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:...	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification Number:: NA → NA Make: NA → NA Number:……..: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other):		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve :	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure -		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K →	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles –	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	connections compensation hoses, etc.):: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA Tyre pressure monitoring system: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: …(°C): 740 °C → 740 °C Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump → Pump Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	exhaust		
(continued)	outlet location, indicated in a Schematic dimensional drawing.): Refer Enclosure 4G → Refer Enclosure 4G Schematic Diagram of Engine cooling system: - → - Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly → FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering Components placed in Engine compartment: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 96; applicable IS 11852 / IS 15986 route Secondary / residual braking	Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Axle spacing in case of multi axle vehicles.: NA → NA Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	whic5h may be coupled: NA → NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Final Drive (4x2 or 4x4 or 6x2 or 6x4 or others) Front / Rear / Both: (4 x 2) Rear → (4 x 2) Rear Rear axle make: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Rear axle ratio: 4.56 → 5.13 Differential type: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2.		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	YYYY – 18.2 :1 Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Parking brake: Hand operated [LHS] → Hand operated [LHS] Service Brake: Foot operated → Foot operated Service Brake: → Type of coupling device for brake: NA → NA Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing surfaces, “E” in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → -- Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	· Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Make: M/s Brakes India Private Limited → M/s Brakes India Private Limited Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others):		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vacuum assisted hydraulic → Vacuum assisted hydraulic Control system & braking wheel: Hydraulic acting on all wheels →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydraulic acting on all wheels Schematic layout indicating method of split of brake system, location of valves, reservoirs, ABS components etc. (Attach drawing and indicate the drawing number): AJ1XXX00 → AJ1XXX00 Anti-Lock braking system Provided (Yes / No / Optional): NA → NA If yes, details of ABS: - → - Make: - → - Category of ABS: - → - Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Modulator: - → - Nos. of Modulators: - → - Make of Modulators: - → - Identification No. / Part No. of Modulator: - → - Brief description and features: - → - Controller: - → - Nos. of Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Height of Center of Gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(mm): - → - Un laden condition: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Laden condition: - → - Slack adjuster (only applicable for M2, M3, N and T category of vehicles.): - → - Front (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Rear (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Front wheel: 132x47.3x10.35 → 132x47.3x10.35 Rear wheel: LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 → LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 Others: NA → NA Type of liner wear indicator (window/ acoustic/ optical/ any other): Window → Window Front: 250 sq.cm → 250 sq.cm Rear: 570 sq.cm → 570 sq.cm Others: NA → NA Front wheel / axle: Option 1: SYY 6XXXYA → Option 1: SYY 6XXXYA Rear wheel / axle: APACHE 4059M → APACHE 4059M Others: NA → NA Whether asbestos or asbestos-free: Asbestos Free → Asbestos Free Front axle (Disc / Drum): DISC → DISC Effective Diameter (mm): 212 → 212 Rear axle (Disc / drum): DRUM → DRUM Effective Diameter (mm): 282 → 282 Other axle (		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Disc / Drum): NA → NA Effective diameter (mm): NA → NA Material (if the braking surface is not ferrous): NA → NA Front wheel: -- → -- Rear wheel: -- → -- Others: -- → -- Make: M/s XXXX YYYY → M/s XXXX YYYY Type (only applicable for M2, M3, N and T category of vehicles.): N Category → N Category Inner diameter of the master cylinder (mm): 22.81 → 22.81 Operating stroke (mm): 41 → 41 Type of supply tank (only applicable for M2, M3, N and T category of vehicles.): → Brake Chamber: Rear → Rear Brake Chamber: Parking → Parking Make and type: NA → NA Make and type: NA → NA Size mm: NA → NA Size mm: NA → NA Inner diameter mm: NA → NA Inner diameter mm: NA → NA Stroke mm: NA → NA Stroke mm: NA → NA Make and Identification No.: MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/XXXX → MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/X	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	XXX Type of expansion (HL / HR): HL → HL Free Length of hoses: FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	L1 – 405, L2 – 360.9 → FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR : L1 – 405, L2 – 360.9 Thickness of lining (mm): 0.6 → 0.6 Nominal bore dia. (mm): 3.1 → 3.1 Type (Visual display/ audible/others): Visual display → Visual display Operation pressure (kg/cm2 / bar / kPa): NA → NA Type of safety device: NA → NA Make: M/s XXXXXYY → M/s XXXXXYY Type (mechanical/spring brake): MECHANICAL → MECHANICAL Acting on Transmission/wheel: WHEEL → WHEEL Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Lining / pad: Lining → Lining Name of producer: M/s XXXXYYY → M/s XXXXYYY Dimension (mm) (D x W x T): LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 → LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 Area (cm 2): 279 sq.cm → 279 sq.cm Material: XYZ 420 → XYZ 420	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diameter of brake drum/disc (mm): 272 → 272 Type: UNFAILED PART OF THE SERVICE BRAKE → UNFAILED PART OF THE SERVICE BRAKE Make: Ms. GULF Oil → Ms. GULF Oil Trade name: DOT 4 → DOT 4 Specification/ grade as		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per Indian standard: DOT 4 → DOT 4 Brake fluid level indicator (Yes/No): Yes → Yes Type (Manual/Auto) Front and Rear: Manual → Manual Lever length: → Pneumatic Brake Actuator: NA → NA Type (as per IS): NA → NA Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Brake Assist System (Fitted : Yes/No): No → No Category of Brake Assist System A / B: -- → -- For category A systems, define the force threshold at which the ratio between pedal force and brake pressure increases: -- → -- For category B systems, define the brake pedal speed which must be achieved in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Motor Vehicle stability function information document: -- → -- System: -- → -- System variants: -- → -- System options: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- System configurations (where appropriate): -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Schematic diagrams of the respective configurations installed on the motor vehicles defined in item 2.1. above with consideration given to the following:: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lift axles: -- → -- Steering axles: -- → -- Anti-lock braking configurations: -- → -- Wheelbase, track, centre of gravity height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → --		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Pneumatic circuits System schematics including anti-lock braking configurations associated with the trailer types defined in paragraph 6.2.1. of this annex.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Vehicle is/ is not equipped to		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tow a trailer with electric braking system (Yes/No): No → No Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2. YYYY Pvt.Ltd Service brake Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others): Vacuum assisted hydraulic → Vacuum assisted		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	hydraulic Auto Slack Adjuster Fitted (Yes / No / Optional): Yes (Automatic wear adjustment) → Yes (Automatic wear adjustment) Anti-Lock braking system Provided(Yes/No/Optional): Optional → Optional Front (Disc / Drum): Disc → Disc Rear (Disc / Drum): Drum → Drum Total braking area (cm²): Front axle-254 sq.cm Rear axle-580 sq.cm → Front axle-254 sq.cm Rear axle-580 sq.cm Parking brake type: Mechanical → Mechanical Secondary brake type: Un failed part of the service brake → Un failed part of the service brake Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 3.26 kg/cm2 Wiper motor: XYZ Ltd. → XYZ Ltd. Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Engine Manufacturer::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:...(min-1): 3300 → 3300 Cut-off point without load (Maximum	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) -		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66°	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method)::	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles –		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Make: XXYY System → XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model: PEAD64XYZS00 → PEAD64XYZS00 Schematic Diagram of Engine cooling system: - → - Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph A list, specifying the parts which constitute the headlamp cleaner and drawings thereof, (e.g. pumps, nozzles, valves, motors and wipers);: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor, Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	only in case of steering column locking systems: Locks the Steering → Locks the Steering Components placed in Engine compartment: - → - Frequency bands (Hz) · Max Output Power (W): Antenna Position at Vehicle, specific conditions for installation and/or use → Antenna Position at Vehicle, specific conditions for installation and/or use Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 96; applicable IS 11852 / IS 15986 route Brake fade, recovery and laden/unladen performance	Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Axle spacing in case of multi axle vehicles.: NA → NA Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA → NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer - TC7719XX000 003XXXF → Refer - TC7719XX000	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L-756XXXX → CM/L-756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Final Drive (4x2 or 4x4 or 6x2 or 6x4 or others) Front / Rear / Both: (4 x 2) Rear → (4 x 2) Rear Rear axle make: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Rear axle ratio: 4.56 → 5.13 Differential type: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2.	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	YYYY – 18.2 :1 Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Parking brake: Hand operated [LHS] → Hand operated [LHS] Service Brake: Foot operated → Foot operated Service Brake: → Type of coupling device for brake: NA → NA Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing surfaces, “E” in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → -- Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	· Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Make: M/s Brakes India Private Limited → M/s Brakes India Private Limited Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others):		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vacuum assisted hydraulic → Vacuum assisted hydraulic Control system & braking wheel: Hydraulic acting on all wheels →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydraulic acting on all wheels Schematic layout indicating method of split of brake system, location of valves, reservoirs, ABS components etc. (Attach drawing and indicate the drawing number): AJ1XXX00 → AJ1XXX00 Anti-Lock braking system Provided (Yes / No / Optional): NA → NA If yes, details of ABS: - → - Make: - → - Category of ABS: - → - Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Modulator: - → - Nos. of Modulators: - → - Make of Modulators: - → - Identification No. / Part No. of Modulator: - → - Brief description and features: - → - Controller: - → - Nos. of Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Height of Center of Gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(mm): - → - Un laden condition: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Laden condition: - → - Slack adjuster (only applicable for M2, M3, N and T category of vehicles.): - → - Front (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Rear (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Front wheel: 132x47.3x10.35 → 132x47.3x10.35 Rear wheel: LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 → LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 Others: NA → NA Type of liner wear indicator (window/ acoustic/ optical/ any other): Window → Window Front: 250 sq.cm → 250 sq.cm Rear: 570 sq.cm → 570 sq.cm Others: NA → NA Front wheel / axle: Option 1: SYY 6XXXYA → Option 1: SYY 6XXXYA Rear wheel / axle: APACHE 4059M → APACHE 4059M Others: NA → NA Whether asbestos or asbestos-free: Asbestos Free → Asbestos Free Front axle (Disc / Drum): DISC → DISC Effective Diameter (mm): 212 → 212 Rear axle (Disc / drum): DRUM → DRUM Effective Diameter (mm): 282 → 282 Other axle (		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Disc / Drum): NA → NA Effective diameter (mm): NA → NA Material (if the braking surface is not ferrous): NA → NA Front wheel: -- → -- Rear wheel: -- → -- Others: -- → -- Make: M/s XXXX YYYY → M/s XXXX YYYY Type (only applicable for M2, M3, N and T category of vehicles.): N Category → N Category Inner diameter of the master cylinder (mm): 22.81 → 22.81 Operating stroke (mm): 41 → 41 Type of supply tank (only applicable for M2, M3, N and T category of vehicles.): → Brake Chamber: Rear → Rear Brake Chamber: Parking → Parking Make and type: NA → NA Make and type: NA → NA Size mm: NA → NA Size mm: NA → NA Inner diameter mm: NA → NA Inner diameter mm: NA → NA Stroke mm: NA → NA Stroke mm: NA → NA Make and Identification No.: MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/XXXX → MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/X	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	XXX Type of expansion (HL / HR): HL → HL Free Length of hoses: FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	L1 – 405, L2 – 360.9 → FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR : L1 – 405, L2 – 360.9 Thickness of lining (mm): 0.6 → 0.6 Nominal bore dia. (mm): 3.1 → 3.1 Type (Visual display/ audible/others): Visual display → Visual display Operation pressure (kg/cm2 / bar / kPa): NA → NA Type of safety device: NA → NA Make: M/s XXXXXYY → M/s XXXXXYY Type (mechanical/spring brake): MECHANICAL → MECHANICAL Acting on Transmission/wheel: WHEEL → WHEEL Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Lining / pad: Lining → Lining Name of producer: M/s XXXXYYY → M/s XXXXYYY Dimension (mm) (D x W x T): LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 → LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 Area (cm 2): 279 sq.cm → 279 sq.cm Material: XYZ 420 → XYZ 420	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diameter of brake drum/disc (mm): 272 → 272 Type: UNFAILED PART OF THE SERVICE BRAKE → UNFAILED PART OF THE SERVICE BRAKE Make: Ms. GULF Oil → Ms. GULF Oil Trade name: DOT 4 → DOT 4 Specification/ grade as		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per Indian standard: DOT 4 → DOT 4 Brake fluid level indicator (Yes/No): Yes → Yes Type (Manual/Auto) Front and Rear: Manual → Manual Lever length: → Pneumatic Brake Actuator: NA → NA Type (as per IS): NA → NA Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Brake Assist System (Fitted : Yes/No): No → No Category of Brake Assist System A / B: -- → -- For category A systems, define the force threshold at which the ratio between pedal force and brake pressure increases: -- → -- For category B systems, define the brake pedal speed which must be achieved in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Motor Vehicle stability function information document: -- → -- System: -- → -- System variants: -- → -- System options: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- System configurations (where appropriate): -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Schematic diagrams of the respective configurations installed on the motor vehicles defined in item 2.1. above with consideration given to the following:: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lift axles: -- → -- Steering axles: -- → -- Anti-lock braking configurations: -- → -- Wheelbase, track, centre of gravity height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → --		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Pneumatic circuits System schematics including anti-lock braking configurations associated with the trailer types defined in paragraph 6.2.1. of this annex.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Vehicle is/ is not equipped to		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tow a trailer with electric braking system (Yes/No): No → No Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2. YYYY Pvt.Ltd Service brake Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others): Vacuum assisted hydraulic → Vacuum assisted		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	hydraulic Auto Slack Adjuster Fitted (Yes / No / Optional): Yes (Automatic wear adjustment) → Yes (Automatic wear adjustment) Anti-Lock braking system Provided(Yes/No/Optional): Optional → Optional Front (Disc / Drum): Disc → Disc Rear (Disc / Drum): Drum → Drum Total braking area (cm²): Front axle-254 sq.cm Rear axle-580 sq.cm → Front axle-254 sq.cm Rear axle-580 sq.cm Parking brake type: Mechanical → Mechanical Secondary brake type: Un failed part of the service brake → Un failed part of the service brake Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 3.26 kg/cm2 Wiper motor: XYZ Ltd. → XYZ Ltd. Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Engine Manufacturer::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) -		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number:: NA → NA Make: NA → NA Number:……..: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): … (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66°	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method)::	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles –		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Make: XXYY System → XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model: PEAD64XYZS00 → PEAD64XYZS00 Schematic Diagram of Engine cooling system: - → - Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph A list, specifying the parts which constitute the headlamp cleaner and drawings thereof, (e.g. pumps, nozzles, valves, motors and wipers);: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor, Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	only in case of steering column locking systems: Locks the Steering → Locks the Steering Components placed in Engine compartment: - → - Frequency bands (Hz) · Max Output Power (W): Antenna Position at Vehicle, specific conditions for installation and/or use → Antenna Position at Vehicle, specific conditions for installation and/or use Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 96; applicable braking standard Brake circuit, warning, control and installation checks	Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Axle spacing in case of multi axle vehicles.: NA → NA Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA → NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer - TC7719XX000 003XXXF → Refer - TC7719XX000	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L-756XXXX → CM/L-756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Final Drive (4x2 or 4x4 or 6x2 or 6x4 or others) Front / Rear / Both: (4 x 2) Rear → (4 x 2) Rear Rear axle make: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Rear axle ratio: 4.56 → 5.13 Differential type: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2.	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	YYYY – 18.2 :1 Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Parking brake: Hand operated [LHS] → Hand operated [LHS] Service Brake: Foot operated → Foot operated Service Brake: → Type of coupling device for brake: NA → NA Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing surfaces, “E” in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → -- Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	· Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Make: M/s Brakes India Private Limited → M/s Brakes India Private Limited Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others):		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vacuum assisted hydraulic → Vacuum assisted hydraulic Control system & braking wheel: Hydraulic acting on all wheels →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydraulic acting on all wheels Schematic layout indicating method of split of brake system, location of valves, reservoirs, ABS components etc. (Attach drawing and indicate the drawing number): AJ1XXX00 → AJ1XXX00 Anti-Lock braking system Provided (Yes / No / Optional): NA → NA If yes, details of ABS: - → - Make: - → - Category of ABS: - → - Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Modulator: - → - Nos. of Modulators: - → - Make of Modulators: - → - Identification No. / Part No. of Modulator: - → - Brief description and features: - → - Controller: - → - Nos. of Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Height of Center of Gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(mm): - → - Un laden condition: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Laden condition: - → - Slack adjuster (only applicable for M2, M3, N and T category of vehicles.): - → - Front (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Rear (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Front wheel: 132x47.3x10.35 → 132x47.3x10.35 Rear wheel: LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 → LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 Others: NA → NA Type of liner wear indicator (window/ acoustic/ optical/ any other): Window → Window Front: 250 sq.cm → 250 sq.cm Rear: 570 sq.cm → 570 sq.cm Others: NA → NA Front wheel / axle: Option 1: SYY 6XXXYA → Option 1: SYY 6XXXYA Rear wheel / axle: APACHE 4059M → APACHE 4059M Others: NA → NA Whether asbestos or asbestos-free: Asbestos Free → Asbestos Free Front axle (Disc / Drum): DISC → DISC Effective Diameter (mm): 212 → 212 Rear axle (Disc / drum): DRUM → DRUM Effective Diameter (mm): 282 → 282 Other axle (		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Disc / Drum): NA → NA Effective diameter (mm): NA → NA Material (if the braking surface is not ferrous): NA → NA Front wheel: -- → -- Rear wheel: -- → -- Others: -- → -- Make: M/s XXXX YYYY → M/s XXXX YYYY Type (only applicable for M2, M3, N and T category of vehicles.): N Category → N Category Inner diameter of the master cylinder (mm): 22.81 → 22.81 Operating stroke (mm): 41 → 41 Type of supply tank (only applicable for M2, M3, N and T category of vehicles.): → Brake Chamber: Rear → Rear Brake Chamber: Parking → Parking Make and type: NA → NA Make and type: NA → NA Size mm: NA → NA Size mm: NA → NA Inner diameter mm: NA → NA Inner diameter mm: NA → NA Stroke mm: NA → NA Stroke mm: NA → NA Make and Identification No.: MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/XXXX → MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/X	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	XXX Type of expansion (HL / HR): HL → HL Free Length of hoses: FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	L1 – 405, L2 – 360.9 → FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR : L1 – 405, L2 – 360.9 Thickness of lining (mm): 0.6 → 0.6 Nominal bore dia. (mm): 3.1 → 3.1 Type (Visual display/ audible/others): Visual display → Visual display Operation pressure (kg/cm2 / bar / kPa): NA → NA Type of safety device: NA → NA Make: M/s XXXXXYY → M/s XXXXXYY Type (mechanical/spring brake): MECHANICAL → MECHANICAL Acting on Transmission/wheel: WHEEL → WHEEL Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Lining / pad: Lining → Lining Name of producer: M/s XXXXYYY → M/s XXXXYYY Dimension (mm) (D x W x T): LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 → LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 Area (cm 2): 279 sq.cm → 279 sq.cm Material: XYZ 420 → XYZ 420	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diameter of brake drum/disc (mm): 272 → 272 Type: UNFAILED PART OF THE SERVICE BRAKE → UNFAILED PART OF THE SERVICE BRAKE Make: Ms. GULF Oil → Ms. GULF Oil Trade name: DOT 4 → DOT 4 Specification/ grade as		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per Indian standard: DOT 4 → DOT 4 Brake fluid level indicator (Yes/No): Yes → Yes Type (Manual/Auto) Front and Rear: Manual → Manual Lever length: → Pneumatic Brake Actuator: NA → NA Type (as per IS): NA → NA Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Brake Assist System (Fitted : Yes/No): No → No Category of Brake Assist System A / B: -- → -- For category A systems, define the force threshold at which the ratio between pedal force and brake pressure increases: -- → -- For category B systems, define the brake pedal speed which must be achieved in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Motor Vehicle stability function information document: -- → -- System: -- → -- System variants: -- → -- System options: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- System configurations (where appropriate): -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Schematic diagrams of the respective configurations installed on the motor vehicles defined in item 2.1. above with consideration given to the following:: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lift axles: -- → -- Steering axles: -- → -- Anti-lock braking configurations: -- → -- Wheelbase, track, centre of gravity height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → --		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Pneumatic circuits System schematics including anti-lock braking configurations associated with the trailer types defined in paragraph 6.2.1. of this annex.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Vehicle is/ is not equipped to		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tow a trailer with electric braking system (Yes/No): No → No Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2. YYYY Pvt.Ltd Service brake Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others): Vacuum assisted hydraulic → Vacuum assisted		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	hydraulic Auto Slack Adjuster Fitted (Yes / No / Optional): Yes (Automatic wear adjustment) → Yes (Automatic wear adjustment) Anti-Lock braking system Provided(Yes/No/Optional): Optional → Optional Front (Disc / Drum): Disc → Disc Rear (Disc / Drum): Drum → Drum Total braking area (cm²): Front axle-254 sq.cm Rear axle-580 sq.cm → Front axle-254 sq.cm Rear axle-580 sq.cm Parking brake type: Mechanical → Mechanical Secondary brake type: Un failed part of the service brake → Un failed part of the service brake Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 3.26 kg/cm2 Wiper motor: XYZ Ltd. → XYZ Ltd. Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Engine Manufacturer::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:...(min-1): 3300 → 3300 Cut-off point without load (Maximum	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	no-load speed)...(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter...(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) -		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66°	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method)::	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles –		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Make: XXYY System → XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model: PEAD64XYZS00 → PEAD64XYZS00 Schematic Diagram of Engine cooling system: - → - Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph A list, specifying the parts which constitute the headlamp cleaner and drawings thereof, (e.g. pumps, nozzles, valves, motors and wipers);: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor, Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	only in case of steering column locking systems: Locks the Steering → Locks the Steering Components placed in Engine compartment: - → - Frequency bands (Hz) · Max Output Power (W): Antenna Position at Vehicle, specific conditions for installation and/or use → Antenna Position at Vehicle, specific conditions for installation and/or use Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 96; IS 11852:2019; GSR 834(E), 11 Nov 2025 2027 braking standard transition	Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Axle spacing in case of multi axle vehicles.: NA → NA Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA	CEA / report needs review	Confirm manufacture date and whether this is a new or existing model. Final notified route and edition still need confirmation.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA → NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Refer - TC7719XX000 003XXXF → Refer - TC7719XX000	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L-756XXXX → CM/L-756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Final Drive (4x2 or 4x4 or 6x2 or 6x4 or others) Front / Rear / Both: (4 x 2) Rear → (4 x 2) Rear Rear axle make: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Rear axle ratio: 4.56 → 5.13 Differential type: Single Reduction, Hypoid, Semi-floating Banjo type → Single Reduction, Hypoid, Semi-floating Banjo type Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2.	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	YYYY – 18.2 :1 Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Parking brake: Hand operated [LHS] → Hand operated [LHS] Service Brake: Foot operated → Foot operated Service Brake: → Type of coupling device for brake: NA → NA Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing surfaces, “E” in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → -- Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	· Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Make: M/s Brakes India Private Limited → M/s Brakes India Private Limited Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others):		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vacuum assisted hydraulic → Vacuum assisted hydraulic Control system & braking wheel: Hydraulic acting on all wheels →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydraulic acting on all wheels Schematic layout indicating method of split of brake system, location of valves, reservoirs, ABS components etc. (Attach drawing and indicate the drawing number): AJ1XXX00 → AJ1XXX00 Anti-Lock braking system Provided (Yes / No / Optional): NA → NA If yes, details of ABS: - → - Make: - → - Category of ABS: - → - Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Modulator: - → - Nos. of Modulators: - → - Make of Modulators: - → - Identification No. / Part No. of Modulator: - → - Brief description and features: - → - Controller: - → - Nos. of Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Height of Center of Gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(mm): - → - Un laden condition: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Laden condition: - → - Slack adjuster (only applicable for M2, M3, N and T category of vehicles.): - → - Front (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Rear (Automatic / Manual): - → - Make of slack adjuster: - → - Model and Type: - → - Front wheel: 132x47.3x10.35 → 132x47.3x10.35 Rear wheel: LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 → LEADING - 314x51.8x5.6 TRAILING – 246x51.8x5.6 Others: NA → NA Type of liner wear indicator (window/ acoustic/ optical/ any other): Window → Window Front: 250 sq.cm → 250 sq.cm Rear: 570 sq.cm → 570 sq.cm Others: NA → NA Front wheel / axle: Option 1: SYY 6XXXYA → Option 1: SYY 6XXXYA Rear wheel / axle: APACHE 4059M → APACHE 4059M Others: NA → NA Whether asbestos or asbestos-free: Asbestos Free → Asbestos Free Front axle (Disc / Drum): DISC → DISC Effective Diameter (mm): 212 → 212 Rear axle (Disc / drum): DRUM → DRUM Effective Diameter (mm): 282 → 282 Other axle (		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Disc / Drum): NA → NA Effective diameter (mm): NA → NA Material (if the braking surface is not ferrous): NA → NA Front wheel: -- → -- Rear wheel: -- → -- Others: -- → -- Make: M/s XXXX YYYY → M/s XXXX YYYY Type (only applicable for M2, M3, N and T category of vehicles.): N Category → N Category Inner diameter of the master cylinder (mm): 22.81 → 22.81 Operating stroke (mm): 41 → 41 Type of supply tank (only applicable for M2, M3, N and T category of vehicles.): → Brake Chamber: Rear → Rear Brake Chamber: Parking → Parking Make and type: NA → NA Make and type: NA → NA Size mm: NA → NA Size mm: NA → NA Inner diameter mm: NA → NA Inner diameter mm: NA → NA Stroke mm: NA → NA Stroke mm: NA → NA Make and Identification No.: MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/XXXX → MAKE: XXX GIRLING TAC No: AB2XX1. CoP No: SYL/29/2020-2021/300000YY60/ADDM/X	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	XXX Type of expansion (HL / HR): HL → HL Free Length of hoses: FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	L1 – 405, L2 – 360.9 → FOR NON ABS. FRONT : L1 – 470, L2 -417.7 REAR : L1 – 405, L2 – 360.9 Thickness of lining (mm): 0.6 → 0.6 Nominal bore dia. (mm): 3.1 → 3.1 Type (Visual display/ audible/others): Visual display → Visual display Operation pressure (kg/cm2 / bar / kPa): NA → NA Type of safety device: NA → NA Make: M/s XXXXXYY → M/s XXXXXYY Type (mechanical/spring brake): MECHANICAL → MECHANICAL Acting on Transmission/wheel: WHEEL → WHEEL Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Lining / pad: Lining → Lining Name of producer: M/s XXXXYYY → M/s XXXXYYY Dimension (mm) (D x W x T): LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 → LEADING - 305x48.8x5.6 TRAILING – 256x48.8x5.6 Area (cm 2): 279 sq.cm → 279 sq.cm Material: XYZ 420 → XYZ 420	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diameter of brake drum/disc (mm): 272 → 272 Type: UNFAILED PART OF THE SERVICE BRAKE → UNFAILED PART OF THE SERVICE BRAKE Make: Ms. GULF Oil → Ms. GULF Oil Trade name: DOT 4 → DOT 4 Specification/ grade as		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	per Indian standard: DOT 4 → DOT 4 Brake fluid level indicator (Yes/No): Yes → Yes Type (Manual/Auto) Front and Rear: Manual → Manual Lever length: → Pneumatic Brake Actuator: NA → NA Type (as per IS): NA → NA Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Brake Assist System (Fitted : Yes/No): No → No Category of Brake Assist System A / B: -- → -- For category A systems, define the force threshold at which the ratio between pedal force and brake pressure increases: -- → -- For category B systems, define the brake pedal speed which must be achieved in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Motor Vehicle stability function information document: -- → -- System: -- → -- System variants: -- → -- System options: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- System configurations (where appropriate): -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Schematic diagrams of the respective configurations installed on the motor vehicles defined in item 2.1. above with consideration given to the following:: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lift axles: -- → -- Steering axles: -- → -- Anti-lock braking configurations: -- → -- Wheelbase, track, centre of gravity height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → --		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Pneumatic circuits System schematics including anti-lock braking configurations associated with the trailer types defined in paragraph 6.2.1. of this annex.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Safety concept statement verifies that	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- Vehicle is/ is not equipped to		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tow a trailer with electric braking system (Yes/No): No → No Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of cylinders: 3 → 3 Displacement (cm^3): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2. YYYY Pvt.Ltd Service brake Type (Mechanical/hydraulic/air/air assisted/vacuum assisted/others): Vacuum assisted hydraulic → Vacuum assisted		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	hydraulic Auto Slack Adjuster Fitted (Yes / No / Optional): Yes (Automatic wear adjustment) → Yes (Automatic wear adjustment) Anti-Lock braking system Provided(Yes/No/Optional): Optional → Optional Front (Disc / Drum): Disc → Disc Rear (Disc / Drum): Drum → Drum Total braking area (cm²): Front axle-254 sq.cm Rear axle-580 sq.cm → Front axle-254 sq.cm Rear axle-580 sq.cm Parking brake type: Mechanical → Mechanical Secondary brake type: Un failed part of the service brake → Un failed part of the service brake Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	→ 3.26 kg/cm2 Wiper motor: XYZ Ltd. → XYZ Ltd. Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Maximum GVW kg (for rigid vehicles): Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B → Refer Annexure T7 – A Details for Recommended load body - Refer Annexure T7 - B Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Declared (Rounded to 3 decimal places): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Engine Manufacturer::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:...(min-1): 3300 → 3300 Cut-off point without load (Maximum	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) -		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66°	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method)::	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles –		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah (Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Make: XXYY System → XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model: PEAD64XYZS00 → PEAD64XYZS00 Schematic Diagram of Engine cooling system: - → - Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph A list, specifying the parts which constitute the headlamp cleaner and drawings thereof, (e.g. pumps, nozzles, valves, motors and wipers);: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor, Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	only in case of steering column locking systems: Locks the Steering → Locks the Steering Components placed in Engine compartment: - → - Frequency bands (Hz) · Max Output Power (W): Antenna Position at Vehicle, specific conditions for installation and/or use → Antenna Position at Vehicle, specific conditions for installation and/or use Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 118; AIS-018 Speed limiting function / device and tamper protection	Control (Forward / semi-forward / normal / others): Forward → Forward Vehicle Max Speed in unladen condition (km/h): Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Vehicle Max Speed in laden condition (km/h): Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade	CEA / report needs review	Confirm Transport vehicle. Unknown fitment / scope is retained for review.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/ XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L-756XXXX → CM/L-	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Control system: NA → NA Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Detailed drawing of the mounting arrangement of Steering control assembly showing vertical / tilt / actual angle.: FZY0XX00 → FZY0XX00 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	:1 Head lamp – upper / lower control: 0 – 3 Switch provided → 0 – 3 Switch provided Ignition cut-off: Not Provided → Not	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Provided Turn signal: Green Color - LH and RH Direction Symbol → Green Color - LH and RH Direction Symbol Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol High beam indicator: Blue Color – High Beam Symbol → Blue Color – High Beam Symbol Electrical charge indicator (Battery charge): Red Color – Battery Symbol → Red Color – Battery Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Front fog light, if provided: Not applicable. → Not applicable. Front fog light, if provided: → Rear fog light, if provided: Not applicable. → Not applicable. Horn: On steering Wheel → On steering Wheel ABS: Red Color – ABS function alert → Red Color –		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	ABS function alert Seat Belt Reminder: Not Applicable → Not Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Others (such as, Airbag, HVAC, Content gauge, LPG / CNG changeover switch etc.,): Not Applicable → Not Applicable Speedometer:: 7.00 R15 LT → 7.00 R15 LT Speedometer:: 215/75 R15 LT → 215/75 R15 LT Type: Electronic → Electronic Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneridge → Minda stoneridge Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneri dge → Minda stoneri dge Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneridge → Minda stoneridge Make, and Identification No. / Part No./ Drawing No.: FL102XXX → FL102XXX		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make, and Identification No. / Part No./ Drawing No.: FL103XXX → FL103XXX Make, and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification No. / Part No./ Drawing No.: FL104XXX → FL104XXX Make, and Identification No. / Part No./ Drawing No.: FL103 XXX → FL103 XXX Make, and Identification No. / Part No./ Drawing No.: FL102XXX → FL102XXX Make, and Identification No. / Part No./ Drawing No.: FL103XXX → FL103XXX Range: 0 - 130 km/h → 0 - 130 km/h Major marking: NA (Digital Indicator) → NA (Digital Indicator) Minor marking: NA (Digital Indicator) → NA (Digital Indicator) Speedometer ratio, Pulses / km RAR: 2440 → 2440 Speedometer ratio, Pulses / km RAR: 2424 → 2424 Speedometer ratio, Pulses / km RAR: 2590 → 2590 Ratio of Speedo drive: 0.60 → 0.60 Ratio of Speedo drive: 0.60 → 0.60 Ratio of Speedo drive: 0.64 → 0.64 Odometer: (only in case of CNG buses): NA → NA Make: -- → -- Type and Identification No. / Part No.: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen pressure) etc., distance between tyre & RUPD RH &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	LH, frame rear width, P1, P2 points.: - → - Controls - Specify method of operation, hand operated – left / right, foot operated – left / right Ref. SS12.1 or corresponding Indian Standard).: - → - Ignition: Hand operated [Rotary Switch RHS] → Hand operated [Rotary Switch RHS] Horn: Hand operated [Horn pad on steering wheel] → Hand operated [Horn pad on steering wheel] Lamps (Head lamp, Tail lamp, Parking lamp and Number plate lamp): Hand operated [Rotary switch – RHS] → Hand operated [Rotary switch – RHS] Turn signal: Hand operated [Swivel arm switch – RHS] → Hand operated [Swivel arm switch – RHS] Transmission shift lever: Hand operated [LHS] → Hand operated [LHS] Wind shield wiper: Hand operated [Swivel arm switch – LHS] → Hand operated [Swivel arm switch – LHS] High beam/low beam: Hand operated [Swivel arm switch – RHS] → Hand operated [Swivel arm switch – RHS]	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Parking brake: Hand operated [LHS] → Hand operated [LHS] Master switch for electrical: NA → NA Hazard warning signal: Hand		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	operated [Push button Switch -LHS] → Hand operated [Push button Switch -LHS] Service Brake: Foot operated → Foot operated Service Brake: → Accelerator Pedal (Floor hinged/hanging type): Foot operated – [Floor hanging Type] → Foot operated – [Floor hanging Type] Others: NA → NA Instrument Panel (Dash Board): → Drawing showing the mounting details, overall size and all control switches with dimensions: BWR0XX01 → BWR0XX01 Instrument Panel Variants with photographs (With / without Airbag, Music system, AC): NA → NA Material used for instrument Panel: PPTYY & PCXXS → PPTYY & PCXXS Instrument Panel mounting (With hardware details): BWRXXYY1 → BWRXXYY1 Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Arrangement of front controls (For M1	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	category as per AIS – 035): NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	surfaces, “E” in mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → -- Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- Drawing showing the parts and arrangement of the foot controls along with dimensions “E”, “H” and “J”: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Type of Speed Governor (SLD / SLF): SLF → SLF Speed Limiting Device (SLD) [For	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vehicles with max. speed above 80 kmph in		
(continued)	unladen condition as per AIS-018]: NA → NA Make: M/s. XXXYYY → M/s. XXXYYY Model: DXM 8.1AP → DXM 8.1AP Identification: TAC No. / BIS License No.: AL Development ID : F6NXYZW1 AL Production ID : F6NXYZW0 → AL Development ID : F6NXYZW1 AL Production ID : F6NXYZW0 SLD approved for speed/s (Annexure to be provided including information for every Vehicle Model / Variants indicating various SLD approved speeds contributed due to declared transmission options): -- → -- Information for Speed alert: NA → NA Details of audible alert (Alert above 80 km/h): -- → -- Details of audible alert Continuous /intermittent.(Alert above 120 km/h): -- → -- if intermittent, Interval between Alerts: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Control system & braking wheel: Hydraulic acting on all wheels → Hydraulic acting on all wheels Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Controller: - → - Nos. of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No For category B systems, define the brake pedal speed which must be achieved in order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Adequate documentation according to annex G (Complex electronics vehicle control systems) provided in respect of the following system(s): ..... Yes / No / Not applicable: -- → -- List of complex functions covered:: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods and tools identified.: -- → -- Information regarding design provisions in case of failure is provided including any error messages, warning signals, partial performance conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986 : 2015 and meets manufacturer’s specifications under non-fault conditions.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electronic Stability Control system as per AIS:133 (Fitted : Yes/No): NO → NO If yes, details of ESC: -- → -- ESC System Technical Documentation. To ensure a vehicle is equipped with an ESC system that meets the definition of "ESC System" in paragraph 2.0, the vehicle manufacturer shall make available to the test agency, upon request, the documentation specified below.: -- → -- System diagram identifying all ESC system hardware.: -- → -- A brief written explanation sufficient to describe the ESC system basic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hardware Id: -- → -- Brief description of failure warning tell–tale: -- → -- Control function (Directional / Roll / Directional and Roll): -- → -- Steering Angle Sensor: -- → -- Make and Country of manufacturer(if imported): -- → -- Identification No. / Part No.: -- → -- Brief description and features: -- → -- Yaw Rate Sensor: -- → -- Make and Country of manufacturer(if imported): -- → -- Identification No. / Part No.: -- → -- Brief description and features: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- Additional information (if applicable) to the application of the directional control and roll-over control functions, for example:: -- → -- Wheelbase, track, centre of gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Drive train options (e.g. retarder): -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Differential type/differential lock(s) (e.g. standard or self-locking, automatic or driver selected): -- → -- Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → -- Limitations on the location of the controller(s): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the system directly controls output variables.: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- Additional information (if applicable) to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the application of the directional control and/or the roll-over control function(s): -- → -- Sensors external to the controller (a) Function; (b) Limitations on the location of the sensors; (c) Identification, e.g. part numbers.: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional features.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mesh & Rev with sliding mesh → Manual gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39 Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rating: 215/75 R15 LT → 215/75 R15 LT Front: Q → Q Front: L → L Front: S → S Front: S → S Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Vehicle Max Speed in unladen condition (km/h): Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Vehicle Max Speed in laden condition (km/h): Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s.	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type Make and type of the control unit:: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ±		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:.....cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA NOx Sensor (Before /After of SCR):: No → No Make:: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH)		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	use by the rescue services [Note: cross-reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Electronic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number (CVN): NA → NA Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph Drawing showing type and controls of dipped beam leveling device with table showing positions of switch for various vehicle loading condition: VFTXYZ507 → VFTXYZ507 Tell-Tale for leveling switch: Yes → Yes Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA Name of the manufacturer of the system (Steering column shaft / Gear shift	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly → FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering VLT to Control Centre: -- → -- No. of speeds offered in blower (s): 4 → 4 No. of speeds offered in blower (s): 4 → 4 No. of speeds offered in blower (s): 4 → 4 Whether air vents have been provided for direction control, If yes, their locations: Yes (4) → Yes (4) Whether air vents have been provided for direction control, If yes, their locations: Yes (4) LH/CLH/CRH/RH → Yes (4) LH/CLH/CRH/RH Whether air vents have been provided	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	for direction control, If yes, their locations: Yes (4) → Yes (4) Whether air vents have been provided for direction control, If yes, their locations: LH/CLH/CRH/RH → LH/CLH/CRH/RH Whether air vents have been provided for direction control, If yes, their locations: LH/CLH/CRH/RH → LH/CLH/CRH/RH		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
AIS-071; applicable category scope Identification of controls, tell-tales and indicators	Control (Forward / semi-forward / normal / others): Forward → Forward Vehicle Max Speed in unladen condition (km/h): Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Vehicle Max Speed in laden condition (km/h): Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L-756XXXX → CM/L-756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L – 670007XXXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS:		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Control system: NA → NA Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Detailed drawing of the mounting arrangement of Steering control assembly showing vertical / tilt / actual angle.: FZY0XX00 → FZY0XX00 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	:1 Head lamp – upper / lower control: 0 – 3 Switch provided → 0 – 3 Switch provided Ignition cut-off: Not Provided → Not Provided Turn signal: Green Color - LH and RH Direction Symbol → Green Color - LH and	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	RH Direction Symbol Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol High beam indicator: Blue Color – High Beam Symbol → Blue Color – High Beam Symbol Electrical charge indicator (Battery charge): Red Color – Battery Symbol → Red Color – Battery Symbol Brake failure: Red Color – Parking and Oil symbol together → Red Color – Parking and Oil symbol together Brake failure: → Front fog light, if provided: Not applicable. → Not applicable. Front fog light, if provided: → Rear fog light, if provided: Not applicable. → Not applicable. Horn: On steering Wheel → On steering Wheel ABS: Red Color – ABS function alert → Red Color –		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	ABS function alert Seat Belt Reminder: Not Applicable → Not Applicable Others (such as, Airbag, HVAC, Content gauge, LPG / CNG changeover switch etc.,): Not Applicable → Not Applicable	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Speedometer:: 7.00 R15 LT → 7.00 R15 LT Speedometer:: 215/75 R15 LT → 215/75 R15 LT Type: Electronic → Electronic Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneridge → Minda stoneridge Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneri dge → Minda stoneri dge Make, and Identification No. / Part No./ Drawing No.: Pricol Limited → Pricol Limited Make, and Identification No. / Part No./ Drawing No.: Minda stoneridge → Minda stoneridge Make, and Identification No. / Part No./ Drawing No.: FL102XXX → FL102XXX Make, and Identification No. / Part No./ Drawing No.: FL103XXX → FL103XXX Make, and		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification No. / Part No./ Drawing No.: FL104XXX → FL104XXX Make, and Identification No. / Part No./ Drawing No.: FL103 XXX → FL103 XXX Make, and Identification No. / Part No./ Drawing No.: FL102XXX → FL102XXX Make, and Identification No. / Part No./ Drawing No.: FL103XXX → FL103XXX Range: 0 - 130 km/h → 0 - 130 km/h Major marking: NA (Digital Indicator) → NA (Digital Indicator) Minor marking: NA (Digital Indicator) → NA (Digital Indicator) Speedometer ratio, Pulses / km RAR: 2440 → 2440 Speedometer ratio, Pulses / km RAR: 2424 → 2424 Speedometer ratio, Pulses / km RAR: 2590 → 2590 Ratio of Speedo drive: 0.60 → 0.60 Ratio of Speedo drive: 0.60 → 0.60 Ratio of Speedo drive: 0.64 → 0.64 Odometer: (only in case of CNG buses): NA → NA Make: -- → -- Type and Identification No. / Part No.: -- → -- Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size, make, unladen	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	pressure) etc., distance between tyre & RUPD RH &		
(continued)	LH, frame rear width, P1, P2 points.: - → - Controls - Specify method of operation, hand operated – left / right, foot operated – left / right Ref. SS12.1 or corresponding Indian Standard).: - → - Ignition: Hand operated [Rotary Switch RHS] → Hand operated [Rotary Switch RHS] Horn: Hand operated [Horn pad on steering wheel] → Hand operated [Horn pad on steering wheel] Lamps (Head lamp, Tail lamp, Parking lamp and Number plate lamp): Hand operated [Rotary switch – RHS] → Hand operated [Rotary switch – RHS] Turn signal: Hand operated [Swivel arm switch – RHS] → Hand operated [Swivel arm switch – RHS] Transmission shift lever: Hand operated [LHS] → Hand operated [LHS] Wind shield wiper: Hand operated [Swivel arm switch – LHS] → Hand operated [Swivel arm switch – LHS] High beam/low beam: Hand operated [Swivel arm switch – RHS] → Hand operated [Swivel arm switch – RHS] Parking brake: Hand operated [LHS] → Hand operated [LHS] Master switch for electrical: NA → NA Hazard warning signal: Hand	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	operated [Push button Switch -LHS] → Hand operated [Push button Switch -LHS] Service Brake: Foot operated → Foot operated Service Brake: → Accelerator Pedal (Floor hinged/hanging type): Foot operated – [Floor hanging Type] → Foot operated – [Floor hanging Type] Others: NA → NA Instrument Panel (Dash Board): → Drawing showing the mounting details, overall size and all control switches with dimensions: BWR0XX01 → BWR0XX01 Instrument Panel Variants with photographs (With / without Airbag, Music system, AC): NA → NA Material used for instrument Panel: PPTYY & PCXXS → PPTYY & PCXXS Instrument Panel mounting (With hardware details): BWRXXYY1 → BWRXXYY1 Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Arrangement of front controls (For M1 category as per AIS – 035): NA → NA Distance between the contour points of the orthogonal projections on to plane “P” of the accelerator pedal and service brake pedal bearing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	surfaces, “E” in mm.: -- → -- Distance between the projection of the service brake pedal on to the reference plane “P”, to the right, “H” in mm.: -- → -- Distance between the projection of the service brake pedal on to the0 reference plane “P”, to the left, “J” in mm.: -- → -- Drawing showing the parts and arrangement of the foot controls along with dimensions “E”, “H” and “J”: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Type of Speed Governor (SLD / SLF): SLF → SLF Speed Limiting Device (SLD) [For vehicles with max. speed above 80 kmph in	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	unladen condition as per AIS-018]: NA → NA Make: M/s. XXXYYY → M/s. XXXYYY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Model: DXM 8.1AP → DXM 8.1AP Identification: TAC No. / BIS License No.: AL Development ID : F6NXYZW1 AL Production ID : F6NXYZW0 → AL Development ID : F6NXYZW1 AL Production ID : F6NXYZW0 SLD approved for speed/s (Annexure to be provided including information for every Vehicle Model / Variants indicating various SLD approved speeds contributed due to declared transmission options): -- → -- Information for Speed alert: NA → NA Details of audible alert (Alert above 80 km/h): -- → -- Details of audible alert Continuous /intermittent.(Alert above 120 km/h): -- → -- if intermittent, Interval between Alerts: -- → -- 1st · Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Control system & braking wheel: Hydraulic acting on all wheels → Hydraulic acting on all wheels Nos. of directly controlled wheel(s): - → - Brief description of failure warning tell–tale: - → - Wheel Speed Sensors: - → - No. of sensors: - → - Make of sensors: - → - Type of sensors: - → - Controller: - → - Nos. of		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Controller: - → - Make of Controller: - → - Identification No. / Part No. of Controller: - → - Brief description and features: - → - Control System & Braking wheel: Hand operated & Rear wheel → Hand operated & Rear wheel Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No For category B systems, define the brake pedal speed which must be achieved in order to activate the Brake Assist System (e.g. pedal stroke speed (mm/s) during a given time interval): -- → -- Adequate documentation according to annex G (Complex electronics vehicle control systems) provided in respect of the following system(s): ..... Yes / No / Not applicable: -- → -- List of complex functions covered:: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods and tools identified.: -- → -- Information regarding design provisions in case of failure is provided including any error messages, warning signals, partial performance conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986 : 2015 and meets manufacturer’s specifications under non-fault conditions.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electronic Stability Control system as per AIS:133 (Fitted : Yes/No): NO → NO If yes, details of ESC: -- → -- ESC System Technical Documentation. To ensure a vehicle is equipped with an ESC system that meets the definition of "ESC System" in paragraph 2.0, the vehicle manufacturer shall make available to the test agency, upon request, the documentation specified below.: -- → -- System diagram identifying all ESC system hardware.: -- → -- A brief written explanation sufficient to describe the ESC system basic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Hardware Id: -- → -- Brief description of failure warning tell–tale: -- → -- Control function (Directional / Roll / Directional and Roll): -- → -- Steering Angle Sensor: -- → -- Make and Country of manufacturer(if imported): -- → -- Identification No. / Part No.: -- → -- Brief description and features: -- → -- Yaw Rate Sensor: -- → -- Make and Country of manufacturer(if imported): -- → -- Identification No. / Part No.: -- → -- Brief description and features: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- Additional information (if applicable) to the application of the directional control and roll-over control functions, for example:: -- → -- Wheelbase, track, centre of gravity		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	height: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Drive train options (e.g. retarder): -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Differential type/differential lock(s) (e.g. standard or self-locking, automatic or driver selected): -- → -- Management of the engine or any other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → -- Limitations on the location of the controller(s): -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the system directly controls output variables.: -- → -- Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control: -- → -- Additional information (if applicable) to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the application of the directional control and/or the roll-over control function(s): -- → -- Sensors external to the controller (a) Function; (b) Limitations on the location of the sensors; (c) Identification, e.g. part numbers.: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional features.: -- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	mesh & Rev with sliding mesh → Manual gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39 Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rating: 215/75 R15 LT → 215/75 R15 LT Front: Q → Q Front: L → L Front: S → S Front: S → S Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Vehicle Max Speed in unladen condition (km/h): Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Vehicle Max Speed in laden condition (km/h): Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s.	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Continental and Thermistor Type → M/s. Continental and Thermistor Type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type Make and type of the control unit:: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal / plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ±		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:.....cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA NOx Sensor (Before /After of SCR):: No → No Make:: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems, description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter (DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number:		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH)		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID :		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	use by the rescue services [Note: cross-reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Electronic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2 Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number (CVN): NA → NA Maximum vehicle speed for continuous operation (km/h) (Only for AC horns): 80 Kmph → 80 Kmph Drawing showing type and controls of dipped beam leveling device with table showing positions of switch for various vehicle loading condition: VFTXYZ507 → VFTXYZ507 Tell-Tale for leveling switch: Yes → Yes Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA Name of the manufacturer of the system (Steering column shaft / Gear shift	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly → FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering VLT to Control Centre: -- → -- No. of speeds offered in blower (s): 4 → 4 No. of speeds offered in blower (s): 4 → 4 No. of speeds offered in blower (s): 4 → 4 Whether air vents have been provided for direction control, If yes, their locations: Yes (4) → Yes (4) Whether air vents have been provided for direction control, If yes, their locations: Yes (4) LH/CLH/CRH/RH → Yes (4) LH/CLH/CRH/RH Whether air vents have been provided	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	for direction control, If yes, their locations: Yes (4) → Yes (4) Whether air vents have been provided for direction control, If yes, their locations: LH/CLH/CRH/RH → LH/CLH/CRH/RH Whether air vents have been provided for direction control, If yes, their locations: LH/CLH/CRH/RH → LH/CLH/CRH/RH		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
AIS-137 Part 3; CMVR 115 Emission family, road-load and worst-case selection	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Max. Recommended Gradeability (Stand-start) – in degrees: 10 degs → 10 degs Max. Recommended Gradeability (Stand-start) – in degrees: 8.5 degs → 8.5 degs Vehicle kerb weight kg: Refer Annexure T7 - B	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Details for Recommended load body Refer Annexure T7 – C → Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Maximum (Including driver) for completely built vehicles: D+2 → D+2 Maximum (Including driver) for completely built vehicles: → Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. /		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube /		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Rear axle ratio: 4.56 → 5.13 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 Type and description (Leaf / Coil / Air / Semi- pneumatic/ Torsion bar): Leaf → Leaf Front: Parabolic spring- Rigid axle suspension → Parabolic spring- Rigid axle suspension Rear: Semi elliptic spring-		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rigid axle suspension → Semi elliptic spring- Rigid axle suspension	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Weight of seats: Driver seat : 11.5 Kg & Co-driver seat : 11Kg → Driver seat : 11.5 Kg & Co-driver seat : 11Kg Description of seats: -- → -- Make: -- → -- Number of seats fitted or capable of being fitted with head restraints, adjustable or not adjustable.: -- → -- Description of the adjustment, displacement and locking systems of the seat or of its parts and a description of occupant protection system against displacement of luggage.: -- → -- Description of seat anchorage: -- → -- Longitudinal / Vertical (If applicable) position of the seats during the tests.: -- → -- Drawings, diagrams and plans of the seats, their anchorages on the vehicle, the floor layout, the adjustment and displacement system of the seats and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	their parts, and their locking devices and of additional occupant protection system against displacement of luggage.: -- → -- In the case of seats fitted with head	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	restraints, the head restraint shall be shown on all drawings, diagrams and photographs.: -- → -- Seat Identification No. / Part No/Drawing Number.: ID Number/ Part Number/D rawing number Weight Description Make (Seat (kg) Manufactu rer/ Vehicle Manufactu rer): -- → -- Description: Make → Make Description: ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) → ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) Description: Weight (kg) → Weight (kg) Brief description and drawings of the		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	seat type, its attachment fittings and its adjustment, displacement and locking systems including the minimum distance between fitting points.: D+1 → D+1 Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed engineering drawing as specified in AIS – 013 or photographs	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Drawings and/or photographs of the locations of the statutory plates and inscriptions and of the vehicle		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	identification number: X4W0XYZ4 / VFT0XYZ2 → X4W0XYZ4 / VFT0XYZ2 Drawings and / or photographs of the official part of the plates and inscriptions (completed example with dimensions): X4WXYZ44 → X4WXYZ44 Location of Vehicle Descriptor Section (VDS) in Vehicle Identification Number (VIN): 4thto 9thCharacter of 17 Characters VIN → 4thto 9thCharacter of 17 Characters VIN Brief description of the Provision for emergency exit provided in case of sealed windows: -- → -- Dimensions of the instructions describing each motion necessary to unlatch and open the exit are provided from the release mechanism.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Brief description of the vehicle type as regards its structure, dimensions, lines and constituent materials:: -- → -- Description of the protective system installed in the vehicle:: -- → -- Description of the protective system installed in the vehicle:: → Description of interior arrangements or fittings that might affect the tests:: -- → -- A detailed description, including photographs and/or drawings, of the vehicle type with respect to the structure, the dimensions, the design and the constituent materials, the side walls of the passenger compartment (exterior and interior), including specific details of the protection system, where applicable:: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- Construction: Integrated / Attached with	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Mounts: 4 Mounts, Fixed CABIN → 4 Mounts, Fixed CABIN 1st ·		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	20.44 Description · Make: ID		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) → ID Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) Description · Make: Weight (kg) → Weight (kg) B 51.0 B49.0 · Pedestrian protection as per AIS-100: The distance, measured longitudinally on a horizontal plane, between the transverse centerline of the front axle & the R point of the driver’s seat is less than 1100 mm (Yes / No).: NA → NA Brief description of failure warning tell–tale: - → - Brief description and features: - → - Brief description and features: - → - Description: -- → -- Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	-- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	version as appropriate.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- A brief written explanation sufficient to describe the ESC system basic operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → -- Brief description of failure warning tell–tale: -- → -- Brief description and features: -- → -- Brief description and features: -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ -- Management of the engine or any		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Component description:: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification: -- → -- Limitations on the location of the controller(s): -- → -- Additional features: -- → -- Modulators: -- → -- General description and function: -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification (if applicable): -- → -- Limitations: -- → -- Electrical Equipment: -- → -- Circuit diagrams: -- → -- Powering methods: -- → -- Pneumatic circuits: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Safety aspects of the electronic system in accordance with Annex S of IS 11852: -- → -- List of complex functions covered::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	-- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	and tools identified.: -- → -- Information regarding		
(continued)	design provisions in case of failure is provided including any error messages, warning signals, partial performance conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986:2015 and meets manufacturer’s specifications under non-fault conditions.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electro-magnetic compatibility: -- → -- Documentation demonstrating compliance with AIS-004(Part 3).: -- → -- Component description: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	features.: -- → -- Modulators (a) General description and function; (b) Identification; (c) Limitations.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- (see annex 4 of this report for test data): -- → -- (see annex 4 of this report for test data): -- → -- Type / Description: Fully built vehicle, D+2, Non AC, 3.39 Ton GVW → Fully built vehicle, D+2, Non AC, 3.39 Ton GVW Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Type / Description: Re-circulating ball type → Re-circulating ball type Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Wiping system (Brief description): Twin blade, Tandem Pattern → Twin blade, Tandem Pattern Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC Overall Length (mm): 4750 → 4750 Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961 Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Weights: Base → Base Weights: Variant 1 → Variant 1	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Weights: Variant 2 → Variant 2 Weights: Variant 3 → Variant 3 Weights: Variant 4 → Variant 4 Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	spare wheel, tools, etc.) (kg): - → - Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% / Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kPa Length and internal diameter...(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s.	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm^2): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump: Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Distribution by ports: NA → NA Volume of crank-case cavity with piston	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/disable after parking: NA → NA Lowest		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	unit for NG fuelling: NA → NA Make(s):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: .....(kg): NA → NA Capacity: Ah		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	°C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: …(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42 Minimum: …(°C): - 5 → - 5 Maximum: …(°C): 120 → 120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA OBD IUPR Family if any: XX+XYY → XX+XYY exhaust outlet location, indicated in a Schematic dimensional drawing.): Refer Enclosure 4G → Refer Enclosure 4G 1.2.11.2.1.1 4.4 · 3. · Calibration Verification Number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Engine speed, engine load, A/F mode, catalyst temperature → Engine speed, engine load, A/F mode, catalyst temperature Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Two Type I cycles → Two Type I cycles Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Type I → Type I Type (Manual / power): Manual → Manual No. of wipers: Two → Two Make: XXYY System →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Wiper arm: Provided → Provided Length: 500mm → 500mm Make: M/s XYZ → M/s XYZ Length: 450 mm → 450 mm Make: BCS (SYNDICATE WIPER SYSTEM) → BCS (SYNDICATE WIPER SYSTEM) Identification: X5XY0258 → X5XY0258 Washer tank: BWXY4501 → BWXY4501 Type: Plastic → Plastic Make: XYZ → XYZ Identification No. / Part No.: X5BXYZ168 → X5BXYZ168 Capacity l: 1.5Ltr → 1.5Ltr Material: H.D.P.E → H.D.P.E Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ Model: PEAD64XYZS00 → PEAD64XYZS00 Nozzle(s): XYZ → XYZ No. Of Nozzles: 2 → 2 Defroster: NA → NA Type: - → - Make: - → - Details of Demisting and / or Defrosting system: - → - Schematic Diagram of Engine cooling system: - → - Schematic Diagram of Heating system: - →		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	- Cross sectional view of complete air handling system including position of louvers (vents) on dash board: - → - Blower identification: - → - Blower Voltage and capacity (Watts): - → - If defrost system is fitted, specify defrost test temperature (-8 +/- 2 deg C / -18 +/- 3 deg C): → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	& related dimensions as per related standards.: → Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA A brief technical description indicating the length of the cleaning period, the consumption of cleaning fluid during the cleaning period and the minimum capacity of the container provided;: -- → -- Description and drawings/photographs	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all subassemblies, which includes an electronic oscillator or switching frequency greater than 9 kHz (like ECU, steering, suspension, Body: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering SYSTEM SOFTWARE (If fitted): NA → NA Make: -- → -- Version: -- → -- Operating System Details with Version: -- → -- DESCRIPTION OF DEVICE (If fitted): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the VAS: - → - A detailed description of the alarm system and of the vehicle parts related to alarm system installed with Block Diagram: → A list of main components comprising the alarm system: - → - Components placed in Engine compartment: - → - Components placed in Passenger compartment: - → - Availability of radio/wireless transmission Yes/No: - → - Details of radio/wireless transmission if any: - → - frequency: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Availability of WPC approval for wireless technology Yes/No: - → - The measures taken against false alarm: - → - Range of protection offered by the device. Details about the detection mechanism (opening of Door, Bonnet, luggage compartment, intrusion etc.): - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details & specification of audible alarm: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details about the panic alarm , if provided: - → - Description of the Immobilizer: - → - A detailed description of the vehicle immobilizer and of the vehicle and the measures		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	taken against inadvertent activation with block diagram.: → The vehicle system(s) on which the immobilizer acts.: - → - Details of radio/wireless transmission if any: - → - Method of setting/ unsetting the device.: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - A list of main components comprising the device: - → - Description: AL Part No. → AL Part No. Description · AL Part No.: Hella Part No → Hella Part No ECU: FXX01000 → FXX01000 ECU · FXX01000: 015.XYZ-01 → 015.XYZ-01 a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
AIS-137 Part 3; CMVR 115 Mass-emission conformity of production	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Max. Recommended Gradeability (Stand-start) – in degrees: 10 degs → 10 degs Max. Recommended Gradeability (Stand-start) – in degrees: 8.5 degs → 8.5 degs Vehicle kerb weight kg: Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C → Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Annexure T7-B → Refer Annexure T7-B Maximum (Including driver) for completely built vehicles: D+2 → D+2 Maximum (Including driver) for completely built vehicles: → Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Rear axle ratio: 4.56 → 5.13 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type and description (Leaf / Coil / Air / Semi- pneumatic/ Torsion bar): Leaf → Leaf Front: Parabolic spring- Rigid axle suspension → Parabolic spring- Rigid axle suspension Rear: Semi elliptic spring-		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rigid axle suspension → Semi elliptic spring- Rigid axle suspension Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Weight of seats: Driver seat : 11.5 Kg & Co-driver seat : 11Kg → Driver seat : 11.5 Kg & Co-driver seat : 11Kg Description of seats: -- → -- Make: -- → -- Number of seats fitted or capable of being fitted with head restraints, adjustable or not adjustable.: -- → -- Description of the adjustment, displacement and locking systems of the seat or of its parts and a description of occupant protection system against displacement of luggage.: -- → -- Description of seat anchorage: -- → -- Longitudinal / Vertical (If applicable) position of the seats during the tests.: --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ -- Drawings, diagrams and plans of the seats, their anchorages on the vehicle, the floor layout, the adjustment and displacement system of the seats and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	their parts, and their locking devices and of additional occupant protection system against displacement of luggage.: -- → -- In the case of seats fitted with head restraints, the head restraint shall be shown on all drawings, diagrams and photographs.: -- → -- Seat Identification No. / Part No/Drawing Number.: ID Number/ Part Number/D rawing number Weight Description Make (Seat (kg) Manufactu rer/ Vehicle Manufactu rer): -- → -- Description: Make → Make Description: ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) → ID Number/ Part Number/D	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) Description: Weight (kg) → Weight (kg) Brief description and drawings of the seat type, its attachment fittings and its adjustment, displacement and locking systems including the minimum distance between fitting points.: D+1 → D+1 Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Drawings and/or photographs of the locations of the statutory plates and inscriptions and of the vehicle		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	identification number: X4W0XYZ4 / VFT0XYZ2 → X4W0XYZ4 / VFT0XYZ2 Drawings and / or photographs of the official part of the plates and inscriptions (completed example with dimensions): X4WXYZ44 → X4WXYZ44 Location of Vehicle Descriptor Section (VDS) in Vehicle Identification Number (VIN): 4thto 9thCharacter of 17 Characters VIN → 4thto 9thCharacter of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	17 Characters VIN Brief description of the Provision for emergency exit provided in case of sealed windows: -- → -- Dimensions of the instructions describing each motion necessary to unlatch and open the exit are provided from the release mechanism.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Brief description of the vehicle type as regards its structure, dimensions, lines and constituent materials:: -- → -- Description of the protective system installed in the vehicle:: -- → -- Description of the protective system installed in the vehicle:: → Description of interior arrangements or fittings that might affect the tests:: -- → -- A detailed description, including	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	photographs and/or drawings, of the vehicle type with respect to the structure, the dimensions, the design and the constituent materials, the side walls of the passenger compartment (exterior and interior), including specific details of the protection system, where applicable:: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- Construction: Integrated / Attached with Mounts: 4 Mounts, Fixed CABIN → 4 Mounts, Fixed CABIN 1st ·		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Description · Make: ID		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) → ID Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) Description · Make: Weight (kg) → Weight (kg) B 51.0 B49.0 · Pedestrian protection as per AIS-100: The distance, measured longitudinally on a horizontal plane, between the transverse centerline of the front axle & the R point of the driver’s seat is less than 1100 mm (Yes / No).: NA → NA Brief description of failure warning	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tell–tale: - → - Brief description and features: - → - Brief description and features: - → - Description: -- → -- Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	version as appropriate.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- A brief written explanation sufficient to describe the ESC system basic operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Brief description of failure warning tell–tale: -- → -- Brief description and features: -- → -- Brief description and features: -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Management of the engine or any		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Component description:: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Software identification: -- → -- Limitations on the location of the controller(s): -- → -- Additional features: -- → -- Modulators: -- → -- General description and function: -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification (if applicable): -- → -- Limitations: -- → -- Electrical Equipment: -- → -- Circuit diagrams: -- → -- Powering methods: -- → -- Pneumatic circuits: -- → -- Safety aspects of the electronic system in accordance with Annex S of IS 11852: -- → -- List of complex functions covered::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	-- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods and tools identified.: -- → -- Information regarding		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	design provisions in case of failure is provided including any error messages, warning signals, partial performance conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986:2015 and meets manufacturer’s specifications under non-fault conditions.: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electro-magnetic compatibility: -- → -- Documentation demonstrating compliance with AIS-004(Part 3).: -- → -- Component description: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	features.: -- → -- Modulators (a) General description and function; (b) Identification; (c) Limitations.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- (see annex 4 of this report for test data): -- → -- (see annex 4 of this report for test data): -- → -- Type / Description: Fully built vehicle, D+2, Non AC, 3.39 Ton GVW → Fully	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	built vehicle, D+2, Non AC, 3.39 Ton GVW Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Type / Description: Re-circulating ball type → Re-circulating ball type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Wiping system (Brief description): Twin blade, Tandem Pattern → Twin blade, Tandem Pattern Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Overall Length (mm): 4750 → 4750 Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961 Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Weights: Base → Base Weights: Variant 1 → Variant 1 Weights: Variant 2 → Variant 2 Weights: Variant 3 → Variant 3 Weights: Variant 4 → Variant 4 Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	spare wheel, tools, etc.) (kg): - → - Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel → Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% / Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) &		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	No: 9 → 9 Material of blades (metal /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type(s) & Identification Number:: NA → NA Make: NA → NA Number:……: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): … (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:.....cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing (Periodic / Continuous): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	°C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: ...(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42 Minimum: ...(°C): - 5 → - 5 Maximum: ...(°C): 120 → 120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump → Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA OBD IUPR Family if any: XX+XYY → XX+XYY exhaust outlet location, indicated in a Schematic dimensional drawing.): Refer Enclosure 4G → Refer Enclosure 4G 1.2.11.2.1.1 4.4 · 3. · Calibration Verification Number:: NA → NA Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Engine speed, engine load, A/F mode, catalyst temperature → Engine speed, engine load, A/F mode, catalyst temperature Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Two Type I cycles → Two Type I cycles Catalyst · P0420 · Oxygen sensor 1 and	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Type I → Type I Type (Manual / power): Manual → Manual No. of wipers: Two → Two Make: XXYY System →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Wiper arm: Provided → Provided Length: 500mm → 500mm Make: M/s XYZ → M/s XYZ Length: 450 mm → 450 mm Make: BCS (SYNDICATE WIPER SYSTEM) → BCS (SYNDICATE WIPER SYSTEM) Identification: X5XY0258 → X5XY0258 Washer tank: BWXY4501 → BWXY4501 Type: Plastic → Plastic Make: XYZ → XYZ Identification No. / Part No.: X5BXYZ168 → X5BXYZ168 Capacity l: 1.5Ltr → 1.5Ltr Material: H.D.P.E → H.D.P.E Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make: XYZ → XYZ Model: PEAD64XYZS00 → PEAD64XYZS00 Nozzle(s): XYZ → XYZ No. Of Nozzles: 2 → 2 Defroster: NA → NA Type: - → - Make: - → - Details of Demisting and / or Defrosting system: - → - Schematic Diagram of Engine cooling system: - → - Schematic Diagram of Heating system: - →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	- Cross sectional view of complete air handling system including position of louvers (vents) on dash board: - → - Blower identification: - → - Blower Voltage and capacity (Watts): - → - If defrost system is fitted, specify defrost test temperature (-8 +/- 2 deg C / -18 +/- 3 deg C): → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar		
(continued)	& related dimensions as per related standards.: → Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA A brief technical description indicating the length of the cleaning period, the consumption of cleaning fluid during the cleaning period and the minimum capacity of the container provided;: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all subassemblies, which includes an electronic oscillator or switching frequency greater than 9 kHz (like ECU, steering, suspension, Body: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly → FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering SYSTEM SOFTWARE (If fitted): NA → NA Make: -- → -- Version: -- → -- Operating System Details with Version: -- → -- DESCRIPTION OF DEVICE (If fitted): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the VAS: - → - A detailed description of the alarm system and of the vehicle parts related to alarm system installed with Block Diagram: → A list of main components comprising the alarm system: - → - Components placed in Engine compartment: - → - Components placed in Passenger compartment: - → - Availability of radio/wireless transmission Yes/No: - → - Details of radio/wireless transmission if any: - → - frequency: - → - Availability of WPC approval for wireless technology Yes/No: - → - The measures taken against false alarm: - → - Range of protection offered by the device. Details about the detection mechanism (opening of Door, Bonnet, luggage compartment, intrusion etc.): - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details about the panic alarm , if provided: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the Immobilizer: - → - A detailed description of the vehicle immobilizer and of the vehicle and the measures		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	taken against inadvertent activation with block diagram.: → The vehicle system(s) on which the immobilizer acts.: - → - Details of radio/wireless transmission if any: - → - Method of setting/ unsetting the device.: - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - A list of main components comprising the device: - → - Description: AL Part No. → AL Part No. Description · AL Part No.: Hella Part No → Hella Part No ECU: FXX01000 → FXX01000 ECU · FXX01000: 015.XYZ-01 → 015.XYZ-01 a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - a. · Make: - → - b. · Type: - → - c. · Serial Number: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	d. · Name and Address of manufacture: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 115; AIS-137 applicable in-service provisions In-service conformity / PUC applicability	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Max. Recommended Gradeability (Stand-start) – in degrees: 10 degs → 10 degs Max. Recommended Gradeability (Stand-start) – in degrees: 8.5 degs → 8.5 degs Vehicle kerb weight kg: Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C → Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Maximum (Including driver) for completely built vehicles: D+2 → D+2 Maximum (Including driver) for completely built vehicles: → Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. /		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Rear axle ratio: 4.56 → 5.13 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 Type and description (Leaf / Coil / Air / Semi- pneumatic/ Torsion bar): Leaf → Leaf Front: Parabolic spring- Rigid axle suspension → Parabolic spring- Rigid axle suspension Rear: Semi elliptic spring-		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rigid axle suspension → Semi elliptic spring- Rigid axle suspension Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color –	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Weight of seats: Driver seat : 11.5 Kg & Co-driver seat : 11Kg → Driver seat : 11.5 Kg & Co-driver seat : 11Kg Description of seats: -- → -- Make: -- → -- Number of seats fitted or capable of being fitted with head restraints, adjustable or not adjustable.: -- → -- Description of the adjustment, displacement and locking systems of the seat or of its parts and a description of occupant protection system against displacement of luggage.: -- → -- Description of seat anchorage: -- → -- Longitudinal / Vertical (If applicable) position of the seats during the tests.: -- → -- Drawings, diagrams and plans of the seats, their anchorages on the vehicle, the floor layout, the adjustment and displacement system of the seats and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	their parts, and their locking devices and of additional occupant protection system against displacement of luggage.: -- → -- In the case of seats fitted with head restraints, the head restraint shall be shown on all drawings, diagrams and photographs.: -- → -- Seat Identification No. / Part No/Drawing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number.: ID Number/ Part Number/D rawing number Weight Description Make (Seat (kg) Manufactu rer/ Vehicle Manufactu rer): -- → -- Description: Make → Make Description: ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) → ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) Description: Weight (kg) → Weight (kg) Brief description and drawings of the seat type, its attachment fittings and its adjustment, displacement and locking systems including the minimum distance between fitting points.: D+1 → D+1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Drawings and/or photographs of the locations of the statutory plates and inscriptions and of the vehicle		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	identification number: X4W0XYZ4 / VFT0XYZ2 → X4W0XYZ4 / VFT0XYZ2 Drawings and / or photographs of the official part of the plates and inscriptions (completed example with dimensions): X4WXYZ44 → X4WXYZ44 Location of Vehicle Descriptor Section (VDS) in Vehicle Identification Number (VIN): 4thto 9thCharacter of 17 Characters VIN → 4thto 9thCharacter of 17 Characters VIN Brief description of the Provision for emergency exit provided in case of sealed windows: -- → -- Dimensions of the instructions describing each motion necessary to unlatch and open the exit are provided from the release mechanism.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	distance between front extremity & FUPD, width of FUPD, ID		
(continued)	marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Brief description of the vehicle type as regards its structure, dimensions, lines and constituent materials:: -- → -- Description of the protective system installed in the vehicle:: -- → -- Description of the protective system installed in the vehicle:: → Description of interior arrangements or fittings that might affect the tests:: -- → -- A detailed description, including photographs and/or drawings, of the vehicle type with respect to the structure, the dimensions, the design and the constituent materials, the side walls of the passenger compartment (exterior and interior), including specific details of the protection system, where applicable:: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- Construction: Integrated / Attached with Mounts: 4 Mounts, Fixed CABIN → 4 Mounts, Fixed CABIN 1st ·	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Description · Make: ID	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) → ID Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) Description · Make: Weight (kg) → Weight (kg) B 51.0 B49.0 · Pedestrian protection as per AIS-100: The distance, measured longitudinally on a horizontal plane, between the transverse centerline of the front axle & the R point of the driver’s seat is less than 1100 mm (Yes / No).: NA → NA Brief description of failure warning tell–tale: - → - Brief description and features: - → - Brief description and features: - → - Description: -- → -- Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	version as appropriate.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- A brief written explanation sufficient to describe the ESC system basic operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → -- Brief description of failure warning tell–tale: -- → -- Brief description and features: -- → -- Brief description and features: -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Management of the engine or any	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	other source(s) of motive power e.g.Torque Management, control of fuel	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	supply etc.: -- → -- Component description:: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification: -- → -- Limitations on the location of the controller(s): -- → -- Additional features: -- → -- Modulators: -- → -- General description and function: -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification (if applicable): -- → -- Limitations: -- → -- Electrical Equipment: -- → -- Circuit diagrams: -- → -- Powering methods: -- → -- Pneumatic circuits: -- → -- Safety aspects of the electronic system in accordance with Annex S of IS 11852: -- → -- List of complex functions covered::		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	-- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods and tools identified.: -- → -- Information regarding	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	design provisions in case of failure is provided including any error messages, warning signals, partial performance	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986:2015 and meets manufacturer’s specifications under non-fault conditions.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electro-magnetic compatibility: -- → -- Documentation demonstrating compliance with AIS-004(Part 3).: -- → -- Component description: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	features.: -- → -- Modulators (a) General description and function; (b) Identification; (c) Limitations.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- (see annex 4 of this report for test data): -- → -- (see annex 4 of this report for test data): -- → -- Type / Description: Fully built vehicle, D+2, Non AC, 3.39 Ton GVW → Fully built vehicle, D+2, Non AC, 3.39 Ton GVW Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Reverse: 4.39 → 4.39		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Type / Description: Re-circulating ball type → Re-circulating ball type Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Wiping system (Brief description): Twin blade, Tandem Pattern → Twin blade, Tandem Pattern	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC Overall Length (mm): 4750 → 4750 Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961 Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Weights: Base → Base Weights: Variant 1 → Variant 1 Weights: Variant 2 → Variant 2 Weights: Variant 3 → Variant 3 Weights: Variant 4 → Variant 4 Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	spare wheel, tools, etc.) (kg): - → - Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel → Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% / Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump: Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/ Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/disable after parking: NA → NA Lowest		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	unit for NG fuelling: NA → NA Make(s):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: .....(kg): NA → NA Capacity: Ah		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	°C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: …(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42 Minimum: …(°C): - 5 → - 5 Maximum: …(°C): 120 → 120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA OBD IUPR Family if any: XX+XYY → XX+XYY exhaust outlet location, indicated in a Schematic dimensional drawing.): Refer Enclosure 4G → Refer Enclosure 4G 1.2.11.2.1.1 4.4 · 3. · Calibration Verification Number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Engine speed, engine load, A/F mode, catalyst temperature → Engine speed, engine load, A/F mode, catalyst temperature Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Two Type I cycles → Two Type I cycles Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Type I → Type I Type (Manual / power): Manual → Manual No. of wipers: Two → Two Make: XXYY System →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Wiper arm: Provided → Provided Length: 500mm → 500mm Make: M/s XYZ → M/s XYZ Length: 450 mm → 450 mm Make: BCS (SYNDICATE WIPER SYSTEM) → BCS (SYNDICATE WIPER SYSTEM) Identification: X5XY0258 → X5XY0258 Washer tank: BWXY4501 → BWXY4501 Type: Plastic → Plastic Make: XYZ → XYZ Identification No. / Part No.: X5BXYZ168 → X5BXYZ168 Capacity l: 1.5Ltr → 1.5Ltr Material: H.D.P.E → H.D.P.E Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ Model: PEAD64XYZS00 → PEAD64XYZS00 Nozzle(s): XYZ → XYZ No. Of Nozzles: 2 → 2 Defroster: NA → NA Type: - → - Make: - → - Details of Demisting and / or Defrosting system: - → - Schematic Diagram of Engine cooling system: - → - Schematic Diagram of Heating system: - →		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	- Cross sectional view of complete air handling system including position of louvers (vents) on dash board: - → - Blower identification: - → - Blower Voltage and capacity (Watts): - → - If defrost system is fitted, specify defrost test temperature (-8 +/- 2 deg C / -18 +/- 3 deg C): → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	& related dimensions as per related standards.: → Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA A brief technical description indicating the length of the cleaning period, the consumption of cleaning fluid during the cleaning period and the minimum capacity of the container provided;: -- → -- Description and drawings/photographs	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all subassemblies, which includes an electronic oscillator or switching frequency greater than 9 kHz (like ECU, steering, suspension, Body: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering SYSTEM SOFTWARE (If fitted): NA → NA Make: -- → -- Version: -- → -- Operating System Details with Version: -- → -- DESCRIPTION OF DEVICE (If fitted): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the VAS: - → - A detailed description of the alarm system and of the vehicle parts related to alarm system installed with Block Diagram: → A list of main components comprising the alarm system: - → - Components placed in Engine compartment: - → - Components placed in Passenger compartment: - → - Availability of radio/wireless transmission Yes/No: - → - Details of radio/wireless transmission if any: - → - frequency: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Availability of WPC approval for wireless technology Yes/No: - → - The measures taken against false alarm: - → - Range of protection offered by the device. Details about the detection mechanism (opening of Door, Bonnet, luggage compartment, intrusion etc.): - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details & specification of audible alarm: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details about the panic alarm , if provided: - → - Description of the Immobilizer: - → - A detailed description of the vehicle immobilizer and of the vehicle and the measures		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	taken against inadvertent activation with block diagram.: → The vehicle system(s) on which the immobilizer acts.: - → - Details of radio/wireless transmission if any: - → - Method of setting/ unsetting the device.: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - A list of main components comprising the device: - → - Description: AL Part No. → AL Part No. Description · AL Part No.: Hella Part No → Hella Part No ECU: FXX01000 → FXX01000 ECU · FXX01000: 015.XYZ-01 → 015.XYZ-01 a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
CMVR 115; AIS-137 applicable diesel provisions Diesel free-acceleration smoke / opacity route	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Max. Recommended Gradeability (Stand-start) – in degrees: 10 degs → 10 degs Max. Recommended Gradeability (Stand-start) – in degrees: 8.5 degs → 8.5 degs Vehicle kerb weight kg: Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C → Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Annexure T7-B → Refer Annexure T7-B Maximum (Including driver) for completely built vehicles: D+2 → D+2 Maximum (Including driver) for completely built vehicles: → Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Rear axle ratio: 4.56 → 5.13 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type and description (Leaf / Coil / Air / Semi- pneumatic/ Torsion bar): Leaf → Leaf Front: Parabolic spring- Rigid axle suspension → Parabolic spring- Rigid axle suspension Rear: Semi elliptic spring-		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rigid axle suspension → Semi elliptic spring- Rigid axle suspension Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Weight of seats: Driver seat : 11.5 Kg & Co-driver seat : 11Kg → Driver seat : 11.5 Kg & Co-driver seat : 11Kg Description of seats: -- → -- Make: -- → -- Number of seats fitted or capable of being fitted with head restraints, adjustable or not adjustable.: -- → -- Description of the adjustment, displacement and locking systems of the seat or of its parts and a description of occupant protection system against displacement of luggage.: -- → -- Description of seat anchorage: -- → -- Longitudinal / Vertical (If applicable) position of the seats during the tests.: --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ -- Drawings, diagrams and plans of the seats, their anchorages on the vehicle, the floor layout, the adjustment and displacement system of the seats and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	their parts, and their locking devices and of additional occupant protection system against displacement of luggage.: -- → -- In the case of seats fitted with head restraints, the head restraint shall be shown on all drawings, diagrams and photographs.: -- → -- Seat Identification No. / Part No/Drawing Number.: ID Number/ Part Number/D rawing number Weight Description Make (Seat (kg) Manufactu rer/ Vehicle Manufactu rer): -- → -- Description: Make → Make Description: ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) → ID Number/ Part Number/D	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) Description: Weight (kg) → Weight (kg) Brief description and drawings of the seat type, its attachment fittings and its adjustment, displacement and locking systems including the minimum distance between fitting points.: D+1 → D+1 Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Drawings and/or photographs of the locations of the statutory plates and inscriptions and of the vehicle		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	identification number: X4W0XYZ4 / VFT0XYZ2 → X4W0XYZ4 / VFT0XYZ2 Drawings and / or photographs of the official part of the plates and inscriptions (completed example with dimensions): X4WXYZ44 → X4WXYZ44 Location of Vehicle Descriptor Section (VDS) in Vehicle Identification Number (VIN): 4thto 9thCharacter of 17 Characters VIN → 4thto 9thCharacter of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	17 Characters VIN Brief description of the Provision for emergency exit provided in case of sealed windows: -- → -- Dimensions of the instructions describing each motion necessary to unlatch and open the exit are provided from the release mechanism.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Brief description of the vehicle type as regards its structure, dimensions, lines and constituent materials:: -- → -- Description of the protective system installed in the vehicle:: -- → -- Description of the protective system installed in the vehicle:: → Description of interior arrangements or fittings that might affect the tests:: -- → -- A detailed description, including	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	photographs and/or drawings, of the vehicle type with respect to the structure, the dimensions, the design and the constituent materials, the side walls of the passenger compartment (exterior and interior), including specific details of the protection system, where applicable:: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- Construction: Integrated / Attached with Mounts: 4 Mounts, Fixed CABIN → 4 Mounts, Fixed CABIN 1st ·		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Description · Make: ID		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) → ID Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) Description · Make: Weight (kg) → Weight (kg) B 51.0 B49.0 · Pedestrian protection as per AIS-100: The distance, measured longitudinally on a horizontal plane, between the transverse centerline of the front axle & the R point of the driver’s seat is less than 1100 mm (Yes / No).: NA → NA Brief description of failure warning	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tell–tale: - → - Brief description and features: - → - Brief description and features: - → - Description: -- → -- Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	version as appropriate.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- A brief written explanation sufficient to describe the ESC system basic operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Brief description of failure warning tell–tale: -- → -- Brief description and features: -- → -- Brief description and features: -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Management of the engine or any		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Component description:: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Software identification: -- → -- Limitations on the location of the controller(s): -- → -- Additional features: -- → -- Modulators: -- → -- General description and function: -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification (if applicable): -- → -- Limitations: -- → -- Electrical Equipment: -- → -- Circuit diagrams: -- → -- Powering methods: -- → -- Pneumatic circuits: -- → -- Safety aspects of the electronic system in accordance with Annex S of IS 11852: -- → -- List of complex functions covered::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	-- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods and tools identified.: -- → -- Information regarding		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	design provisions in case of failure is provided including any error messages, warning signals, partial performance conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986:2015 and meets manufacturer’s specifications under non-fault conditions.: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electro-magnetic compatibility: -- → -- Documentation demonstrating compliance with AIS-004(Part 3).: -- → -- Component description: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	features.: -- → -- Modulators (a) General description and function; (b) Identification; (c) Limitations.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- (see annex 4 of this report for test data): -- → -- (see annex 4 of this report for test data): -- → -- Type / Description: Fully built vehicle, D+2, Non AC, 3.39 Ton GVW → Fully	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	built vehicle, D+2, Non AC, 3.39 Ton GVW Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Type / Description: Re-circulating ball type → Re-circulating ball type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Wiping system (Brief description): Twin blade, Tandem Pattern → Twin blade, Tandem Pattern Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Overall Length (mm): 4750 → 4750 Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961 Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Weights: Base → Base Weights: Variant 1 → Variant 1 Weights: Variant 2 → Variant 2 Weights: Variant 3 → Variant 3 Weights: Variant 4 → Variant 4 Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	spare wheel, tools, etc.) (kg): - → - Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel → Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% / Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) &		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	No: 9 → 9 Material of blades (metal /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type(s) & Identification Number:: NA → NA Make: NA → NA Number:……: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): … (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:.....cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing (Periodic / Continuous): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	°C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: ...(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42 Minimum: ...(°C): - 5 → - 5 Maximum: ...(°C): 120 → 120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump → Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA OBD IUPR Family if any: XX+XYY → XX+XYY exhaust outlet location, indicated in a Schematic dimensional drawing.): Refer Enclosure 4G → Refer Enclosure 4G 1.2.11.2.1.1 4.4 · 3. · Calibration Verification Number:: NA → NA Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Engine speed, engine load, A/F mode, catalyst temperature → Engine speed, engine load, A/F mode, catalyst temperature Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Two Type I cycles → Two Type I cycles Catalyst · P0420 · Oxygen sensor 1 and	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Type I → Type I Type (Manual / power): Manual → Manual No. of wipers: Two → Two Make: XXYY System →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Wiper arm: Provided → Provided Length: 500mm → 500mm Make: M/s XYZ → M/s XYZ Length: 450 mm → 450 mm Make: BCS (SYNDICATE WIPER SYSTEM) → BCS (SYNDICATE WIPER SYSTEM) Identification: X5XY0258 → X5XY0258 Washer tank: BWXY4501 → BWXY4501 Type: Plastic → Plastic Make: XYZ → XYZ Identification No. / Part No.: X5BXYZ168 → X5BXYZ168 Capacity l: 1.5Ltr → 1.5Ltr Material: H.D.P.E → H.D.P.E Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make: XYZ → XYZ Model: PEAD64XYZS00 → PEAD64XYZS00 Nozzle(s): XYZ → XYZ No. Of Nozzles: 2 → 2 Defroster: NA → NA Type: - → - Make: - → - Details of Demisting and / or Defrosting system: - → - Schematic Diagram of Engine cooling system: - → - Schematic Diagram of Heating system: - →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	- Cross sectional view of complete air handling system including position of louvers (vents) on dash board: - → - Blower identification: - → - Blower Voltage and capacity (Watts): - → - If defrost system is fitted, specify defrost test temperature (-8 +/- 2 deg C / -18 +/- 3 deg C): → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar		
(continued)	& related dimensions as per related standards.: → Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA A brief technical description indicating the length of the cleaning period, the consumption of cleaning fluid during the cleaning period and the minimum capacity of the container provided;: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all subassemblies, which includes an electronic oscillator or switching frequency greater than 9 kHz (like ECU, steering, suspension, Body: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly → FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering SYSTEM SOFTWARE (If fitted): NA → NA Make: -- → -- Version: -- → -- Operating System Details with Version: -- → -- DESCRIPTION OF DEVICE (If fitted): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the VAS: - → - A detailed description of the alarm system and of the vehicle parts related to alarm system installed with Block Diagram: → A list of main components comprising the alarm system: - → - Components placed in Engine compartment: - → - Components placed in Passenger compartment: - → - Availability of radio/wireless transmission Yes/No: - → - Details of radio/wireless transmission if any: - → - frequency: - → - Availability of WPC approval for wireless technology Yes/No: - → - The measures taken against false alarm: - → - Range of protection offered by the device. Details about the detection mechanism (opening of Door, Bonnet, luggage compartment, intrusion etc.): - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details about the panic alarm , if provided: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the Immobilizer: - → - A detailed description of the vehicle immobilizer and of the vehicle and the measures		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	taken against inadvertent activation with block diagram.: → The vehicle system(s) on which the immobilizer acts.: - → - Details of radio/wireless transmission if any: - → - Method of setting/ unsetting the device.: - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - A list of main components comprising the device: - → - Description: AL Part No. → AL Part No. Description · AL Part No.: Hella Part No → Hella Part No ECU: FXX01000 → FXX01000 ECU · FXX01000: 015.XYZ-01 → 015.XYZ-01 a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - a. · Make: - → - b. · Type: - → - c. · Serial Number: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	d. · Name and Address of manufacture: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
Applicable AIS-137 / notified consumption measurement procedure Fuel consumption and CO2 declaration	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Max. Recommended Gradeability (Stand-start) – in degrees: 10 degs → 10 degs Max. Recommended Gradeability (Stand-start) – in degrees: 8.5 degs → 8.5 degs Vehicle kerb weight kg: Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C → Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Maximum (Including driver) for completely built vehicles: D+2 → D+2 Maximum (Including driver) for completely built vehicles: → Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. /		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Rear axle ratio: 4.56 → 5.13 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 Type and description (Leaf / Coil / Air / Semi- pneumatic/ Torsion bar): Leaf → Leaf Front: Parabolic spring- Rigid axle suspension → Parabolic spring- Rigid axle suspension Rear: Semi elliptic spring-		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rigid axle suspension → Semi elliptic spring- Rigid axle suspension Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color –	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Weight of seats: Driver seat : 11.5 Kg & Co-driver seat : 11Kg → Driver seat : 11.5 Kg & Co-driver seat : 11Kg Description of seats: -- → -- Make: -- → -- Number of seats fitted or capable of being fitted with head restraints, adjustable or not adjustable.: -- → -- Description of the adjustment, displacement and locking systems of the seat or of its parts and a description of occupant protection system against displacement of luggage.: -- → -- Description of seat anchorage: -- → -- Longitudinal / Vertical (If applicable) position of the seats during the tests.: -- → -- Drawings, diagrams and plans of the seats, their anchorages on the vehicle, the floor layout, the adjustment and displacement system of the seats and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	their parts, and their locking devices and of additional occupant protection system against displacement of luggage.: -- → -- In the case of seats fitted with head restraints, the head restraint shall be shown on all drawings, diagrams and photographs.: -- → -- Seat Identification No. / Part No/Drawing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number.: ID Number/ Part Number/D rawing number Weight Description Make (Seat (kg) Manufactu rer/ Vehicle Manufactu rer): -- → -- Description: Make → Make Description: ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) → ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) Description: Weight (kg) → Weight (kg) Brief description and drawings of the seat type, its attachment fittings and its adjustment, displacement and locking systems including the minimum distance between fitting points.: D+1 → D+1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Drawings and/or photographs of the locations of the statutory plates and inscriptions and of the vehicle		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	identification number: X4W0XYZ4 / VFT0XYZ2 → X4W0XYZ4 / VFT0XYZ2 Drawings and / or photographs of the official part of the plates and inscriptions (completed example with dimensions): X4WXYZ44 → X4WXYZ44 Location of Vehicle Descriptor Section (VDS) in Vehicle Identification Number (VIN): 4thto 9thCharacter of 17 Characters VIN → 4thto 9thCharacter of 17 Characters VIN Brief description of the Provision for emergency exit provided in case of sealed windows: -- → -- Dimensions of the instructions describing each motion necessary to unlatch and open the exit are provided from the release mechanism.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	distance between front extremity & FUPD, width of FUPD, ID		
(continued)	marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Brief description of the vehicle type as regards its structure, dimensions, lines and constituent materials:: -- → -- Description of the protective system installed in the vehicle:: -- → -- Description of the protective system installed in the vehicle:: → Description of interior arrangements or fittings that might affect the tests:: -- → -- A detailed description, including photographs and/or drawings, of the vehicle type with respect to the structure, the dimensions, the design and the constituent materials, the side walls of the passenger compartment (exterior and interior), including specific details of the protection system, where applicable:: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- Construction: Integrated / Attached with Mounts: 4 Mounts, Fixed CABIN → 4 Mounts, Fixed CABIN 1st ·	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Description · Make: ID	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) → ID Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) Description · Make: Weight (kg) → Weight (kg) B 51.0 B49.0 · Pedestrian protection as per AIS-100: The distance, measured longitudinally on a horizontal plane, between the transverse centerline of the front axle & the R point of the driver’s seat is less than 1100 mm (Yes / No).: NA → NA Brief description of failure warning tell–tale: - → - Brief description and features: - → - Brief description and features: - → - Description: -- → -- Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	version as appropriate.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- A brief written explanation sufficient to describe the ESC system basic operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → -- Brief description of failure warning tell–tale: -- → -- Brief description and features: -- → -- Brief description and features: -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Management of the engine or any	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	other source(s) of motive power e.g.Torque Management, control of fuel	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	supply etc.: -- → -- Component description:: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification: -- → -- Limitations on the location of the controller(s): -- → -- Additional features: -- → -- Modulators: -- → -- General description and function: -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification (if applicable): -- → -- Limitations: -- → -- Electrical Equipment: -- → -- Circuit diagrams: -- → -- Powering methods: -- → -- Pneumatic circuits: -- → -- Safety aspects of the electronic system in accordance with Annex S of IS 11852: -- → -- List of complex functions covered::		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	-- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods and tools identified.: -- → -- Information regarding	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	design provisions in case of failure is provided including any error messages, warning signals, partial performance	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986:2015 and meets manufacturer’s specifications under non-fault conditions.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electro-magnetic compatibility: -- → -- Documentation demonstrating compliance with AIS-004(Part 3).: -- → -- Component description: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	features.: -- → -- Modulators (a) General description and function; (b) Identification; (c) Limitations.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- (see annex 4 of this report for test data): -- → -- (see annex 4 of this report for test data): -- → -- Type / Description: Fully built vehicle, D+2, Non AC, 3.39 Ton GVW → Fully built vehicle, D+2, Non AC, 3.39 Ton GVW Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Reverse: 4.39 → 4.39		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Type / Description: Re-circulating ball type → Re-circulating ball type Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Wiping system (Brief description): Twin blade, Tandem Pattern → Twin blade, Tandem Pattern	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC Overall Length (mm): 4750 → 4750 Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961 Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Weights: Base → Base Weights: Variant 1 → Variant 1 Weights: Variant 2 → Variant 2 Weights: Variant 3 → Variant 3 Weights: Variant 4 → Variant 4 Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	spare wheel, tools, etc.) (kg): - → - Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel → Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% / Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump: Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/ Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/disable after parking: NA → NA Lowest		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	unit for NG fuelling: NA → NA Make(s):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: .....(kg): NA → NA Capacity: Ah		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	°C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: …(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42 Minimum: …(°C): - 5 → - 5 Maximum: …(°C): 120 → 120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA OBD IUPR Family if any: XX+XYY → XX+XYY exhaust outlet location, indicated in a Schematic dimensional drawing.): Refer Enclosure 4G → Refer Enclosure 4G 1.2.11.2.1.1 4.4 · 3. · Calibration Verification Number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Engine speed, engine load, A/F mode, catalyst temperature → Engine speed, engine load, A/F mode, catalyst temperature Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Two Type I cycles → Two Type I cycles Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Type I → Type I Type (Manual / power): Manual → Manual No. of wipers: Two → Two Make: XXYY System →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Wiper arm: Provided → Provided Length: 500mm → 500mm Make: M/s XYZ → M/s XYZ Length: 450 mm → 450 mm Make: BCS (SYNDICATE WIPER SYSTEM) → BCS (SYNDICATE WIPER SYSTEM) Identification: X5XY0258 → X5XY0258 Washer tank: BWXY4501 → BWXY4501 Type: Plastic → Plastic Make: XYZ → XYZ Identification No. / Part No.: X5BXYZ168 → X5BXYZ168 Capacity l: 1.5Ltr → 1.5Ltr Material: H.D.P.E → H.D.P.E Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ Model: PEAD64XYZS00 → PEAD64XYZS00 Nozzle(s): XYZ → XYZ No. Of Nozzles: 2 → 2 Defroster: NA → NA Type: - → - Make: - → - Details of Demisting and / or Defrosting system: - → - Schematic Diagram of Engine cooling system: - → - Schematic Diagram of Heating system: - →		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	- Cross sectional view of complete air handling system including position of louvers (vents) on dash board: - → - Blower identification: - → - Blower Voltage and capacity (Watts): - → - If defrost system is fitted, specify defrost test temperature (-8 +/- 2 deg C / -18 +/- 3 deg C): → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	& related dimensions as per related standards.: → Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA A brief technical description indicating the length of the cleaning period, the consumption of cleaning fluid during the cleaning period and the minimum capacity of the container provided;: -- → -- Description and drawings/photographs	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all subassemblies, which includes an electronic oscillator or switching frequency greater than 9 kHz (like ECU, steering, suspension, Body: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering SYSTEM SOFTWARE (If fitted): NA → NA Make: -- → -- Version: -- → -- Operating System Details with Version: -- → -- DESCRIPTION OF DEVICE (If fitted): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the VAS: - → - A detailed description of the alarm system and of the vehicle parts related to alarm system installed with Block Diagram: → A list of main components comprising the alarm system: - → - Components placed in Engine compartment: - → - Components placed in Passenger compartment: - → - Availability of radio/wireless transmission Yes/No: - → - Details of radio/wireless transmission if any: - → - frequency: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Availability of WPC approval for wireless technology Yes/No: - → - The measures taken against false alarm: - → - Range of protection offered by the device. Details about the detection mechanism (opening of Door, Bonnet, luggage compartment, intrusion etc.): - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details & specification of audible alarm: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details about the panic alarm , if provided: - → - Description of the Immobilizer: - → - A detailed description of the vehicle immobilizer and of the vehicle and the measures		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	taken against inadvertent activation with block diagram.: → The vehicle system(s) on which the immobilizer acts.: - → - Details of radio/wireless transmission if any: - → - Method of setting/ unsetting the device.: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - A list of main components comprising the device: - → - Description: AL Part No. → AL Part No. Description · AL Part No.: Hella Part No → Hella Part No ECU: FXX01000 → FXX01000 ECU · FXX01000: 015.XYZ-01 → 015.XYZ-01 a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
AIS-137 Part 3 OBD interface, fault detection and calibration identification	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied	CEA / report needs review	Resolve test inputs and referenced source annexes before concluding whether a new test is needed.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Max. Recommended Gradeability (Stand-start) – in degrees: 10 degs → 10 degs Max. Recommended Gradeability (Stand-start) – in degrees: 8.5 degs → 8.5 degs Vehicle kerb weight kg: Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C → Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Annexure T7-B → Refer Annexure T7-B Maximum (Including driver) for completely built vehicles: D+2 → D+2 Maximum (Including driver) for completely built vehicles: → Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022-2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022-2023/300002XXX	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26 kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Rear axle ratio: 4.56 → 5.13 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type and description (Leaf / Coil / Air / Semi- pneumatic/ Torsion bar): Leaf → Leaf Front: Parabolic spring- Rigid axle suspension → Parabolic spring- Rigid axle suspension Rear: Semi elliptic spring-		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rigid axle suspension → Semi elliptic spring- Rigid axle suspension Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color – Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Weight of seats: Driver seat : 11.5 Kg & Co-driver seat : 11Kg → Driver seat : 11.5 Kg & Co-driver seat : 11Kg Description of seats: -- → -- Make: -- → -- Number of seats fitted or capable of being fitted with head restraints, adjustable or not adjustable.: -- → -- Description of the adjustment, displacement and locking systems of the seat or of its parts and a description of occupant protection system against displacement of luggage.: -- → -- Description of seat anchorage: -- → -- Longitudinal / Vertical (If applicable) position of the seats during the tests.: --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	→ -- Drawings, diagrams and plans of the seats, their anchorages on the vehicle, the floor layout, the adjustment and displacement system of the seats and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	their parts, and their locking devices and of additional occupant protection system against displacement of luggage.: -- → -- In the case of seats fitted with head restraints, the head restraint shall be shown on all drawings, diagrams and photographs.: -- → -- Seat Identification No. / Part No/Drawing Number.: ID Number/ Part Number/D rawing number Weight Description Make (Seat (kg) Manufactu rer/ Vehicle Manufactu rer): -- → -- Description: Make → Make Description: ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) → ID Number/ Part Number/D	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) Description: Weight (kg) → Weight (kg) Brief description and drawings of the seat type, its attachment fittings and its adjustment, displacement and locking systems including the minimum distance between fitting points.: D+1 → D+1 Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Drawings and/or photographs of the locations of the statutory plates and inscriptions and of the vehicle		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	identification number: X4W0XYZ4 / VFT0XYZ2 → X4W0XYZ4 / VFT0XYZ2 Drawings and / or photographs of the official part of the plates and inscriptions (completed example with dimensions): X4WXYZ44 → X4WXYZ44 Location of Vehicle Descriptor Section (VDS) in Vehicle Identification Number (VIN): 4thto 9thCharacter of 17 Characters VIN → 4thto 9thCharacter of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	17 Characters VIN Brief description of the Provision for emergency exit provided in case of sealed windows: -- → -- Dimensions of the instructions describing each motion necessary to unlatch and open the exit are provided from the release mechanism.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side, distance between front extremity & FUPD, width of FUPD, ID		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Brief description of the vehicle type as regards its structure, dimensions, lines and constituent materials:: -- → -- Description of the protective system installed in the vehicle:: -- → -- Description of the protective system installed in the vehicle:: → Description of interior arrangements or fittings that might affect the tests:: -- → -- A detailed description, including	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	photographs and/or drawings, of the vehicle type with respect to the structure, the dimensions, the design and the constituent materials, the side walls of the passenger compartment (exterior and interior), including specific details of the protection system, where applicable:: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- Construction: Integrated / Attached with Mounts: 4 Mounts, Fixed CABIN → 4 Mounts, Fixed CABIN 1st ·		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Description · Make: ID		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) → ID Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) Description · Make: Weight (kg) → Weight (kg) B 51.0 B49.0 · Pedestrian protection as per AIS-100: The distance, measured longitudinally on a horizontal plane, between the transverse centerline of the front axle & the R point of the driver’s seat is less than 1100 mm (Yes / No).: NA → NA Brief description of failure warning	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tell–tale: - → - Brief description and features: - → - Brief description and features: - → - Description: -- → -- Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	version as appropriate.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- A brief written explanation sufficient to describe the ESC system basic operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Brief description of failure warning tell–tale: -- → -- Brief description and features: -- → -- Brief description and features: -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Management of the engine or any		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	other source(s) of motive power e.g.Torque Management, control of fuel supply etc.: -- → -- Component description:: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Software identification: -- → -- Limitations on the location of the controller(s): -- → -- Additional features: -- → -- Modulators: -- → -- General description and function: -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification (if applicable): -- → -- Limitations: -- → -- Electrical Equipment: -- → -- Circuit diagrams: -- → -- Powering methods: -- → -- Pneumatic circuits: -- → -- Safety aspects of the electronic system in accordance with Annex S of IS 11852: -- → -- List of complex functions covered::		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	-- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods and tools identified.: -- → -- Information regarding		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	design provisions in case of failure is provided including any error messages, warning signals, partial performance conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986:2015 and meets manufacturer’s specifications under non-fault conditions.: -- → --	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electro-magnetic compatibility: -- → -- Documentation demonstrating compliance with AIS-004(Part 3).: -- → -- Component description: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	features.: -- → -- Modulators (a) General description and function; (b) Identification; (c) Limitations.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- (see annex 4 of this report for test data): -- → -- (see annex 4 of this report for test data): -- → -- Type / Description: Fully built vehicle, D+2, Non AC, 3.39 Ton GVW → Fully	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	built vehicle, D+2, Non AC, 3.39 Ton GVW Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1 Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67 Reverse: 4.39 → 4.39		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Type / Description: Re-circulating ball type → Re-circulating ball type	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Wiping system (Brief description): Twin blade, Tandem Pattern → Twin blade, Tandem Pattern Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Overall Length (mm): 4750 → 4750 Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961 Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Weights: Base → Base Weights: Variant 1 → Variant 1 Weights: Variant 2 → Variant 2 Weights: Variant 3 → Variant 3 Weights: Variant 4 → Variant 4 Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	spare wheel, tools, etc.) (kg): - → - Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel → Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% / Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) &		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3 Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	No: 9 → 9 Material of blades (metal /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump:	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger / Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA → NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type(s) & Identification Number:: NA → NA Make: NA → NA Number:……: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): … (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:.....cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA → NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing (Periodic / Continuous): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137):	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling /disable after parking: NA → NA Lowest		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control unit for NG fuelling: NA → NA Make(s):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW (manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: ……(kg): NA → NA Capacity: Ah		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA → NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	°C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: ...(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42 Minimum: ...(°C): - 5 → - 5 Maximum: ...(°C): 120 → 120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump → Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA OBD IUPR Family if any: XX+XYY → XX+XYY exhaust outlet location, indicated in a Schematic dimensional drawing.): Refer Enclosure 4G → Refer Enclosure 4G 1.2.11.2.1.1 4.4 · 3. · Calibration Verification Number:: NA → NA Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Engine speed, engine load, A/F mode, catalyst temperature → Engine speed, engine load, A/F mode, catalyst temperature Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Two Type I cycles → Two Type I cycles Catalyst · P0420 · Oxygen sensor 1 and	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Type I → Type I Type (Manual / power): Manual → Manual No. of wipers: Two → Two Make: XXYY System →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Wiper arm: Provided → Provided Length: 500mm → 500mm Make: M/s XYZ → M/s XYZ Length: 450 mm → 450 mm Make: BCS (SYNDICATE WIPER SYSTEM) → BCS (SYNDICATE WIPER SYSTEM) Identification: X5XY0258 → X5XY0258 Washer tank: BWXY4501 → BWXY4501 Type: Plastic → Plastic Make: XYZ → XYZ Identification No. / Part No.: X5BXYZ168 → X5BXYZ168 Capacity l: 1.5Ltr → 1.5Ltr Material: H.D.P.E → H.D.P.E Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make: XYZ → XYZ Model: PEAD64XYZS00 → PEAD64XYZS00 Nozzle(s): XYZ → XYZ No. Of Nozzles: 2 → 2 Defroster: NA → NA Type: - → - Make: - → - Details of Demisting and / or Defrosting system: - → - Schematic Diagram of Engine cooling system: - → - Schematic Diagram of Heating system: - →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	- Cross sectional view of complete air handling system including position of louvers (vents) on dash board: - → - Blower identification: - → - Blower Voltage and capacity (Watts): - → - If defrost system is fitted, specify defrost test temperature (-8 +/- 2 deg C / -18 +/- 3 deg C): → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar		
(continued)	& related dimensions as per related standards.: → Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA A brief technical description indicating the length of the cleaning period, the consumption of cleaning fluid during the cleaning period and the minimum capacity of the container provided;: -- → -- Description and drawings/photographs of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all subassemblies, which includes an electronic oscillator or switching frequency greater than 9 kHz (like ECU, steering, suspension, Body: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly → FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering SYSTEM SOFTWARE (If fitted): NA → NA Make: -- → -- Version: -- → -- Operating System Details with Version: -- → -- DESCRIPTION OF DEVICE (If fitted): NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the VAS: - → - A detailed description of the alarm system and of the vehicle parts related to alarm system installed with Block Diagram: → A list of main components comprising the alarm system: - → - Components placed in Engine compartment: - → - Components placed in Passenger compartment: - → - Availability of radio/wireless transmission Yes/No: - → - Details of radio/wireless transmission if any: - → - frequency: - → - Availability of WPC approval for wireless technology Yes/No: - → - The measures taken against false alarm: - → - Range of protection offered by the device. Details about the detection mechanism (opening of Door, Bonnet, luggage compartment, intrusion etc.): - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details about the panic alarm , if provided: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Description of the Immobilizer: - → - A detailed description of the vehicle immobilizer and of the vehicle and the measures		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	taken against inadvertent activation with block diagram.: → The vehicle system(s) on which the immobilizer acts.: - → - Details of radio/wireless transmission if any: - → - Method of setting/ unsetting the device.: - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - A list of main components comprising the device: - → - Description: AL Part No. → AL Part No. Description · AL Part No.: Hella Part No → Hella Part No ECU: FXX01000 → FXX01000 ECU · FXX01000: 015.XYZ-01 → 015.XYZ-01 a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - a. · Make: - → - b. · Type: - → - c. · Serial Number: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	d. · Name and Address of manufacture: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
AIS-137 Part 3; applicable diesel provisions SCR reagent / inducement and dosing system	Base fuel type used for vehicle Type approval including blend level (Ethanol, Bio-diesel etc): Diesel (Maximum 7% bio-diesel blend) → Diesel (Maximum 7% bio-diesel blend) Name and address of engine manufacturing plant (In case of imported vehicles, above details shall be supplied for importer also): Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX → Ashok Leyland Limited XXXXXX, XXXXXXX - XXXXXX Max. Recommended Gradeability (Stand-start) – in degrees: 10 degs → 10 degs Max. Recommended Gradeability (Stand-start) – in degrees: 8.5 degs → 8.5 degs Vehicle kerb weight kg: Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C → Refer Annexure T7 - B Details for Recommended load body Refer Annexure T7 – C	CEA / report needs review	Confirm SCR fitted. Unknown fitment / scope is retained for review.

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Other axle: NA → NA Gross combination weight kg (applicable for articulated / Tractor Trailer combination vehicles): NA → NA Front axle: NA → NA Rear axle: NA → NA Trailer axle / other axles: NA → NA If applicable, maximum mass of trailer whic5h may be coupled: NA →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	NA Unbraked trailer: NA → NA Reference mass (for vehicles with GVW less than or equal to 3.5 ton) (kg.): Refer Annexure T7-B → Refer Annexure T7-B Maximum (Including driver) for completely built vehicles: D+2 → D+2 Maximum (Including driver) for completely built vehicles: → Make: APOLLO → APOLLO Make: CEAT → CEAT Make: APOLLO → APOLLO Make: CEAT → CEAT Brand(s) name(s) and/or Trade description(s): APOLLO Endumaxx LT → APOLLO Endumaxx LT Brand(s) name(s) and/or Trade description(s): CEAT WINMILE X3-R → CEAT WINMILE X3-R Brand(s) name(s) and/or Trade description(s): Endu Maxx LT → Endu Maxx LT	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Brand(s) name(s) and/or Trade description(s): CEAT MILAZE LT → CEAT MILAZE LT Tyres - Rolling Sound Emissions, Adhesion on Wet Surfaces, Rolling Resistance as per AIS:142: Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 → Compliance to Wet Surfaces & Rolling Resistance as per AIS 142 Tyre rolling Resistance: 7.84 N/kN → 7.84 N/kN Tyre rolling Resistance: 7.86 N/kN → 7.86 N/kN Tyre		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	rolling Resistance: 8.13 N/kN → 8.13 N/kN Tyre rolling Resistance: 9.61 N/kN → 9.61 N/kN Front: C2 → C2 Category of use (Normal / Special / Snow/Severe Snow): Normal → Normal Tyre Tread Pattern along with Drawing: Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer – SHL / 16 /	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	2022 – 2023 / 300002XXXX / TA / XXXX → Refer – SHL / 16 / 2022 – 2023 / 300002XXXX / TA / XXXX Tyre Tread Pattern along with Drawing: Refer - TC7719XX000 003XXXF → Refer - TC7719XX000 003XXXF Tyre Tread Pattern along with Drawing: Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX → Refer - SHL/16/2022- 2023/300002XXX X/TA/XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 756XXXX → CM/L - 756XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L - 372XXXX → CM/L - 372XXXX Identification: TAC No. / BIS License No. / E- Marking: CM/L- 756XXXX → CM/L- 756XXXX Identification: TAC No. /		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	BIS License No. / E- Marking: CM/L – 670007XXXX CM/L - 372XXXX → CM/L – 670007XXXX CM/L - 372XXXX Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Front: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Rear: 2 → 2 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Spare wheel: 1 → 1 Identification: TAC No. / BIS License No. / E- Marking: NA → NA Others (For articulated / combination vehicles): → Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 Q Radial Tube → 7.00 R15 LT 12 PR 114/109 Q Radial Tube Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 7.00 R15 LT 12 PR 114/109 L Radial Tube → 7.00 R15 LT 12 PR 114/109 L Radial Tube Tyre type (Radial/cross ply) (with Tube /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Tyre type (Radial/cross ply) (with Tube / Tubeless), size designation including ply rating, Speed rating, Load rating or Load index. Use symbols as per IS: 15633 / IS: 15636 as may be applicable.: 215/75 R15 LT 115/113 S	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Radial, Tubeless → 215/75 R15 LT 115/113 S Radial, Tubeless Other (for articulated/combination vehicles): NA → NA Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 349 mm → 349 mm Static rolling radius (only applicable for M2, M3, N and T category of vehicles.): 325 mm → 325 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 362 mm → 362 mm Dynamic rolling radius, mm, as per IS 15633 / IS 15636: 339 mm → 339 mm Front, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rear, kg/cm2: 2.91 kg/cm2 → 2.91 kg/cm2 Rear, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Other: NA → NA Front, kg/cm2: 3.61 kg/cm2 → 3.61 kg/cm2 Front, kg/cm2: 2.56 kg/cm2 → 2.56 kg/cm2 Rear, kg/cm2: 6.77 kg/cm2 → 6.77 kg/cm2 Rear, kg/cm2: 3.26 kg/cm2 → 3.26	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	kg/cm2 Other: NA → NA Tyre Pressure Monitoring System (TPMS) / Run Flat Warning System as applicable for Run-Flat tyres - as per AIS-110 (If Provided): NA → NA Make (If Applicable): NA → NA Brief Description of the system: NA → NA Temporary Spare Wheel / RFT- as per AIS-110 (If Provided): NA → NA Type as per AIS-110: NA → NA Make: NA → NA Size: NA → NA Load and speed rating: NA → NA Load and speed rating: → Recommended max speed in Unladen condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Recommended max speed in Unladen condition: → Recommended max speed Laden condition: Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF → Limited to 80 Kmph, 60 Kmph & 40 Kmph with SLF Type: Fully		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	synchronized 5 - forward gears with a Constant mesh reverse gear → Fully synchronized 5 - forward gears with a Constant mesh reverse gear Model name (if any): XXX-XXX → XXX-XXX Model name (if any): YYYY-YY → YYYY-YY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Gear shifting control system (Sketch showing gear shifting arrangement): → No. of gears: Forward 5 + 1 Rev → Forward 5 + 1 Rev Gear ratio: 4.62 → 4.62 Gear ratio: 4.60 → 4.60 Gear ratio: 0.70 → 0.70 Gear ratio: 0.70 → 0.70 Rear axle ratio: 4.56 → 5.13 Type of steering gear (Re-circulating ball / Worm & Roller / Rack & Pinion / Others): Re-circulating ball type → Re-circulating ball type Make: 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd → 1. XXXX India Pvt.Ltd 2. YYYY India Pvt.Ltd Steering gear ratio: 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 → 1. XXXX – 17.8:1 2. YYYY – 18.2 :1 Type and description (Leaf / Coil / Air / Semi- pneumatic/ Torsion bar): Leaf → Leaf Front: Parabolic spring- Rigid axle suspension → Parabolic spring- Rigid axle suspension Rear: Semi elliptic spring-		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Rigid axle suspension → Semi elliptic spring- Rigid axle suspension Fuel Gauge: LED Bars → LED Bars Engine coolant temperature: Red Color – Thermometer Symbol → Red Color –	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Thermometer Symbol Engine low oil pressure: Red Color – Oil Tank Symbol → Red Color – Oil Tank Symbol Weight of seats: Driver seat : 11.5 Kg & Co-driver seat : 11Kg → Driver seat : 11.5 Kg & Co-driver seat : 11Kg Description of seats: -- → -- Make: -- → -- Number of seats fitted or capable of being fitted with head restraints, adjustable or not adjustable.: -- → -- Description of the adjustment, displacement and locking systems of the seat or of its parts and a description of occupant protection system against displacement of luggage.: -- → -- Description of seat anchorage: -- → -- Longitudinal / Vertical (If applicable) position of the seats during the tests.: -- → -- Drawings, diagrams and plans of the seats, their anchorages on the vehicle, the floor layout, the adjustment and displacement system of the seats and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	their parts, and their locking devices and of additional occupant protection system against displacement of luggage.: -- → -- In the case of seats fitted with head restraints, the head restraint shall be shown on all drawings, diagrams and photographs.: -- → -- Seat Identification No. / Part No/Drawing	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number.: ID Number/ Part Number/D rawing number Weight Description Make (Seat (kg) Manufactu rer/ Vehicle Manufactu rer): -- → -- Description: Make → Make Description: ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) → ID Number/ Part Number/D rawing number (Seat Manufactu rer/ Vehicle Manufactu rer) Description: Weight (kg) → Weight (kg) Brief description and drawings of the seat type, its attachment fittings and its adjustment, displacement and locking systems including the minimum distance between fitting points.: D+1 → D+1		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Installation drawing showing rear extremity of vehicle, chassis rear overhang, chassis cross section details. tyre details (size,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	make, unladen pressure) etc., distance between tyre & RUPD RH & LH, frame rear width, P1, P2 points.: - → - Make: M/s. XXXXYYYY → M/s. XXXXYYYY Material (Metallic / Plastic etc.): Metallic → Metallic Nominal thickness mm: 1.0mm → 1.0mm Capacity, Litre: 40 Lts, 30 Lts → 40 Lts, 30 Lts Detailed drawing of the fuel tank assembly with material specifications: FHSXXXXY FHSXXXXY → FHSXXXXY FHSXXXXY Detailed drawing indicating the position / location of the fuel tank(s) in the vehicle.: A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) → A5LXXX00 (50 LTS) A5LXXX00 (40 LTS) Identification No. / Part No.: FHS00544 FfffXXXcdHXXS00744 → FHS00544 FfffXXXcdHXXS00744 Detailed engineering drawing as specified in AIS – 013 or photographs showing the mounting details with dimensions.: NA → NA Cross sectional Drawing of Gear shift	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	lever (Knob): FXA0XXYY (Non AC) / FXA0XXYY (AC) → FXA0XXYY (Non AC) / FXA0XXYY (AC) Drawings and/or photographs of the locations of the statutory plates and inscriptions and of the vehicle		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	identification number: X4W0XYZ4 / VFT0XYZ2 → X4W0XYZ4 / VFT0XYZ2 Drawings and / or photographs of the official part of the plates and inscriptions (completed example with dimensions): X4WXYZ44 → X4WXYZ44 Location of Vehicle Descriptor Section (VDS) in Vehicle Identification Number (VIN): 4thto 9thCharacter of 17 Characters VIN → 4thto 9thCharacter of 17 Characters VIN Brief description of the Provision for emergency exit provided in case of sealed windows: -- → -- Dimensions of the instructions describing each motion necessary to unlatch and open the exit are provided from the release mechanism.: -- → -- In the case of special device, full description and/or drawing of the front under run protection (including mountings and fittings). & material details, distance of P1 & P2 points from center & tyres.: -- → -- Installation drawing of the FUPD with dimensions like cross section, distance between tyre & FUPD RH side & LH side,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	distance between front extremity & FUPD, width of FUPD, ID		
(continued)	marking of FUPD, Width measured from the outermost points of access steps to driver cabin etc.: -- → -- Brief description of the vehicle type as regards its structure, dimensions, lines and constituent materials:: -- → -- Description of the protective system installed in the vehicle:: -- → -- Description of the protective system installed in the vehicle:: → Description of interior arrangements or fittings that might affect the tests:: -- → -- A detailed description, including photographs and/or drawings, of the vehicle type with respect to the structure, the dimensions, the design and the constituent materials, the side walls of the passenger compartment (exterior and interior), including specific details of the protection system, where applicable:: -- → -- Drawing / Images of position & arrangement of Engine: -- → -- Drawing/ Images of Under bonnet (Engine house) Components.: -- → -- Construction: Integrated / Attached with Mounts: 4 Mounts, Fixed CABIN → 4 Mounts, Fixed CABIN 1st ·	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Gear Ratio / RSB: 4.62 → 4.62 1st · Gear Ratio / AVTEC: 4.60 → 4.60 1st · Overall Ratio / RSB: 21.53 → 23.70 1st · Overall Ratio / AVTEC: 21.44 → 21.44 2nd · Gear Ratio / RSB: 2.68 → 2.68 2nd · Gear Ratio / AVTEC: 2.69 → 2.69 2nd · Overall Ratio / RSB: 12.67 → 12.67 2nd · Overall Ratio / AVTEC: 12.71 → 12.71 3rd · Gear Ratio / RSB: 1.51 → 1.51 3rd · Gear Ratio / AVTEC: 1.49 → 1.49 3rd · Overall Ratio / RSB: 7.34 → 7.34 3rd · Overall Ratio / AVTEC: 7.25 → 7.25 4th · Gear Ratio / RSB: 1.0 → 1.0 4th · Gear Ratio / AVTEC: 1.0 → 1.0 4th · Overall Ratio / RSB: 4.56 → 4.56 4th · Overall Ratio / AVTEC: 4.56 → 4.56 5th · Gear Ratio / RSB: 0.70 → 0.70 5th · Gear Ratio / AVTEC: 0.71 → 0.71 5th · Overall Ratio / RSB: 3.64 → 3.64 5th · Overall Ratio / AVTEC: 3.67 → 3.67 6th: - → - Over drive: - → - Reverse · Gear Ratio / RSB: 4.39 → 4.39 Reverse · Gear Ratio / AVTEC: 4.38 → 4.38 Reverse · Overall Ratio / RSB: 20.46 → 20.46 Reverse · Overall Ratio / AVTEC: 20.44 → 20.44 Description · Make: ID	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) → ID Number/ Part Number/Drawing number (Seat Manufacturer/ Vehicle Manufacturer) Description · Make: Weight (kg) → Weight (kg) B 51.0 B49.0 · Pedestrian protection as per AIS-100: The distance, measured longitudinally on a horizontal plane, between the transverse centerline of the front axle & the R point of the driver’s seat is less than 1100 mm (Yes / No).: NA → NA Brief description of failure warning tell–tale: - → - Brief description and features: - → - Brief description and features: - → - Description: -- → -- Tyre dimensions: NA → NA Temporary-use spare wheel/tyre dimensions:: NA → NA Vehicle meets the technical requirements of AIS 110: Yes/No: No → No Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	version as appropriate.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- A brief written explanation sufficient to describe the ESC system basic operational characteristics.: -- → -- Logic diagram: -- → -- Understeer information.: -- → -- Static Stability Factor: -- → -- Make and Country of manufacturer(if imported): -- → -- Software Id / version: -- → -- Brief description of failure warning tell–tale: -- → -- Brief description and features: -- → -- Brief description and features: -- → -- System identification including software level identifier: -- → -- List of motor vehicles by description and configuration that are covered by the information document: -- → -- Wheel type (single or twin) and tyre type (e.g. structure, category of use, size): -- → -- Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): -- → -- Management of the engine or any	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	other source(s) of motive power e.g.Torque Management, control of fuel	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	supply etc.: -- → -- Component description:: -- → -- Sensor(s) External to the Controller: -- → -- Function: -- → -- Limitations on the location of the sensors: -- → -- Identification (e.g. part numbers): -- → -- Controller(s): -- → -- General description and function: -- → -- Functionality of internal sensors (if applicable): -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification: -- → -- Limitations on the location of the controller(s): -- → -- Additional features: -- → -- Modulators: -- → -- General description and function: -- → -- Hardware identification (e.g. part numbers): -- → -- Software identification (if applicable): -- → -- Limitations: -- → -- Electrical Equipment: -- → -- Circuit diagrams: -- → -- Powering methods: -- → -- Pneumatic circuits: -- → -- Safety aspects of the electronic system in accordance with Annex S of IS 11852: -- → -- List of complex functions covered::		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	-- → -- Manufacturer’s documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the operational status may be checked.: -- → -- A list of input and output variables, including their working range, is provided.: -- → -- Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non- fault conditions.: -- → -- System architecture, design methods and tools identified.: -- → -- Information regarding	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	design provisions in case of failure is provided including any error messages, warning signals, partial performance	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	conditions, back-up modes and/or removal of high- level functions.: -- → -- Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.: -- → -- Document reference and date of inspection:: -- → -- Vehicle complies with all other performance requirements of IS 15986:2015 and meets manufacturer’s specifications under non-fault conditions.: -- → -- Under fault conditions, vehicle response corresponds to that described in the manufacturer’s documents / failure analysis and safety concept is verified.: -- → -- Electro-magnetic compatibility: -- → -- Documentation demonstrating compliance with AIS-004(Part 3).: -- → -- Component description: -- → -- Controller(s) (a) General description and function; (b) Identification e.g. part numbers; (c) Limitations on the location of the controller(s); (d) Additional		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	features.: -- → -- Modulators (a) General description and function; (b) Identification; (c) Limitations.: -- → -- Manufacturer’s documents explain the functionality and safety concept of the system and describe how the	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	operational status may be checked.: -- → -- Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.: -- → -- (see annex 4 of this report for test data): -- → -- (see annex 4 of this report for test data): -- → -- Type / Description: Fully built vehicle, D+2, Non AC, 3.39 Ton GVW → Fully built vehicle, D+2, Non AC, 3.39 Ton GVW Make: Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. → Ashok Leyland Limited Hosur Factory Unit I, 175 Hosur Sipcot Industrial Complex, Rajaji Nagar, Hosur– 635 126. Model name / identification: ALXYZABC0 → ALXYZABC0 Type: Compression Ignition → Compression Ignition Bore x stroke (mm): 80 X 98 → 80 X 98 No. of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	cylinders: 3 → 3 Displacement (cm³): 1468 cc → 1468 cc Compression ratio: 16.5±1.0 : 1 → 16.5±1.0 : 1 Max. Engine output (KW @ rpm): 55.1	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Kw @ 3300 rpm → 55.1 Kw @ 3300 rpm Max. Torque (Nm @ rpm): 170Nm @ 1600 to 2400 rpm → 170Nm @ 1600 to 2400 rpm Air cleaner type: Dry type → Dry type Type of Fuel: Diesel → Diesel Make and model: RSB Transmission (T210R) → RSB Transmission (T210R) Make and model: AVTEC (A5M210) → AVTEC (A5M210) Type: Manual gearbox with fully synchro mesh & Rev with sliding mesh → Manual gearbox with fully synchro mesh & Rev with sliding mesh No. of gears: 5 Forward and 1 Reverse → 5 Forward and 1 Reverse 1st: 4.62 → 4.62 1st: 4.60 → 4.60 1st: 21.53 → 21.53 1st: 21.44 → 21.44 2nd: 2.68 → 2.68 2nd: 2.69 → 2.69 2nd: 12.67 → 12.67 2nd: 12.71 → 12.71 3rd: 1.51 → 1.51 3rd: 1.49 → 1.49 3rd: 7.34 → 7.34 3rd: 7.25 → 7.25 4th: 1.0 → 1.0 4th: 1.0 → 1.0 4th: 4.56 → 4.56 4th: 4.56 → 4.56 5th: 0.70 → 0.70 5th: 0.71 → 0.71 5th: 3.64 → 3.64 5th: 3.67 → 3.67		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Reverse: 4.39 → 4.39		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Reverse: 20.36 → 20.36 Reverse: 4.38 → 4.38 Reverse: 20.34 → 20.34 Drive Axle (Front / Rear / All): Rear → Rear Front axle ratio: NA → NA Type / Description: Re-circulating ball type → Re-circulating ball type Steering Gear Ratio: 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd → 1. XXXX Pvt.Ltd 2. YYYY Pvt.Ltd Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Wheels and tyres: Apollo → Apollo Wheels and tyres: Ceat → Ceat Tyre size designation including ply rating: 7.00 R15 LT 12 PR → 7.00 R15 LT 12 PR Tyre size designation including ply rating: 215/75 R15 LT → 215/75 R15 LT Front: Radial, Tube Type → Radial, Tube Type Front: Radial, Tubeless → Radial, Tubeless Front: 3.61 kg/cm2 → 3.61 kg/cm2 Front: 2.56 kg/cm2 → 2.56 kg/cm2 Rear: 6.77 kg/cm2 → 6.77 kg/cm2 Rear: 3.26 kg/cm2 → 3.26 kg/cm2 Wiping system (Brief description): Twin blade, Tandem Pattern → Twin blade, Tandem Pattern	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Material: Metallic → Metallic Capacity (l): 50 lts , 40 lts → 50 lts , 40 lts Dimensions: 2490WB → 2490WB		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Dimensions: 2640WB → 2640WB Wheel base (mm): 2490 → 2490 Wheel base (mm): 2640 → 2640 Overall width (mm): 1842 → 1842 Overall Length (mm): CC → CC Overall Length (mm): 4750 → 4750 Overall Length (mm): 4988 → 4988 Overall Length (mm): FSD / HSD → FSD / HSD Overall Length (mm): 4925 → 4925 Overall Length (mm): 5225 → 5225 Overall height (mm): 1961 → 1961 Overall height (mm): 1948 → 1948 Maximum gradeability in 1st gear: GVW 2890 kg → GVW 2890 kg Maximum gradeability in 1st gear: GVW 3390 kg → GVW 3390 kg Maximum gradeability in 1st gear: 10 Deg → 10 Deg Maximum gradeability in 1st gear: 8.5 Deg → 8.5 Deg Weights: Base → Base Weights: Variant 1 → Variant 1 Weights: Variant 2 → Variant 2 Weights: Variant 3 → Variant 3 Weights: Variant 4 → Variant 4 Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With spare wheel, tools, etc.) (kg): - → - Kerb weight with 90% fuel (With		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	spare wheel, tools, etc.) (kg): - → - Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA Description of energy converters and power plant. (In the case of a vehicle that can run either on petrol, diesel, etc., or also in combination with another fuel, items shall be repeated.1): Diesel → Diesel Engine Manufacturer:: Ashok Leyland Limited → Ashok Leyland Limited Manufacturer's engine model code (as marked on the engine, or other means of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	identification): Model Code: ALXYZABC0 Enclosure – 4A → Model Code: ALXYZABC0 Enclosure – 4A Location of engine model code and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	engine serial number on block / crankcase (drawing to be attached): Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A → Engine Model Code Location - On Engine Head Cover Engine Serial Number Location - On Engine Block Refer Enclosure – 4A Working principle: positive ignition / compression- ignition, four-stroke / two-stroke / rotary cycle (DI/IDI) (NA/TC/TCIC/Any other): Compression Ignition, four-stroke, DI, TCIC → Compression Ignition, four-stroke, DI, TCIC Number, arrangement of cylinders:: 3, Inline → 3, Inline Bore----------(mm): 80 → 80 Stroke: ----------(mm): 98 → 98 Firing order:: 1-2-3 → 1-2-3 Engine capacity:----------(cm3): 1468 → 1468 Volumetric compression ratio (specify	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerance): 17.5±1.0 : 1 → 17.5±1.0 : 1 Drawings of combustion chamber and piston crown and, in the case of positive ignition engine, piston rings:: Refer Enclosure – 4B → Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure – 4B Vehicle with Defeat Device Yes / No: No → No If yes – Details of the Defeat device: NA → NA Normal engine idling speed (specify tolerance):: 700 ± 100 rpm → 700 ± 100 rpm Carbon monoxide in % and HC (n hexane equivalent) in PPM content by volume in the exhaust gas with the engine idling,(for SI engines only): NA → NA High idle engine speed (specify tolerance): -: NA → NA High Idle Lambda value (1± 0.03) or as specified by the vehicle manufacturer): NA → NA Maximum net power:..... kW at min-1: 55.1 KW @ 3100 ± 50 rpm → 55.1 KW @ 3100 ± 50 rpm Maximum permitted engine speed as prescribed by the manufacturer- min-1: 3500 rpm → 3500 rpm Maximum net torque: .....Nm at: -------min-1 ...(manufacturer's declared value): 170 Nm @ 1600 – 2400 rpm → 170 Nm @ 1600 – 2400 rpm Maximum rated speed (Specify	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	tolerance): 3100 rpm ± 50 rpm → 3100 rpm ± 50 rpm Minimum rated speed (Specify tolerance): 1000 rpm ± 100 rpm → 1000 rpm ± 100 rpm Engine Power Table: → Type of fuel used : Mono		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fuel / Bi-Fuel / Flex Fuel / Dual Fuel: Mono Fuel → Mono Fuel Fuel type: Diesel / Gasoline / LPG / CNG / Biomethane / Bio-H methane / HCNG / Biogas / LNG / Ethanol ((E85) / 2 (E100)) /ED (95) Flex fuel Methanol M15 / M100 / MD95 / Biodiesel up to 100% / Hydrogen.: Diesel → Diesel Maximum amount of biofuel acceptable in fuel (manufacturer's declared value): per cent by volume: 7% → 7% By fuel injection (compression-ignition only): (Yes/No): Yes → Yes System description:: Unit Pump Common Rail Direct Injection system → Unit Pump Common Rail Direct Injection system Working principle: direct-injection/pre-chamber/swirl Chamber: Direct Injection → Direct Injection Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Plunger Type & FHL0XYZ0 → Plunger Type & FHL0XYZ0 Maximum fuel delivery in (mm3 /Stroke) or (kg/hr) with tolerance at max net	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	pressure versus engine speed: 13.5 ± 1 kg/hr @ 3300 rpm → 13.5 ± 1 kg/hr @ 3300 rpm Injection advance curve (Specify the tolerances):: ECU Controlled → ECU Controlled Governor: ECU Controlled → ECU Controlled Cut-off point under load:…(min-1): 3300 → 3300 Cut-off point without load (Maximum no-load speed)…(min-1): 3600 ± 100 rpm → 3600 ± 100 rpm Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s) & Identification Number:: Multi-Hole Injectors (7 holes) FKXXYYZZ → Multi-Hole Injectors (7 holes) FKXXYYZZ Opening pressure (kPa) (specify the tolerances) or characteristic diagram: 15000 to 23000 kPa → 15000 to 23000 kPa Length and internal diameter…(mm): Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3 Internal Diameter – 3mm → Length (Pump to Rail) - 130± 3 Length (Rail to Injectors) - 193.5± 3	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Internal Diameter – 3mm Common rail make & type: M/s. XXXXYYY, Hot Forged Rail Type → M/s. XXXXYYY, Hot Forged Rail Type Cold start system: NA → NA Make(s):: NA → NA Type(s) &		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: NA → NA Description:: NA → NA Auxiliary starting aid: NA → NA Make(s):: NA → NA Type(s):: NA → NA System description:: NA → NA Electronic controlled injection: (Yes/No): Yes → Yes Make(s):: M/s. XXXXYYY → M/s. XXXXYYY Type(s):: Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU → Common Rail, Direct Injection with Pressure Sensor, Controlled by ECU Description of the system (in the case of systems other than continuous injection, give equivalent details):: NA → NA Make and type of the control unit:: M/s. XYZX and Electronic Control Unit → M/s. XYZX and Electronic Control Unit Make and type of the fuel regulator:: M/s. XYZX and Fuel metering Pump with in-built Inlet Metering Valve → M/s. XYZX and Fuel metering Pump with in-built Inlet Metering	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Valve Make and type of air-flow sensor:: M/s. Continental and Active type → M/s. Continental and Active type Make and type of fuel distributor:: M/s. Delphi-TVS and Metering type → M/s. Delphi-TVS and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Metering type Make and type of throttle housing:: M/s. Padmini and Butterfly Valve type → M/s. Padmini and Butterfly Valve type Make and type of water temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air temperature sensor:: M/s. Continental and Thermistor Type → M/s. Continental and Thermistor Type Make and type of air pressure sensor:: M/s. Continental and Piezo Resistance Type → M/s. Continental and Piezo Resistance Type By fuel injection (positive ignition only): (Yes/No): No → No Working principle: intake manifold (single/multi- point)/direct injection/other (specify): NA → NA Make(s) of Injection Pump :: NA → NA Type(s) & Identification Number of Injection Pump:: NA → NA System description (in the case of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	systems other than continuous injection give equivalent details):: NA → NA Make and type of the control unit:: NA → NA Make and type of the fuel regulator:: NA → NA Make and type of the air-flow sensor:: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make and type of the micro-switch:: NA → NA Make and type of the throttle housing:: NA → NA Make and type of the water temperature sensor:: NA → NA Make and type of the air temperature sensor:: NA → NA Injectors: Opening pressure (Specify the tolerances):,..... (kPa) or characteristic diagram:: NA → NA Make(s):: NA → NA Type(s):: NA → NA Identification Number: NA → NA Maximum fuel delivery in (kg/hr) with tolerance at max net power engine speed (min-1) or characteristic diagram: If boost control is supplied, state the characteristics fuel delivery and boost pressure versus engine speed: NA → NA Cut-off point under load (min-1): NA → NA Injection timing:: NA → NA Cold start system:: NA → NA Operating principle(s):: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Operating limits/settings(Specify the tolerances): NA → NA Pressure, (Specify the tolerances).....(kPa) or characteristic diagram:: 320 ± 50 kPa → 320 ± 50 kPa Rated voltage:...V, positive/negative ground: 12V, Negative Ground → 12V, Negative Ground		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Generator/ Alternator: Alternator → Alternator Type:: DC – Alternator with Voltage regulator → DC – Alternator with Voltage regulator Nominal output: ...VA: 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps → 1205 VA (For Non AC Variants) 850 Amps 1745 VA (For AC Variants) 120 Amps Ignition: NA → NA Make(s):: NA → NA Type(s):: NA → NA Working principle:: NA → NA Ignition advance curve or map(Specify the tolerances):: NA → NA Static ignition timing (Specify the tolerances):.....degrees before TDC: NA → NA Spark plugs: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Gap setting----------(mm): NA → NA Nominal resistance (kilo ohm) (if resistive type): NA → NA Ignition coil(s): NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA Ignition condenser: NA → NA Make: NA → NA Type: NA → NA Identification number: NA → NA EMI suppressor cap / Device / Electronic unit: NA → NA Make: NA → NA Type (Resistive/Capacitive): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification number: NA → NA Nominal resistance (kilo ohm): NA → NA Terminology and Drawing of interference Suppression equipment: NA → NA H.T. Cable: NA → NA Make and Place: NA → NA Type (Resistive/Non-resistive): NA → NA Length (mm) (if resistive type): NA → NA Outside dia. (mm) (if resistive type): NA → NA Nominal resistance kilo ohm, (if resistive type): NA → NA Radiator drawing(s): Refer enclosure 4C2 & 4C4 → Refer enclosure 4C2 & 4C4 Radiator drawing(s): Refer enclosurev 4C1 & 4C3 → Refer enclosurev 4C1 & 4C3	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make: Mahle Behr → Mahle Behr Make: Tata Toyo → Tata Toyo Type (s): Cross flow → Cross flow Relief valve pressure setting: 0.9 kg/cm2 → 0.9 kg/cm2 Fan characteristics (Fan power, kW) Enclose the fan power curve corresponding to full load (v/s engine speed).: Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) → Electric Fan – 0.449 kW (NON AC), 0.5165 kW (AC) Make (s): M/s. XXXX → M/s. XXXX Number of blades and Identification/Part No: 9 → 9 Material of blades (metal /		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	plastic): Plastic → Plastic Type(s) [Fixed / Viscous / Electrical driven]: Electrical driven → Electrical driven Drive ratio: NA → NA Fan diameter (mm): 360 → 360 Max. Speed of fan (in rev/min): 3200 ± 150 (Non-AC) 3400 ± 150 (AC) → 3200 ± 150 (Non-AC) 3400 ± 150 (AC) Radiator core open area (cm²): 1906 cm2 → 1906 cm2 Cooling system: liquid/air: Liquid → Liquid Nominal setting of the engine temperature control o mechanism (C)::: 85 deg C → 85 deg C	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Liquid: Yes → Yes Nature of liquid:: Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L → Option 1: Water + 70 % Antifreeze coolant capacity: approx..8 L Option 2 : Water + 60 % Antifreeze coolant capacity: approx..8 L Circulating pump(s): (Yes/No): Yes → Yes Characteristics of Circulating pump: Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm → Pump Discharge of 130 lpm @ 0.9 Bar at rotational speed of 4177 ± 25 rpm Make(s):: XXXXX		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Fasteners Ltd → XXXXX Fasteners Ltd Type(s):: Impeller Type → Impeller Type Drive ratio(s):: 1: 1.12 → 1: 1.12 Description of the fan and its drive mechanism:: Electric Fan , Motor Driven → Electric Fan , Motor Driven Blower: (Yes/No): No → No Characteristics of blower: NA → NA Make(s):: NA → NA Type(s):: NA → NA Drive ratio(s):: NA → NA Pressure charger: (Yes/No) Turbocharger	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/ Supercharger: Yes, Turbo charger → Yes, Turbo charger Make(s):: M/s. XYZ, India → M/s. XYZ, India Type(s) & Identification Number:: Waste Gate with Vacuum Actuator, X3M1XXXX → Waste Gate with Vacuum Actuator, X3M1XXXX Description of the system (maximum charge pressure:.....(kPa), waste-gate, if applicable): Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa → Waste gate turbocharger with vacuum actuator Maximum Charge pressure - 145 kPa Inter-cooler: (Yes/No): Yes → Yes Type: air-air / air-water: Air – Air → Air – Air Make :: Mahle Behr → Mahle Behr Make :: Tata Toyo → Tata Toyo		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Identification Number:: B1V42XXX → B1V42XXX Identification Number:: B1V42YYY → B1V42YYY Air pressure drop across the intercooler: 15 kPa → 15 kPa Intake depression at rated engine speed and at 100 per cent load (Specify location of measurement) Minimum allowable:... (kPa) Maximum allowable:... (kPa): Minimum : 0 kPa Maximum : 6.0 kPa	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(at 150 mm from TC compressor inlet) → Minimum : 0 kPa Maximum : 6.0 kPa (at 150 mm from TC compressor inlet) Description and drawings of inlet pipes and their accessories (plenum chamber, heating device, additional air intakes, etc.):: Intake manifold with Electronic throttle valve Refer Enclosure 4D → Intake manifold with Electronic throttle valve Refer Enclosure 4D Intake manifold description including dimensional drawings and/or photographs):: Aluminium Intake manifold with pressure sensor Refer Enclosure 4D → Aluminium Intake manifold with pressure sensor Refer Enclosure 4D Air filter, Dimensional drawings: Refer Enclosure 4E → Refer Enclosure 4E		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Make(s):: M/s. Fleet guard → M/s. Fleet guard Type(s) & Identification Number:: Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 → Dry Type, Air Filter Assembly – B0EXXX01 Air Filter Element – FG80XXX0 Intake silencer, Dimensional drawing: NA → NA Make(s):: NA → NA Type(s) & Identification Number:: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Description and Dimensional drawing of the exhaust manifold:: Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F → Cast iron manifold with adaptation for Turbocharger and EGR inlet Refer Enclosure 4F Description and drawing with dimensions of the exhaust system:(Description with general arrangement of exhaust system along with its routing indicating the lengths of exhaust pipe, tail pipe and: Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after LNT, Differential Pressure Sensor provided across DPF and		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	O2 sensors provided before and after LNT-DPF assembly → Exhaust system with Lean NOx Trap (LNT), Diesel Particulate Filter (DPF) and Passive Selective Catalytic Reduction Catalyst (pSCR). Temperature Sensors are provided before and after	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	LNT, Differential Pressure Sensor provided across DPF and O2 sensors provided before and after LNT-DPF assembly Effective volume of exhaust-system (specify the tolerance & range) in liters (from exhaust manifold / TC outlet to tail pipe end), Enclose the exhaust system dimensional drawing and indicate the volume of each parts clearly: 13.05 ± 1 litre Refer Enclosure 4G → 13.05 ± 1 litre Refer Enclosure 4G Description and drawing of the exhaust Muffler/Silencer:: NA → NA Type(s) & Identification Number:: NA → NA Make: NA → NA Number:.....: NA → NA Internal diameter of exhaust pipe (mm)(min): diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G → diameter 50.8 mm 1.5 mm thickness Refer Enclosure 4G Minimum distance between exhaust pipe(s) and the fuel		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	line: >150 mm → >150 mm Auxiliary Noise shields for compliance to IS 3028 and / OR AIS-020 (If Provided): NA → NA Material: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Layout of noise shield / Photographs / Diagram Showing arrangements indicating fitment on vehicle.: NA → NA Maximum allowable exhaust back pressure at rated engine speed and at 100 per cent load (location of measurement): ... (kPa): 30.0 kpa (150 mm after Turbocharger) → 30.0 kpa (150 mm after Turbocharger) Minimum cross-sectional areas of inlet and outlet ports:: Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm → Inlet – 31.5 mm x 40.5 mm Outlet – 24.0 mm x 28.0 mm Inlet (mm): 8.00 → 8.00 Exhaust (mm): 8.00 → 8.00 Opening: 17° bTDC → 17° bTDC Closing: 51° aBDC → 51° aBDC Opening: 66° bBDC → 66° bBDC Closing: 25° aTDC → 25° aTDC Transfer: NA → NA Opening: NA → NA Closing: NA → NA Reference and/or setting ranges (Specify the tolerances):: On Cold Engine → On Cold Engine Inlet: 0.3 to 0.4 mm → 0.3 to 0.4 mm Exhaust: 0.3 to 0.4 mm → 0.3 to 0.4 mm		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Distribution by ports: NA → NA Volume of crank-case cavity with piston at TDC: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Reed valve fitted (Yes / No): No → No Description of inlet ports, scavenging and exhaust ports with corresponding timing.: NA → NA Device for recycling crankcase gases (description and drawings):: Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J → Closed Loop Crankcase ventilation with external oil separator Refer Enclosure 4J Additional pollution control devices - Lean NOx Trap/SCR/Lean NOx Catalyst (if any, and if not covered by another heading:: Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) → Lean NOx Trap (LNT) and Diesel Particulate Filter (DPF) integrated and Passive Selective catalytic reduction (pSCR) Catalytic converter: (Yes/No) DOC+DPF / DOC+SDPF / LNT+DPF /LNT+SDPF / GPF / Any Other: Yes, LNT+DPF → Yes, LNT+DPF Catalytic converter Make Canner / Loader: Canner - M/s. Sharda Motor		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Industries Limited (SMIL) Loader - M/s. BASF → Canner - M/s. Sharda Motor Industries Limited (SMIL)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Loader - M/s. BASF Catalytic converter Identification Number: FZL1XXYY → FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Number of catalytic converters and elements (provide the information below for each separate unit:: Catalytic Converter – 1No Element – 1 No. → Catalytic Converter – 1No Element – 1 No. Dimensions (mm) and shape of the catalytic converter(s) (volume, etc.):: Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G → Dimension - 102.2 mm x 187.0 mm x 105.0 mm (LXBxH) Volume - 1.6 Litre Shape - Elliptical Refer Enclosure 4G Type of catalytic action:: Adsorption, desorption and regeneration → Adsorption, desorption and regeneration Regeneration type		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	for LNT (periodic / continuous):: Continuous → Continuous Total charge of precious metal: (g/vehicle): 6.68 g/Vehicle → 6.68 g/Vehicle Relative concentration (%): Pt: Rh:Pd: 90% : 5% : 5% → 90% : 5% : 5% Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite. → Structure: Oval shape, Honey Comb type Material: Cordierite. Cell density:……cells per sq. inch / cm: 300 CPSI → 300 CPSI Type of casing for catalytic converter(s):: Gap Bulk Density (GBD) Type → Gap Bulk Density (GBD) Type Diagram indicating the arrangement and Positioning of the catalytic converter(s) (place and reference distances in the exhaust system):: Refer Enclosure 4G → Refer Enclosure 4G Heat shield: (Yes/No): Yes → Yes Principle and Characteristics:: Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	downstream SCR catalyst for the reduction of NOX during normal lean operation. → Passive selective catalytic reduction (pSCR) is a promising approach for the control of NOX emissions in lean burn systems. By periodically decreasing the AFR to fuel-rich levels, ammonia (NH3) can be produced and stored on a downstream SCR catalyst for the reduction of NOX during normal lean operation. Make:: Canner - M/s. XXXX Loader - M/s. YYYY → Canner - M/s. XXXX Loader - M/s. YYYY Type and Identification number:: Type - Passive NOx Reduction and FWH0YYYY → Type - Passive NOx Reduction and FWH0YYYY Total charge of precious metal:: 211 g/vehicle (Cu Wash coat 90%) → 211 g/vehicle (Cu Wash coat 90%) Substrate (structure and material):: Structure: Oval shape, Honey Comb type Material: Cordierite → Structure: Oval shape, Honey Comb type Material: Cordierite Relative Concentration (%):: 0%:0%:0% (Pt:Pd:Rh) → 0%:0%:0% (Pt:Pd:Rh) Dimensions and shape of the SCR (volume, etc.):: Dimension: 102.2 mm x 187.0 mm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical → Dimension: 102.2 mm x 187.0 mm x 105.0 mm (LxBxH) Volume: 1.6 Litre Shape : Elliptical Unique Identification (Serial No or Month and year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift) Dosing ECU:: No → No Make:: NA → NA Identification number:: NA → NA Calibration Identification number:: NA → NA Urea Dosing Unit:: No → No Dosing Unit:: NA → NA Make:: NA → NA Type and Identification number:: NA → NA Supply Unit:: NA → NA Make:: NA → NA Type and Identification number: NA → NA Dosing Injector:: NA → NA Make:: NA → NA Type and Idetification number :: NA → NA NOx Sensor (Before /After of SCR):: No → No Make:: NA → NA Type and Identification number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Ad Blue Tank:: No → No Tank Capacity (l) with minimum-maximum :: NA → NA Location :: NA → NA Regeneration systems/method of exhaust after-treatment systems,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	description:: Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system → Periodic regeneration triggered by DPF soot load threshold, estimated with pressure difference across DPF. Exhaust temperature increase for DPF regeneration triggered by post-injection and after burning of fuel in exhaust system The number of Type I operating cycles, or equivalent engine test bench cycles, between two cycles where regenerative phases occur under the conditions equivalent to Type I test (Distance "D" in t AIS 137:: 65 cycles → 65 cycles Number of Type I operating cycles for Regeneration (Distance “d” in AIS 137): 2 cycles → 2 cycles Description of method employed to determine the number of cycles	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	between two cycles where regenerative phases occur:: Soot load determined by pressure difference across diesel particulate filter		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(DPF) → Soot load determined by pressure difference across diesel particulate filter (DPF) Parameters to determine the level of loading required before regeneration occurs (i.e. temperature, pressure etc.):: Engine Soot Model → Engine Soot Model Description of method used to load system in the test procedure described in AIS 137: Type I Operating cycle on chassis dynamometer → Type I Operating cycle on chassis dynamometer Normal operating temperature range (°C):: 400 to 500 °C → 400 to 500 °C Consumable reagents (where appropriate): NA → NA Type and concentration of reagent needed for catalytic action (where appropriate):: NA → NA International standard (where appropriate):: NA → NA Frequency of reagent refill: continuous/maintenance (where appropriate):: NA → NA Normal operating temperature range (°C):: NA → NA Oxygen sensor: (Yes/No): Yes → Yes Type: Wide band lambda → Wide band	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	lambda Location of oxygen sensor:: Upstream of LNT and downstream of DPF → Upstream of LNT and downstream		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	of DPF Control range of oxygen sensor (Specify the tolerances):: 0.8 to 9 λ → 0.8 to 9 λ Make of oxygen sensor:: XXXXXXXX → XXXXXXXX Identification / Part number: YYYYYYY → YYYYYYY Air injection: (Yes/No): No → No Type (pulse air, air pump, etc.) & Identification Number:: NA → NA Exhaust gas recirculation (EGR): (Yes/No): Yes (Integrated with EGR Cooler) → Yes (Integrated with EGR Cooler) Make and number of EGR’s :: M/s. Paxxxx and One → M/s. Paxxxx and One Type (Cooled / Non-cooled/Progressive/ On-Off/ Any Other): Cooled, Progressive → Cooled, Progressive Identification Number:: FEW0XXYY → FEW0XXYY Characteristics (flow rate, etc.):: Flow rate 101 kg/hr (Maximum) → Flow rate 101 kg/hr (Maximum) Water cooled system: (Yes/No): Yes → Yes EGR Cooler Make and Identification Number: M/s. XXXXXX and FEW0YYYY → M/s. XXXXXX and FEW0YYYY	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Evaporative emission control system: (Yes / No) Type: Make and Identification/Part number:: No → No Detailed description of the devices and their state of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	tune:: NA → NA Drawing with dimensions of the evaporative control system (including fuel hoses length and diameter):: NA → NA Drawing with dimensions of the carbon canister:: NA → NA Canister Make :: NA → NA Canister Identification/ Part number :: NA → NA Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): NA → NA Mass of dry charcoal:.....g: NA → NA Canister working capacity:.....g: NA → NA Canister bed volume:.....litre: NA → NA Canister Purge and vent strategy description:: NA → NA Purge valve make and Identification Number: NA → NA Fuel tank pressure control valve : (Yes/No): NA → NA Make and Identification Number: NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Operating pressure (kPa): NA → NA Liquid fuel hoses material :: NA → NA Schematic drawing with dimensions of the fuel tank with indication of capacity and material:: NA → NA Drawing of the heat shield between tank and exhaust system:: NA → NA Regeneration system/method. Description and/or		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	drawing (Periodic / Continuous): NA → NA Particulate trap: (Yes/No): Yes → Yes Number of Particulate trap and elements (provide the information below for each separate unit:: Particulate Trap - 1No. Element - 1No. → Particulate Trap - 1No. Element - 1No. Make and Identification Number Canner & Loader: Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY → Canner - M/s. Sharda Motor Industries Limited (SMIL) Loader - M/s. BASF FZL1XXYY Unique Identification (Serial No. or Month and Year of Manufacturing or Batch No. or any other means of identification): XXX / DDMMYY / X (Serial No. / Date / Shift) → XXX / DDMMYY / X (Serial No. / Date / Shift)	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Dimensions and shape of the particulate trap (capacity):: Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre → Dimension : 102.2 mm x 187.0 mm x 141.0 mm (LXBxH) Shape : Oval shape Capacity : 2.1 Litre Type and design of particulate trap: Volume in litres: Type : Periodic		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Regeneration Type Design : Cross flow type design Volume : 2.1 Litres → Type : Periodic Regeneration Type Design : Cross flow type design Volume : 2.1 Litres Location of the particulate trap (reference distances in the exhaust line):: Refer Enclosure 4G → Refer Enclosure 4G Total charge of precious metal:: 0.81 g/vehicle → 0.81 g/vehicle Relative concentration (%): Pt:Pd:Rh: 90% : 10% : 0 → 90% : 10% : 0 Substrate (structure and material):: Structure - Oval shape, Honey Comb type Material - Silicon Carbide → Structure - Oval shape, Honey Comb type Material - Silicon Carbide Cell density (CPSI): 300 CPSI → 300 CPSI Electronic Control Unit (ECU): Yes → Yes	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Make & number of ECU’s: M/s. XXXX & One → M/s. XXXX & One Identification number: Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ → Delphi ID : XSSSZ 5.1XX, AL Development ID : F6NXXYYZ AL Production ID : F6NXXYYZ Calibration Identification Number: P15D4BS6UP80LNT1 → P15D4BS6UP80LNT1 Calibration		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Verification Number: Version → Version Calibration Verification Number: 7.00 R15 → 7.00 R15 Calibration Verification Number: 215/75 R15 LT → 215/75 R15 LT Calibration Verification Number: Version 00 → Version 00 Calibration Verification Number: A4 9D CF BZ → A4 9D CF BZ Calibration Verification Number: 9D 56 DA 0Y → 9D 56 DA 0Y On-board-diagnostic (OBD) system: ((Yes/No)): Yes → Yes Written description and/or drawing of the Malfunction Indicator (MI):: Refer Enclosure 4K → Refer Enclosure 4K List and purpose of all components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Written description (general working principles) for:: Refer Enclosure 4K → Refer Enclosure 4K Positive ignition engines: NA → NA Catalyst monitoring:: NA → NA Misfire detection:: NA → NA Oxygen sensor monitoring:: NA → NA Other components monitored by the OBD system:: NA → NA Compression-ignition engines: Yes → Yes Catalyst monitoring: Refer Enclosure 4K → Refer Enclosure 4K Particulate traps		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	monitoring:: Refer Enclosure 4K → Refer Enclosure 4K Electronic fueling system monitoring:: Refer Enclosure 4K → Refer Enclosure 4K NOx control system monitoring (SCR/LNT/NOx absorber):: Refer Enclosure 4K → Refer Enclosure 4K Other components monitored by the OBD system:: Refer Enclosure 4K → Refer Enclosure 4K Criteria for MI activation (fixed number of driving cycles or statistical method):: Refer Enclosure 4K → Refer Enclosure 4K List of all OBD output codes and formats used (with explanation of each):: Refer Enclosure 4K → Refer Enclosure 4K OBD Communication protocol standard: Refer Enclosure 4K → Refer Enclosure 4K A description of the type and number of	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	the pre- conditioning cycles used for the original type approval of the vehicle.: 3 EUDC → 3 EUDC A description of the type of the OBD demonstration cycle used for the original type-approval of the vehicle for the component monitored by the OBD system.: Refer Enclosure 4K → Refer Enclosure 4K A comprehensive document describing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	all sensed components with the strategy for fault detection and MI activation (fixed number of driving cycles or statistical method), including a list of relevant secondary sensed parameters for each component monitored by the OBD system. A list of all OBD output codes and format used (with an explanation of each) associated with individual emission related power-train components and individual non-emission related components, where monitoring of the component is used to determine MI activation. In particular, a comprehensive explanation for the data given in service $05 Test ID $21 to FF and the data given in service $06 shall be provided. In the case of vehicle types that use a communication link in accordance with ISO 15765-4 "Road vehicles – Diagnostics on Controller Area Network (CAN) – Part 4: Requirements for emissions- related systems", a	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	comprehensive explanation for the data given in service $06 Test ID $00 to FF, for each OBD monitor ID supported, shall be provided.: Refer		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Enclosure 4K → Refer Enclosure 4K The information required by this paragraph may, for example, be defined by completing a table as follows, which shall be attached to this annex:: Refer Enclosure 4K → Refer Enclosure 4K OBD vehicle family if any: AL XYZ engine family → AL XYZ engine family Parameters defining the OBD family (to be attached) (in reference with AIS 137): Refer OBD Family - Annexure → Refer OBD Family - Annexure Torque limiter (Yes/No) (Applicable for OBD): No → No Description of the torque limiter activation:: NA → NA Description of the full load curve limitation:: NA → NA Other systems (description and operation):: NA → NA Systems to ensure the correct operation of NOx control measures: NA → NA 2: NA → NA Vehicle with permanent deactivation of the driver inducement, for use by the rescue services [Note: cross- reference to be determined]. : (Yes/No): NA → NA Activation of the creep mode disable after restart /disable after fueling	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	/disable after parking: NA → NA Lowest		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	concentration of the active ingredient present in the reagent that does not activate the warning system (CD min): %(vol): NA → NA When appropriate, manufacturer reference of the Documentation for installing in a vehicle the systems to ensure the correct operation of NOx control measures: NA → NA Details of any devices designed to influence fuel economy (if not covered by other items):: NA → NA LPG fuelling system: (Yes/No): No → No Electronic engine management control unit for LPG fuelling: NA → NA Make(s):: NA → NA Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to LPG or back:: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.): NA → NA NG fuelling system: (Yes/No): No → No Approval number: NA → NA Electronic engine management control	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	unit for NG fuelling: NA → NA Make(s):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation:: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to NG or back: NA → NA System layout (electrical connections, vacuum connections, compensation hoses, etc.):: NA → NA Hydrogen fuelling system: (Yes/No): No → No Electronic engine management control unit for hydrogen fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to hydrogen or back:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA H2NG fuelling system: (Yes/No): No → No		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Percentage of hydrogen in the fuel (maximum specified by the manufacturer): NA → NA Electronic engine management control unit for H2NG fuelling: NA → NA Make(s):: NA → NA Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA Emission-related adjustment possibilities:: NA → NA Further documentation: NA → NA Description of the safeguarding of the catalyst at switch- over from petrol to H2NG or back:: NA → NA System lay-out (electrical connections, vacuum connections compensation hoses, etc.):: NA → NA Electric motor(Traction): No → No Type (winding, excitation):: NA → NA Maximum hourly output: kW(manufacturer’s declared value): NA → NA Maximum net power:kW(manufacturer’s declared value): NA → NA Maximum 30 minutes power: kW	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	(manufacturer’s declared value): NA → NA Operating voltage: V: NA → NA Make and Identification Number:: → Battery (Traction): NA → NA Number of cells:: NA → NA Mass: .....(kg): NA → NA Capacity: Ah		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	(Amp-hours): NA → NA Position:: NA → NA Engines or motor combinations: NA → NA Hybrid Electric Vehicle: (Yes/No): No → No Category of Hybrid Electric vehicle Off Vehicle Charging/Not Off Vehicle Charging9: NA → NA Operating mode switch: with/without: NA → NA Selectable modes: NA → NA Pure electric: (Yes/No): NA → NA Pure fuel consuming: (Yes/No): NA → NA Hybrid modes: (Yes/No) (if yes, short description): NA → NA Description of the energy storage device: (traction battery, capacitor, flywheel/generator...): NA → NA Make(s):: NA → NA Type(s) :: NA → NA Identification number:: NA → NA Kind of electrochemical couple:: NA → NA Energy: ..... (for battery: voltage and capacity Ah in 2 h, for capacitor: J): NA →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	NA Charger: on board/external/without: NA → NA Electric machines (describe each type of electric machine separately): NA → NA Make:: NA → NA Type:: NA → NA Primary use: traction motor/generator: NA → NA When used as traction motor: mono motor / multi motors (number):: NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Maximum power: kW: NA → NA Working principle:: NA → NA Direct current/alternating current/number of phases:: NA → NA Separate excitation/series/compound: NA → NA Synchronous/asynchronous: NA → NA Control unit: NA → NA Make:: NA → NA Type :: NA → NA Identification number:: NA → NA Power controller: NA → NA Make:: NA → NA Type:: NA → NA Identification number:: NA → NA Vehicle electric range..... km (according to AIS 137):: NA → NA Manufacturer’s recommendation for preconditioning:: NA → NA CO2 (g/km) (Applicable for category M1 with GVW<3.5 T): NA → NA Declared : (rounded to 3 decimal	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	places): NA → NA Fuel consumption (l/100 km) for Petrol, LPG or Diesel and (kg/100km) for CNG and (kWh/100 km) for Electric Driven Vehicles(Applicable for category M1 with GVW<3.5 T) (rounded to 3 decimal places)*: NA → NA Fuel Equivalent Fuel Consumption (Actual Fuel): NA → NA Petrol Equivalent Fuel Consumption (Petrol Equivalent): NA → NA *calculated on the declared CO2: NA → NA CO2 Reducing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	technologies Available (Yes/No): No → No Regenerative braking: NA → NA Start-Stop System: NA → NA Tyre pressure monitoring system: NA → NA 6 or more Speed Transmission: NA → NA Any other technology (manufacture to enclose detail): NA → NA Electric energy consumption for electric vehicles Wh/km: NA → NA Electric energy consumption for pure electric vehicles Wh/km: NA → NA Electric energy consumption for externally chargeable hybrid electric vehicles: NA → NA Electric energy consumption (condition A, combined) Wh/km: NA → NA Electric energy consumption (condition	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	B, combined) Wh/km: NA → NA Electric energy consumption (weighted combined) ….Wh/km: NA → NA Temperatures permitted by the manufacturer: NA → NA Cooling system: Yes → Yes Liquid cooling: Yes → Yes Maximum temperature at outlet: …….(°C): 100 °C → 100 °C Air cooling: NA → NA Reference point:: NA → NA Maximum temperature at reference point: …….(°C): NA → NA Maximum outlet temperature of the inlet intercooler: .. (°C): 45 °C → 45		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	°C Maximum exhaust temperature at the point in the exhaust pipe(s) adjacent to the outer flange(s) of the exhaust manifold: …(°C): 740 °C → 740 °C Minimum: .. (°C): - 8 → - 8 Maximum: .. (°C): 42 → 42 Minimum: …(°C): - 5 → - 5 Maximum: …(°C): 120 → 120 Lubrication system: Yes → Yes Description of the system: Forced Lubrication system → Forced Lubrication system Position of the lubricant reservoir:: Sump beneath crankcase → Sump beneath crankcase Feed system (by pump/injection into intake/mixing with fuel, etc.): Pump →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Pump Lubricating pump: Yes → Yes Make(s):: M/s. ZXWA → M/s. ZXWA Type(s):: XXYY Type → XXYY Type Mixture with fuel: No → No Percentage:: → Oil cooler: (Yes/No): Yes → Yes Drawing(s):..., or: Enclosure 4L → Enclosure 4L Make(s):: M/s. XXYY or Any other equivalent → M/s. XXYY or Any other equivalent Type(s):: DONUT → DONUT Moment of inertia of engine flywheel:: 0.1212 kg/m2 → 0.1212 kg/m2 Additional moment of inertia with no gear engaged:: 0.4565 kg/m2 → 0.4565 kg/m2		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Transmission Control Unit (ECU) (Yes/No):: No → No Type and Identification Number :: NA → NA Calibration Identification Number: NA → NA Calibration Verification Number (CVN): NA → NA OBD IUPR Family if any: XX+XYY → XX+XYY exhaust outlet location, indicated in a Schematic dimensional drawing.): Refer Enclosure 4G → Refer Enclosure 4G 1.2.11.2.1.1 4.4 · 3. · Calibration Verification Number:: NA → NA	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Engine speed, engine load, A/F mode, catalyst temperature → Engine speed, engine load, A/F mode, catalyst temperature Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Two Type I cycles → Two Type I cycles Catalyst · P0420 · Oxygen sensor 1 and 2 signals · Difference between sensor 1 and sensor 2 signals · 3rd cycle: Type I → Type I Type (Manual / power): Manual → Manual No. of wipers: Two → Two Make: XXYY System →		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	XXYY System Type: 12V → 12V Identification mark: X5XYZ100 → X5XYZ100 Rated voltage: 12V → 12V Number of sweep Frequencies: 2 → 2 Highest sweep frequency (Cycles / min): 62±8 rpm → 62±8 rpm	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Lowest sweep frequency (Cycles / min): 40±5 rpm → 40±5 rpm Wiper arm: Provided → Provided Length: 500mm → 500mm Make: M/s XYZ → M/s XYZ Length: 450 mm → 450 mm Make: BCS (SYNDICATE WIPER SYSTEM) → BCS (SYNDICATE WIPER SYSTEM) Identification: X5XY0258 → X5XY0258 Washer tank: BWXY4501 → BWXY4501 Type: Plastic → Plastic Make: XYZ → XYZ Identification No. / Part No.: X5BXYZ168 → X5BXYZ168 Capacity l: 1.5Ltr → 1.5Ltr Material: H.D.P.E → H.D.P.E Washer tank motor / Washer pump: ELECTRIC MOTOR WITH PUMP → ELECTRIC MOTOR WITH PUMP Make: XYZ → XYZ Model: PEAD64XYZS00 → PEAD64XYZS00 Nozzle(s): XYZ → XYZ No. Of Nozzles: 2 → 2 Defroster: NA → NA Type: - → - Make: - → - Details of Demisting and / or Defrosting system: - → - Schematic Diagram of Engine cooling system: - → - Schematic Diagram of Heating system: - →		

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	- Cross sectional view of complete air handling system including position of louvers (vents) on dash board: - → - Blower identification: - → - Blower Voltage and capacity (Watts): - → - If defrost system is fitted, specify defrost test temperature (-8 +/- 2 deg C / -18 +/- 3 deg C): → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F	(continued)	(continued)
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point,	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar & related dimensions as per related standards.: → Drawing indicating seat back angle, seat travel, H point, Rake angle, F point, steering wheel position, Driver’s vision points, Angle obstruction of ‘A’ pillar		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	& related dimensions as per related standards.: → Description and sketch showing the detail positions of Tell-Tale and seating layout (For M category vehicles): NA → NA A brief technical description indicating the length of the cleaning period, the consumption of cleaning fluid during the cleaning period and the minimum capacity of the container provided;: -- → -- Description and drawings/photographs	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	of the shapes and constituent materials of the part of the body forming the engine compartment and the part of the passenger compartment nearest to it:: -- → -- Drawings or photographs of the position of the metal components housed in the engine compartment (e.g. heating appliances, spare wheel, air filter, steering mechanism, etc.):: -- → -- List of all subassemblies, which includes an electronic oscillator or switching frequency greater than 9 kHz (like ECU, steering, suspension, Body: -- → -- List of all Electrical components, which include Broadband EMI sources (like HAVC Motor,		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Wiper Motor, Window motor and Horn etc.) The list shall include Device name, Make/s and Part No. /Type ID/Drawing No: -- → -- Name of the manufacturer of the system (Steering column shaft / Gear shift control / Transmission) on which the Protective Device functions: XXYY Steering System → XXYY Steering System Type/Part no./Identification number(ID)/Drawing No.of the device on which the Protective Device functions: FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly →	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	FZY02XYZ – Upper Column FZY04XYZ – Intermediate column BDJ24XYZ- Steering assembly System locked by Protective Device (s) a) Locks the Steering (or) b) Locks the Transmission (or) c) Locks the Gearshift Control d) Locks the Brakes Applicable only in case of steering column locking systems: Locks the Steering → Locks the Steering SYSTEM SOFTWARE (If fitted): NA → NA Make: -- → -- Version: -- → -- Operating System Details with Version: -- → -- DESCRIPTION OF DEVICE (If fitted): NA → NA		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	Description of the VAS: - → - A detailed description of the alarm system and of the vehicle parts related to alarm system installed with Block Diagram: → A list of main components comprising the alarm system: - → - Components placed in Engine compartment: - → - Components placed in Passenger compartment: - → - Availability of radio/wireless transmission Yes/No: - → - Details of radio/wireless transmission if any: - → - frequency: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Availability of WPC approval for wireless technology Yes/No: - → - The measures taken against false alarm: - → - Range of protection offered by the device. Details about the detection mechanism (opening of Door, Bonnet, luggage compartment, intrusion etc.): - → - Details & specification of audible alarm: - → - Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details & specification of audible alarm: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Details & specification of optical alarm: - → - Details & specification of radio alarm if fitted: - → - Method of setting/ unsetting the device: - → - Type of key used for unsetting (Mechanical Key / electronic device): - → - Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - Exit delay in Sec: - → - Entry Delay in sec: - → - Details of status display (0.5cd): - → - Details of self check , automatic failure indication: - → - Details about the panic alarm , if provided: - → - Description of the Immobilizer: - → - A detailed description of the vehicle immobilizer and of the vehicle and the measures		
Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
(continued)	taken against inadvertent activation with block diagram.: → The vehicle system(s) on which the immobilizer acts.: - → - Details of radio/wireless transmission if any: - → - Method of setting/ unsetting the device.: - → -	(continued)	(continued)

Rule / standard / subject	CEA parameters · tested → proposed	Report / CEA source	Justification and logic
	Number of interchangeable codes, if applicable(Supplier to provide undertaking regarding incorporation of rolling code): - → - A list of main components comprising the device: - → - Description: AL Part No. → AL Part No. Description · AL Part No.: Hella Part No → Hella Part No ECU: FXX01000 → FXX01000 ECU · FXX01000: 015.XYZ-01 → 015.XYZ-01 a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - a. · Make: - → - b. · Type: - → - c. · Serial Number: - → - d. · Name and Address of manufacture: - → - Example of Engine No: HJEZXYZ891 → HJEZXYZ891		

Table 11 of AIS-007 (Revision 5)	
DETAILS OF LOCATION OF CHASSIS NUMBER AND CODE FOR MONTH AND YEAR OF MANUFACTURE AS PER RULE 122 OF CMVR	
Name of the Vehicle Manufacturer & Address:	Ashok Leyland Ltd, 1, Sardar Patel Road, Guindy, Chennai – 600 032.
Name of the basic model:	**BADA DOST i9 TNZX**
Name of Variants, if any:	BADA DOST i8 TNZX
Place of Embossing or etching the Chassis Number (Vehicle Identification Number). Supporting details by drawing or pictures may be provided if necessary.	On LH frame between Bump Stop Mounting bracket and Rear Suspension Rear mounting bracket Assy. 500XYZVA5A
Place of PIN Verification plate. Supporting details by drawing or pictures may be provided if necessary	NA
Place of Local information (Importer Plate)- if machine imported. Supporting details by drawing or pictures may be provided if necessary	NA

Manufacturer signature: _____ Test agency: _____

Code for Year of Production: (for 30 years): Digit 10 in VIN

YEAR	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
CODE	T	V	W	X	Y	1	2	3	4	5	6	7	8	9	A
YEAR	2041	2042	2043	2044	2045	2046	2047	2048	2049	2050	2051	2052	2053	2054	2055
CODE	B	C	D	E	F	G	H	J	K	L	M	N	P	R	S

Code for Month of Production: (for 30 years): Digit 12 in VIN

	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
JAN	S	T	V	W	X	Y	1	2	3	4	5	6	7	N	9
FEB	R	S	T	V	W	X	Y	A	B	C	D	E	F	L	W
MAR	P	R	S	T	V	W	X	Y	A	B	C	D	E	K	V
APR	N	P	R	S	T	V	W	X	Y	A	B	C	D	J	T
MAY	M	N	P	R	S	T	V	W	X	Y	A	B	C	F	S
JUN	L	M	N	P	R	S	T	V	W	X	Y	A	B	D	P
JUL	K	L	M	N	P	R	S	T	V	W	X	Y	A	C	N
AUG	J	K	L	M	N	P	R	S	T	V	W	X	Y	B	M
SEP	H	J	K	L	M	N	P	R	S	T	V	W	X	Z	L
OCT	G	H	J	K	L	M	N	P	R	S	T	V	W	Y	K
NOV	D	G	H	J	K	L	M	N	P	R	S	T	V	W	J
DEC	V	W	X	Y	1	2	3	4	5	6	7	8	9	A	B
	2041	2042	2043	2044	2045	2046	2047	2048	2049	2050	2051	2052	2053	2054	2055
JAN	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R
FEB	Y	A	B	C	D	E	F	G	H	J	K	L	M	N	P
MAR	X	Y	A	B	C	D	E	F	G	H	J	K	L	M	N
APR	W	X	Y	A	B	C	D	E	F	G	H	J	K	L	M
MAY	V	W	X	Y	A	B	C	D	E	F	G	H	J	K	L
JUN	T	V	W	X	Y	A	B	C	D	E	F	G	H	J	K
JUL	S	T	V	W	X	Y	A	B	C	D	E	F	G	H	J
AUG	R	S	T	V	W	X	Y	A	B	C	D	E	F	G	H
SEP	P	R	S	T	V	W	X	Y	A	B	C	D	E	D	G
OCT	N	P	R	S	T	V	W	X	Y	A	B	C	D	C	D
NOV	M	N	P	R	S	T	V	W	X	Y	A	B	C	B	C
DEC	C	D	E	F	G	H	J	K	L	M	N	P	R	S	T

Height of the Chassis number (Vehicle Identification Number)	Min. 7 mm on Chassis frame; Min. 4 mm on Manufacturers plate
Example of Engine No	HJEZXYZ891
Example of Chassis No. (VIN) with Month & Year of Manufacture	XYZABCGCD0JRDX737

Manufacturer signature: _____ Test agency: _____

February			
March			
April			
May			
June			
July			
August			
September			
October			
November			
December			

Position of the code for month of production in the Chassis number :	
Position of the code for year of production in the Chassis number :	
Height of the Chassis number (Vehicle Identification Number) :	

Example of Engine/Motor No. :-

Example of Chassis No. (Vehicle Identification Number) with Month & Year of Manufacture:-

Table 12 of AIS-007 (Revision 5)

DETAILS OF CHANGE IN TECHNICAL SPECIFICATION

Manufacturers name and address :		
Name of the model and variants:		
CMVR Certificate No.	Date	Specification No.

Manufacturer signature: _____ Test agency: _____

Table 12 of AIS-007 (Revision 5)

Details of change in technical specification

Continuation document ALP-829383b7-T12-R3-C · TEST – Bada Dost rear axle 5.13

Engineering draft. Manufacturer and test-agency acceptance / signatures remain pending.

Clause / parameter	Existing specification	Proposed specification	Tables / reason
Rear axle ratio	4.56	5.13	Tables 3, 7 · Engineer edited
1st · Overall Ratio / RSB	21.53	23.70	Tables 3 · Derived by 2ad14450722f4ffeb2f9232526c16b93